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Supporting active ageing before retirement: a systematic review and meta-analysis of workplace physical activity interventions targeting older employees

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Supporting active ageing before retirement: a systematic review and meta-analysis of workplace physical activity interventions targeting older employees

Dafna Merom[1], Fiona Stanaway[2], Klaus Gebel[3,2] Joanna J. Sweeting[2], Anne Tiedemann[4], Shirin J. Mumu[1], Ding Ding[2]

1. School of Health Science, Western Sydney University, Penrith, New South Wales, Australia
2. Sydney School of Public Health, University of Sydney, Camperdown, New South Wales, Australia
3. Australian Centre for Public and Population Health Research, School of Public Health, Faculty of Health, University of Technology Sydney, Ultimo, New South Wales, Australia
4. Sydney School of Public Health, Institute of Musculoskeletal Health, The University of Sydney, Camperdown, New South Wales, Australia

Corresponding Author:

A/Prof Ding Ding (Melody)

Prevention Research Collaboration, Sydney School of Public Health

Level 6 The Hub, The Charles Perkins Centre (D17)

The University of Sydney, Camperdown NSW 2006

Tel : +61-2- 8627 1842 F: | **F** +61 2 9036 3184 **E** melody.ding@sydney.edu.au

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Abstract

Objective: With the growing representation of older adults in the workforce the health and fitness of older employees are critical to support Active Ageing policies. This systematic review aimed to characterise and evaluate the effects on physical activity (PA) and fitness outcomes of workplace PA interventions targeting older employees.

Design: Five online databases were searched for articles published from inception to December 2019. Eligible studies were of any experimental design, included employees aged ≥50 years, had PA as an intervention component and reported PA-related outcomes.

Results: Titles and abstracts of 8168 records were screened and 18 unique interventions were included (3,309 participants). Twelve studies were RCTs. Seven interventions targeted multiple risk factors (n=1,640), involving screening for cardiovascular disease risk factors, but had a non-specific description of the PA intervention. Four interventions targeted nutrition and PA (n=1,127), and seven (n=235) focused only on PA. Interventions overwhelmingly targeted aerobic PA, compared to only four interventions targeting strength and/or balance (n=106) all of which were pilot trials. No studies involved screening for falls/injury risk and only two interventions targeted employees of low socioeconomic status. Computation of effect sizes (ES) was only possible in a maximum of three RCTs per outcome. ESs were medium for PA behaviour (ES=0.25 95% CI: -0.07 to 0.56), muscle strength (ES=0.27, 95%CI: -0.26- 0.80), cardiorespiratory fitness (ES=0.28, 95%CI: -22 to 0.78) and flexibility (ES=0.50, 95% CI: -0.04- 1.05) and high for balance (ES=1.08, 95% CI: 0.19 to 1. 97). GRADE criteria-rated quality of evidence was 'low' due to high risk of bias, imprecision and inconsistency.

Conclusions: The lack of high-quality effective workplace PA interventions contrasts the importance and urgency to improve the health and fitness in this population. Future

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interventions should incorporate strength and balance training and screening of falls/injury risk in multi risk factors approaches.

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https://www.crd.york.ac.uk/prospero/display_record.php?RecordID=84863)

Key words: older employees, Physical activity, exercise, workplace, interventions, systematic review, meta-analyses

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Strengths and Limitations of this study

- This systematic review was registered a priori in PROSPERO.
- We followed PRISMA reporting guidelines, the Cochrane Collaboration and the GRADE approach to rate the quality of evidence.
- We used five databases from inception and a backward citation tracking of all included studies.
- Pooled effect sizes that were not expressed in similar units, although they measured the same concept, may have artificially increased the heterogeneity of the effect.

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1. INTRODUCTION

Population ageing poses individual, social, economic and political challenges and is predicted to accelerate even further in the 21st century [1]. The WHO's "Active Ageing" concept is a leading global policy strategy for successful ageing [2], particularly influencing retirement policies towards maintaining engagement with paid employment. In the developed world, strategies have been implemented to promote economic activities among aged workers, including incentives for late retirement, penalties for early retirement, and an increase in or abolishment of the age of mandatory retirement [3]. Older retirement age means that more employees beyond the age of 55 years will remain in the workforce [4].

Because of age-related declines, an older workforce requires special considerations regarding occupational health, safety and productivity [5]. In-depth reviews on ageing at work [6, 7] identified a number of physiological, sensory and psychological differences between older workers and their younger counterparts, including reduced muscle strength, endurance, trunk flexibility, structural and sensorineural degeneration which affect balance, vision and hearing, aerobic capacity, heat tolerance, increased anthropometric risks, psychological exhaustion, anxiety and depression. Further, older employees suffer from greater prevalence of comorbidities, musculoskeletal problems, and sickness-related absence and severe injuries [6, 7].

It is well established that regular physical activity (PA) is effective in attenuating age-related physiological declines in all body systems, and in preventing and managing many age-related chronic and musculoskeletal conditions [8-10]. Hence, maintaining good PA habits is likely to benefit older employees in terms of health as well as work productivity and safety. Hence, recently physical employment standards for older employees have been suggested [7]. The

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workplace has long been considered a good setting for health promotion and many PA interventions have been conducted in this setting, with the first systematic review of their effectiveness published in 1998 [11]. However, we have not identified any systematic reviews of workplace PA interventions that specifically targeted older employees. PA interventions targeting older adults should additionally include activities to promote muscle mass, strength, and balance, which are important for the prevention of sarcopenia, functional disabilities, falls and injuries [12]. A recent systematic review by Poscia et al. synthesised the evidence on health promotion interventions targeting older employees [13]. However, this review examined a variety of health promotion initiatives and have missed interventions that focused on PA, by only using 'fitness' and 'capacity' as search terms. Given that PA guidelines for older adults include a range of physical outcomes relevant to health beyond cardiorespiratory fitness or capacity, such as muscle strength, balance and flexibility [14, 15], it is important to include broader terms when reviewing health promotion programs. Furthermore, the main outcomes included in the review by Poscia and colleagues [13] were health status, wellbeing and work productivity, not PA measures. Thus, the effectiveness of workplace interventions on PA and fitness outcomes of older employees is uncertain.

The objectives of this systematic review were to: i) identify and characterise workplace PA interventions delivered to employees aged ≥50 years; ii) assess the methodological quality of the studies; iii) assess the effect of interventions on PA behaviour and fitness outcomes highlighted in PA guidelines for older adults.

2. METHODS

This review followed guidance published by the Centre for Reviews and Dissemination, the Cochrane Collaboration [16], and PRISMA guidelines [17] (see Supplementary file 3) and

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was registered with the International Prospective Register of Systematic Reviews (PROSPERO; registration number CRD42018084863 available at https://www.crd.york.ac.uk/prospero/display_record.php?RecordID=84863)

2.1 Inclusion and exclusion criteria

We included studies that involved PA interventions delivered at workplaces to employees aged ≥50 years. Studies that included a wider age range were eligible for inclusion if the mean age of participants was at least 50 years. Interventions could include those related to aerobic PA (e.g., walking), stretching, balance, muscle strength, yoga, tai chi, Pilates, gym workout, sport or any other form of PA/exercise. Studies that targeted other health behaviours in addition to PA were only included if PA was one of the key components of the intervention and a PA-related outcome was reported. Studies with any experimental design (pre-post with or without a comparison group, nonrandomised trials, RCTs and cluster RCTs) were included because of the small number of randomised controlled trials (RCT) anticipated [13].

Studies were excluded if the PA intervention was not delivered in the workplace (e.g., community, health practice), or if the interventions targeted the whole workplace with a wider age range and did not include specific PA-related outcomes.

2.2 Search strategies

We systematically searched five databases from inception to 23rd of December 2017. We then updated our search on the 13th of June 2019 to include all 2018 and again on the 17th of February 2020 to include all 2019. Included databases were: Medline, PreMedline, PsycInfo, CINAHL, and the Cochrane Controlled Register of Trials (CENTRAL). Specific search strategies were developed for each database, using a combination of search terms and subject

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headings where applicable. All electronic searches included MeSH terms and keywords for workplace (e.g. occupations, worksite), PA (e.g., walking, postural balance, muscle strength) and intervention (e.g., health promotion, program, RCT) and were limited to humans, and age (PsycInfo: 40+ all others: 45+) (see Supp1). One reviewer (JS) retrieved the entries from the database using EndNote library and screened the titles and abstracts to identify potential papers for inclusion as well as running backward citation tracking of potential studies selected for full text review. Two reviewers (JS, DM) independently conducted the final selection of papers based on full text. Disagreements were discussed and resolved by consensus.

2.3 Data extraction

Three data extraction formats were used: A table summarising the included studies was created using the subheadings: source, study design, population characteristics, study duration, type of intervention, comparison group, and outcomes. A second table to address research aim 1 described characteristics of each intervention in terms of recruitment, content, intervention deliverer and process outcomes based on data extracted by DM and SM. A third table summarised the quality of the included papers based on the Cochrane risk of bias tool for RCTs and additional domains from the ROBINS-I tool for non-randomised studies with a comparison group [18]. The domains included were randomisation sequence generation, allocation concealment, blinding of participants and personnel, blinding of outcome assessment, incomplete outcome data and selective outcome reporting. The first two domains were only assessed in RCTs. In non-randomised studies with a control group the additional domains of confounding and selection bias were assessed.

2.4 Quality assessments and analysis

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Each domain was assessed independently by two authors (DD, FS) who assigned a rating of either 'low', 'high' or 'unclear' risk of bias. Disagreement was resolved by consensus or, where needed, by consultation with a third reviewer (DM). We used Comprehensive Meta-Analysis software (Version 2, Biostat, Englewood, New Jersey, USA) to conduct random effects meta-analyses for each outcome accounting for homogeneity between studies. The standardised mean difference (SMD) (Hedges' g) was calculated for each meta-analysis, standardised by postscore SD (or its estimate) and calculated using the pre- and post- means and SD or, when unavailable, the mean change score and SD. Effect sizes were categorised as small (≤0.20), medium (> 0.20-0.5) or large (≥0.8) [19]. For trials with moderate-to-vigorous intensity PA as the outcome, we used data from accelerometers when possible. For studies reporting energy expenditure as the outcome, we converted the value to minutes of moderate-intensity PA using the population mean body weight and 4 Metabolic Equivalents for Task (METs) for moderate-intensity PA. Statistical heterogeneity was determined by the I^2 and χ^2 tests. We followed the recommendation of the Cochrane Collaboration which states that tests for funnel plot asymmetry should be used only when there are at least 10 studies included in the meta-analysis [16]. Since we had no more than 3 studies for each outcome, we did not assess publication bias. Finally, we used the Grading of Recommendations Assessment, Development and Evaluation (GRADE) approach to rate the overall quality of evidence for each outcome from 'very low' to 'high' [20, 21], including a narrative summary of findings for outcomes without meta-analysis [21]. GRADE ratings are based on the domains of risk of bias in included studies, indirectness of evidence, imprecision, inconsistency and likelihood of publication bias.

Patient and Public Involvement

Neither patients nor the public were involved in the design, conduct, reporting or dissemination of this research.

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3. RESULTS

3.1. Study selections and characteristics

Our search resulted in 8,168 records which were screened by title and abstract. Of these, 153 were further assessed for eligibility based on full text (Figure 1). Nineteen studies met the inclusion criteria, however, two publications referred to the same intervention [22, 23], resulting in 18 unique interventions.

Insert Figure 1

The characteristics of the studies are summarised in Table 1. Ten studies were RCTs [24-33], and two were cluster RCTs - one involving only two worksites [34] and the other several university units [35]. There were also two non-randomised studies [36, 37] and four pre-and post-evaluations with no comparison group [38-41].

3.2 Participants

In total, 3,309 participants (of those 2085 participated in RCTs) were included in the 18 worksite PA interventions: six in the USA [24, 26-28, 35, 41], two in the UK [30, 38], two in Australia [33, 39] and one each in Taiwan [37], Japan [36], Poland [29], the Netherlands [22, 25], Switzerland [34], Canada [31], Sweden [32], and Italy [40].

Five interventions were delivered in academic institutions [24, 29, 31, 33, 35], four in hospitals or medical centres [25, 28, 40, 41], and four were delivered to factory employees [30, 36-38]. One intervention was delivered to transport workers (drivers) [39], and four were delivered to office workers [27, 32-34]. Most studies (n=12) targeted both genders, with a high percentage of male participants (>60%) in transport [39] and IT workplaces [27], and fewer male participants (5% - 25%) in academic / administrative [24] and medical occupations [25, 31, 41]. Three interventions targeted females only [26, 28, 29] and one males only [36].

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Table 1: Characteristics of the studies that met the inclusion criteria in worksite older employees' physical activity (PA) interventions arranged by design and year

	Design Sample	Population	Duration	Intervention Description	Comparison	Physical activity related outcomes
Bassey et al. 1983 [30]	RCT n=108	UK, factory floor workers (blue collar) from light industrial company, age 55-60, 51% males	12 weeks	PA focus; prescribed aerobic (walking) program with a goal to increase cardiorespiratory fitness	No intervention	1) Physical condition - heart rate (HR) at walking 4.8km/hr 2) Sustained $HR_{4.8km/h}^{-1}$ for at least 7 minutes 3) Daily total minutes walked at $HR_{4.8km/h}^{-1}$ 4) Step count
Sharpe et al. 1992 [35]	Cluster RCT n=250	USA, employees from 'support and academic staff" at the University of Michigan, age 50-69, 53% males	12 months	Multiple risk factor program; after health risk screening a face-to-face meeting with health promoter; optional: walking groups at work/ or use facilities.	Health screening, but no exposure to health promotion program	1) Self-report frequency item with a 5-point response format -which was not explained or referenced. 2) Self–reported action taken to improve fitness in the past year 3) Precursors of exercise behaviours (self-efficacy to exercise 3 times a week and intention to exercise vigorously
Hughes et al. 2011 [24]	RCT (3 arms) n=423	USA, older employees from "support and academic" staff at the University of Illinois, Chicago Age ≥ 55 years 18% males	12 months	Multiple risk factor program to; 1) Enhance Wellness one-on-one coaching program (COACH) 2) a web-based health promotion program (RealAge)	Printed health promotion materials, programs and services offered by university or by community organisations	1) Self -report minutes of moderate-intensity and minutes in vigorous-intensity Using the Behavioral Risk Factor Surveillance System (typical week -7 questions) 2) The Rapid Assessment of Physical Activity (RAPA) for meeting PA recommendations based on 9 items (aerobic muscle training and flexibility, response: yes=1, no=0 combined to total score)
Strijk et al. 2011 [22] & 2012 [25]	RCT n=730	Netherlands, workers from two academic hospitals Age ≥ 45 years Mean age 52.3±4.9 25% males	6 months	PA & nutrition: Vital@work Yoga session 60 minutes and workout classes 30 minutes plus home based aerobic session	General healthy lifestyle written information – diet, physical activity and relaxation	1) The frequency, duration and intensity of participation in commuting, household, occupation and leisure-time PA using the SQUASH [a] questionnaire: 2) Minutes spent on sport-from SQUASH 3) Accelerometer minutes in MVPA on sub-sample (GTM1 Actigraph) 4) Aerobic capacity (VO_2 max) was estimated using the UKK [b] 2 km walk test.

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	Design Sample	Population	Duration	Intervention Description	Comparison	Physical activity related outcomes
Palumbo et al. 2012 [26]	RCT n=14	USA, female nurses from an academic medical centre in Vermont Age ≥ 49 years 0% males	15 weeks	PA focus: Once a week Tai-chi guided classes at work;45 minutes and unguided home-based practice 10-mins 4 times a week	No intervention	1) Flexibility of trunk "Sit-and-Reach" test 2) Strength: isometric knee extensor test with dynamometer 3) Balance-Functional Reach test (cm)
Cook et al. 2015 [27]	RCT n=278	USA, IT employees from two major global IT companies in California and Boston Age – 50 to 68 years 67% males	12 weeks	Multiple risk factor web-based educational program "HealthyPast50" with no human contact.	No intervention Wait-listed	1) Exercise habits: based on Godin LTPA score measuring frequency per week exercising at 3 levels (mild, moderate strenuous) and the number multiplied by intensity factor –reported frequency at each level and in total 2) Exercise self-efficacy score (8-items) 4-point scale 3) Exercise intention
Low et al. 2015 [28]	RCT n=62	USA, female employees of a busy community hospital, North Carolina, 40-65 years old; mean age 52 ± 6.3. 0% males	6 months	Multiple risk factor program to reduce CVD [c] risk as in control, plus weekly motivational sessions by mail or telephone for goal setting	Risk reduction educational classes; Free access to gym on site and walking group	1) Frequency – days per week of exercising 2) Minutes per session of exercising 3) Level of intensity: no exercise, leisurely, moderate, or vigorous exercise (reported as %) 4) Readiness to change exercise
Granacher et al. 2011 [34]	Cluster RCT n=32	Switzerland, sedentary office workers from two large companies in Basel, Age ≥50 37% males	8 weeks	PA focus; progressive balance and strength training at the office using exercise charts to perform 3 times a day; each session lasts 8 minutes	No intervention	1) Balance (static) postural control using balance platform, standing on one leg 30 seconds eyes open– displacement of COP[d] 2) Gait variability and speed on special treadmill 3) Force jumping height measured on a force platform 4) Maximal isometric & isokinetic torque (60° / and rate of torque development (RTD) of the plantar flexor using force platform

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	Design Sample	Population	Duration	Intervention Description	Comparison	Physical activity related outcomes
Kocur et al. 2017 [29]	RCT n=44	Poland, female administrative and academic office workers from 4 higher education institutions in Poznań Age 50-60 years 0% males	12 weeks	PA focus - Nordic walking training program 3 times a week for 1 hour each	No intervention	1) Perceived Pain Threshold of upper body (PPT, kg/cm^2) (i.e., the minimum force that can be applied which induced the feeling of pain) using an electronic pressure algometry 2) Flexibility (shoulders) using back scratch: measuring how close the hands can be brought together behind the back
Chopp-Hurley et al. 2017 [31]	RCT n=24	Canada, Hamilton University employees (McMaster) with clinical osteoarthritis Age > 50 years 21% men	12 weeks	PA focus – Specific Osteoarthritis exercises: static leg strengthening (e.g., yoga poses) 3-4 times a week 7-8 AM	No intervention	1) Hip and knee strengths measured by dynamometer the peak extension and flexion torque out of 5 trials were recorded normalised to body weight 2) 6-minutes walking test 3) Lower limb strength: chair sit to stand during 30 seconds 4) Timed –up-and go test 5) 40meter fast paced (time to complete)
Bergman et al. 2018 [32]	RCT n=80	Sweden, Umea office workers (13 companies). Overweight and obese Age 40-67 years (Mean age 51yrs) 45% males	13 months	PA focus – treadmill workstation recommended to walk 1hr a day at moderate-intensity, but no jogging	Working as usual at their sit-and stand working desk	1) Daily walking time at weekdays and weekend measured by ActiPAL accelerometer 2) Number of steps ActiPAL 3) Time spent in moderate to vigorous PA by Actigraph accelerometer 4) PA bouts of more than 10mins on weekdays and weekend
Qi et al. 2019 [33]	RCT n=40	Queensland, Australia University insufficiently active office workers Age 55-77 (mean 59.6) 22% males	12 weeks	PA focus Taichi movements (10) performed with resistance band (Theraband) 3x per week 45min per session on-site. Employee could choose from 8 different time slots	Active control: Taichi only without the theraband in different room to intervention. Similar number of sessions were offered	1) Lower limb strength using chair sit to stand during 30 seconds 2) Balance Functional Reach test (cm) 3) Hand grip strength 4) 2-minute walk test (work capacity)

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	Design Sample	Population	Duration	Intervention Description	Comparison	Physical activity related outcomes
Chen et al. 2016 [37]	Quasi-experiment (individual allocation) n=108	Taiwan, workers from small- and medium-scale enterprises, Age ≥ 50 years 45% males	6 months	Multiple risk factors; onsite educational workshops during first 4 weeks + meetings with OHN[e] to set goals, plans and on-site group support	Only educational workshops in the first 4 weeks	1) Frequency: number of times per week doing physical activity using Taiwan Longitudinal Study on Aging - no information on the PA domains asked (e.g., leisure, work etc) 2) Sedentary time (hrs/d)
Arao et al. 2007 [36]	Quasi-experiment (cluster allocation) n=197	Japan, employees working in 5 sites of two factories in Tokyo who had at least 1 CVD[c] risk Age 40-59 years (mean age=55) 0% females	6 months	PA & nutrition (LiSM[f]-PAN) multi-components program; individual counselling 5x10-min around goals; social and environmental support	Feedback from the medical check-ups plus recommendations on diet and physical activity including printed materials on exercise, healthy diet and cooking	1) Self-report energy expenditure (kcal/week) derived from the leisure time exercise was assessed by the Kuopio Ischaemic Heart Disease Risk Factor Study 2) Maximum oxygen uptake 3) VO_2max (ml/kg/min) from a sub-maximum bicycle test (Astrad) 4) Stage of change for exercise
Abbas et al. 2015 [38]	Pre-post n=665	UK, low-paid local government employees from socially and economically deprived areas in NE England, Age ≥40 years, mean age 50.5±6.4 37% males	9 months	Multiple risk factor health screening staging risk level and referrals to exercise, weight management, smoking cessation, promotion of mental health and alcohol reduction	Not applicable	1) Participation in aerobic exercise dichotomised to not meeting recommendation (less than 5 times a week of less than 30-minutes session) 2) Referred to exercise question on doing exercise outside work (report on % before and after)
Naug et al. 2016 [39]	Pre-post pilot trial n=33	Australia, bus drivers from two depots of South East Queensland Age: 50-68 years 64% males	6 weeks	PA & nutrition, Group educational sessions; harm of sitting, healthy eating & exercise; PA - pedometer to track steps	Not applicable	1) Exercise levels – no report on questionnaire type; researcher driven classification: a) none (no exercise), b) moderate level (e.g. 30-40min walking twice a week or tennis once a week, c) intense (e.g., gym 4-7 times a week or cycling 5 days/week), sedentary behaviour (hrs/week)

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	Design Sample	Population	Duration	Intervention Description	Comparison	Physical activity related outcomes
Scapellato et al. 2018 [40]	Pre-post n=167	Padua, Italy healthcare workers at risk of CVD Mean age 50±7.3 31.7% males	6 months	PA & nutrition, brochure on exercise and motivational counselling on site at baseline and mid-term by phone	Not applicable	1) PA MET[h] based on a 4-day diary about the type of activity, frequency (day per week) and duration (minutes)
Edman et al. 2019 [41]	Pre-post pilot trial n=54	Philadelphia, USA healthcare hospital workers volunteered with at least one CVD risk Mean age 53.3±10 5% males	6 months 12 weeks cycles with option to continue	Multiple risk factor program - six sessions of Health Coaching f2f or by phone including goal setting, education, motivational strategies	Not applicable	1) Self-reported change in number of sessions per week exercising aerobically for at least 20 minutes 2) Self-reported number of times per week doing muscle strengthening exercise for at least 20 minutes

[a] SQUASH –Short Questionnaire to ASsess Health- enhancing physical activity; [b] UKK- The Finnish Urho Kaleva Kekkonen walking test; [c] CVD – cardiovascular disease; [d] COP – Centre of Pressure; [e] OHN – Occupational health nurse; [f] LiSM – Lifestyle Modification; [g] GHQ – General Health Questionnaire; [h] MET- Metabolic Equivalent

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3. 3 Characteristics of PA interventions

3.3.1 PA types and targeted fitness area

Table 2 summarises the characteristics of the interventions (more details are provided in Supplementary Table 1). Interventions overwhelmingly focused on aerobic PA and other types of PA relevant for older people, such as balance, strength and flexibility, were rarely addressed.

Seven studies addressed multiple cardiovascular risk factors [24, 27, 28, 35, 37, 38, 41], four targeted PA and nutrition [25, 36, 39, 40] and seven focused only on PA [26, 29-34]. Nearly all multiple risk factor interventions used a generic description of the PA intervention such as 'exercise on their own' [35], 'exercise training' [28], referrals to exercise classes [38], or choosing PA goals [24, 27]. In the studies by Hughes et al. [24] and Low et al. [28] the interventions also offered on-site self-managed activities such as walking groups or using workplace exercise facilities.

In addition, in these multiple risk factor interventions, some forms of 'risk assessment', such as a nurses' check-up for CVD risks, were applied before the targeted behaviour was selected by or for participants [38] or self-appraisal of risk using questionnaires [24, 27], or assessment carried out during the first educational session [37, 41]. No study assessed risk of falls or injuries.

The PA and nutrition interventions focused on aerobic regimens such as steps accumulation [25, 36, 39], counselling to shape PA goals [40] or an aerobic workout on site [25]. The study by Strijk et al. [25] also offered yoga sessions, an activity that is recognised as multi-dimensional and which enhances muscle strength, balance and mobility, but these fitness outcomes were not assessed [42].

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The PA-only interventions either focused on aerobic regimens, such as use of a treadmill workstation [32], self-managed aerobic walking [30] or Nordic walking [29]. Although Nordic walking is primarily aerobic activity, it can also increase upper body strength and flexibility (Table 2), we therefore consider the study multi-dimensional [29]. Two studies offered tai-chi classes, which is also considered multi-dimensional PA (tai-chi) [43]. One study invited older office employees to novel tai chi combined with theraband exercises, adding progressive strength training to tai-chi [33]. The other study offered tai-chi classes to nurses as part of a wellness program [26]. The study by Granacher et al. offered progressive strength and balance exercises performed in the office. At the beginning of each week the trainer introduced the series of exercises to be performed three times a day, five days a week [34].

3.3.2 Delivery modes

Three delivery modes were noted (Table 2). In eight studies (mostly PA only interventions) the PA intervention was delivered on site during working hours [25, 26, 28, 29, 32-34], or on site but before working hours [31], in six studies participants were counselled in the workplace in face-to-face meetings [35-37], or by telephone calls (COACH-arm) [24], or a combination of both [40, 41], but the PA sessions took place in employees' discretionary time. In four studies the PA was self-managed with little or no supervision [27, 30, 38, 39]. The study by Hughes et al. [24] had two intervention arms; one involved personal contact with students trained in behaviour change coaching ("COACH"), while the other arm was self-managed through a web-based intervention ("RealAge") with no other contact [24]. The study by Cook et al. [27] was also a web-based self-managed intervention (HealthyPast50).

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3.3.3 Intervention duration and compliance

Eight interventions were short-term, ranging from 6 to 15 weeks [26, 27, 29-31, 33, 34, 39]. Seven interventions lasted between 5 and 9 months [25, 28, 36-38, 40, 41], and three interventions lasted up to 13 months [24, 32, 35]. All studies provided data related to intervention retention rates (i.e., withdrawals) with the exception of one pre-post study [38]. Seven studies did not report intervention compliance (e.g. attendance at classes, recorded logs). Studies of short duration also reported higher compliance (>75%) compared to studies with medium or long duration. Four studies reported on program implementation fidelity [24, 26, 38, 40].

3.3.4 Intervention reach

Program reach (i.e., proportion of study population recruited to the trials) varied substantially. When recruitment methods involved a targeted strategy (TS) based on prior 'screening for eligibility criteria', a strategy reported in five studies [30, 36-38, 40], intervention reach was highly variable. In the Japanese study, 92% of the targeted eligible employees were recruited [36], while two interventions in the UK [30, 38], the Taiwanese study [37] and the Italian study [40] using the same strategy achieved much lower recruitment rates (ranging from 20% to 35%). In the ten studies that used workplace advertisement as the method of recruitment [22, 24, 26-29, 31, 33, 35, 41], only two studies reported on the reach of the program: Strijk et al. invited all employees listed in the targeted age range (n=3756) and indicated that 27% of those agreed to participate [25]. Cook et al. [27] offered a financial incentive ($25) for completing the baseline survey, which 96% of the employees aged >50 years did, resulting in 276 participants [27]. The rest of the studies either did not report or could not estimate reach because key information, such as the number of eligible potential participants exposed to the advertisement, was missing.

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Table 2: Summary of the characteristics of workplace physical activity interventions delivered to older employees (18 studies)

	Multiple risk factors including physical activity							Physical activity & diet				Physical activity only						
Intervention characteristics	Abbas 2015	Chen 2016	Cook 2015	Hughes 2011	Low 2015	Sharp 1992	Edman 2019	Arao 2007	Strijk 2011 2012	Naug 2016	Scapel-lato 2018	Bassey 1983	Grana-cher 2011	Palum-bo 2012	Kocur 2017	Chop Hurley 2017	Berg-man 2018	Qi 2019
Aerobic workout, walking, steps accumulation				X	X	X		X	X	X		X	X				X	
Nordic walking															X			
Balance exercise													X					
Muscle strength													X			X		X
Flexibility																		
Multi-dimensional PA									Yoga					Tai-chi		Yoga poses		Tai-Chi+
Generic description 'exercise goals', referral to class	X	X	X	X	X	X	X				X			X				
Delivery mode [a]	SM	IC	SM	IC/SM	OS	IC	IC	IC	OS	SM	IC	SM	OS	OS	OS	OS	OS	OS
Duration [b]	M	M	S	L	M	L	M	M	M	S	M	S	S	S	S	S	L	S
Recruitment method [c]	TS	TS	WA	WA	WA	WA	WA	TS	WA	I	TS	TS	I	WA	WA	WA	I	WA
Process outcomes [d]	R 64% C - no report F	R 96% C-no report Q	R 80% C 72%	R_{arm1} 91% R_{arm2} 51% C_{arm1} 97% C_{arm2} 57% F	R 28% C Means ses/wk very poor	R 63% C 73%	R 70% C Mean ses/per trial 7.5 out of 12	R 95% C 55%	R 80% C 73%	R 64% C- no report Q	R 53% C – no report F	R 54% C 33%	R 100% C 99%	R 71% C 82% F	R 91% C- no report	R 75% C Means ses/wk 1.2 (poor)	R 85% C – no report	R 75% C- no report
PA-related outcomes [e]	NA	+	+	-	-	-	NA	+	NA	-	NA	+	$+_1$	+	$+_1$	$+_1$	+	+

[a] Delivery mode: OS-on site PA sessions, SM - self-managed program, IC -individualised counselling
[b] Duration: S -short ≤ 15 weeks, M - medium 5-9 months, L - long >9 months
[c] Recruitment: TS-targeted strategy (e.g., screening), WA – worksite advertisement emails, posters, word of mouth inviting participation, I –information sessions and volunteers to take part
[d] Process outcome: R - retention of intervention participants, C - compliance % from ideal target, F - fidelity of intervention delivery, Q - qualitative comments
[e] NA = if no comparison group + = significant between groups positive effect/s in favour of interventions, $+_1$ = significant between groups positive effect in favour of intervention only in one PA-related outcome "-" = no significant between group effect on PA measure

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3.4 Study quality

Included RCTs and non-randomised studies were mainly of poor quality (Supplementary Table 2). In 5 of the RCTs the randomisation was poorly described and in 7 of the RCTs the concealment of allocation was poorly described. Only five out of 12 RCTs were deemed at a 'low risk' of selection bias [25, 27, 31, 32]. Performance bias is expected due to inability to blind participants, as would be expected in trials involving behaviour modification. Four of the 12 RCTs relied only on self-reported PA-related outcomes [24, 27, 28, 35], resulting in a high risk of ascertainment bias. Eight trials included objective measures of PA or fitness outcomes; of those, four involved less than 40 participants [26, 29, 31, 34]. An additional major concern was the risk of selection bias due to attrition given that most trials (78%) had incomplete outcome data, and intention-to-treat analysis was done in only five RCTs [25, 27, 31, 32]. The two included studies that were not RCTs had a high risk of bias in three, respectively four out of the five domains.

3. 5 Effectiveness of interventions

Table 3 summarises the effect of the interventions on PA behaviour and fitness outcomes, including pooled effect sizes where meta-analysis was possible, also presented in Supp Figure 2. Overall, no pooled outcomes showed statistically significant differences between intervention and control groups.

Three studies that examined PA/exercise frequency [27, 28, 37] had a pooled SMD [27, 28, 37] of 0.25 (95% CI -0.07 to 0.56) and moderate heterogeneity (I^2=53%, p=0.19). The study that was not included in the meta-analysis also reported non-significant differences between groups [35]. Time spent in moderate-to-vigorous PA was measured by self-report in three studies [24, 25, 36] and objectively by accelerometers in two studies [25, 32]. The SMD was

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computed from the studies using accelerometers [25, 32] and the study by Arao et al. [36] used self-report) because all these studies presented results at 6-month follow-up. The pooled SMD was 0.22 (95%CI -0.05 to 0.50) with acceptable heterogeneity (I^2=46%, p=0.16). Furthermore, the study by Hughes with accelerometer-measured outcomes also had non-significant findings [24]. Step count (data not shown) was assessed in two studies; a significant between-group difference in step count per day was reported by Bergman et al. [32] which was equivalent to SMD of 0.22. In the study by Bassey et al. [30], step count per day was available for only half the sample, but the author did not report the results by allocation because they found no between-group differences.

Aerobic fitness using VO_2 max was assessed in two studies [25, 36] with a pooled SMD of 0.28 (95% CI -0.22 to 0.78) and high heterogeneity (I^2=82%, p=0.02); in the study by Strijk et al. [25] a small proportion of the sample took the test, introducing performance bias. In the study by Arao and colleagues [36], the participants were not randomised, introducing potential selection bias and confounding.

Two small-scale studies (n=57) reported the effect of the intervention on muscle strength (i.e., plantar strength and knee flexor) with an SMD of 0.27 (95%CI -0.26 to 0.80) [31, 34]. Another study included isometric knee extensor strength as an outcome, but did not provide data, suggesting selective reporting [26]. Three small-scale studies (n=46) reported on static balance (measured by postural control [34] and functional reach [26, 33]) with a significant high pooled effect size (SMD=1.02, 95% CI: 0.19 to 1.97) and high heterogeneity (I^2=71.0%, p=0.069). Two studies (n=58) examined the effect of the intervention on flexibility in trials of tai-chi [26] and Nordic walking [29] and found a moderate effect size (0.50) without heterogeneity, but that was also statistically non-significant (95% CI -0.04 to 1.05).

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GRADE ratings of the certainty or quality of evidence for each outcome were low, mostly due to the high risk of bias of included studies, high heterogeneity (inconsistency), and imprecision (all pooled Cis, apart from balance, included 0 and were statistically non-significant).

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Table 3: Pooled effect sizes for PA behaviour and fitness outcomes with GRADE rating of evidence in trials with comparison group

PA outcomes References	Pooled effect size Standardised Mean Difference	Numbers (studies)	Quality of evidence (GRADE)	Comments related to GRADE and SMD
Frequency PA /exercise (per week) [27,28,37]	SMD=0.25 95% CI: -0.07 to 0.56 Heterogeneity=54%, *p*=0.114	N=448 (3 studies)	Very Low ⊕OOO Serious methodological limitations; serious indirectness; inconsistency with studies not included	All interventions were multiple risk factors and were self-managed; the exact nature of the exercise intervention and the goals set for people were unclear, limiting attribution of increases to the intervention. The RCT by Sharpe et al.[35] was not included as it was not possible to extract SMD from the mixed model.
Moderate-to-vigorous physical activity (MVPA) [25,32,36]	SMD=0.22, 95% CI: -0.05 to 0.50 Heterogeneity=46.2%, *p*=0.16	N=404 (3 studies)	Low ⊕⊕OO Two RCTs used accelerometer data and one self-report non-randomised trial and therefore reduced overall quality	Additional study by Hughes et al. [24] was not included as it was not possible to extract SMD from mixed model with interactions term. The study reported no between-groups significant effect.
Aerobic fitness, VO_2 max [25,36]	SMD=0.28 95% CI: -0.22 to 0.78 Heterogeneity=82%, *p*=0.019	N=389 (2 studies)	Low ⊕⊕OO Serious methodological limitations; inconsistency	Strijik et al. [25] had high loss for this measure and high risk of performance & ascertainment bias; Arao et al. [36] was a non-randomised controlled trial with high risk of bias due to confounding.
Balance [26,33,34]	SMD=1.02 95% CI: 0.19 to 1.97 Heterogeneity=71%, *p*=0.069	N=86 (3 studies)	Low ⊕⊕OO Serious methodological limitations; serious imprecision; inconsistency	Granacher et al.[33] used balance platforms which has greater precision and also measures horizontal and vertical planes. Palumbo et al. [26] and Qi et al.[33] only measured horizontal displacement (functional reach).
Muscle strength [31,34]	SMD=0.27 95% CI: -0.26 to 0.80 Heterogeneity=0%, *p*=0.380	N=57 (2 studies)	Low ⊕⊕OO Serious methodological limitations; serious imprecision	The trial by Palumbo et al [26] was not included despite listing isometric knee extension because the results were not presented (selective reporting). Chopp-Hurley et al. [31] recruited older employees with osteoarthritis; Granacher et al. [34] did not provide information to judge the trial quality;
Flexibility [26, 29]	SMD =0.50 95% CI: -0.05 to 1.05 Heterogeneity=0%, *p*=0.366	N=58 (2 studies)	Low ⊕⊕OO Serious methodological limitations; imprecision	In both RCTs [by Plumbo et al [26] and Kocur et al [29] the randomisation process was unclear; small sample sizes and high risk of performance bias and selective reporting

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4. DISCUSSION

To our knowledge this is the first systematic review and meta-analysis of workplace PA interventions specifically designed for older employees. Although some interventions demonstrated a significant effect on one fitness outcome, collectively we found no definitive evidence on effectiveness as pooled effects remained non-significant. The methodological quality of the studies was generally low. Only one pre-post study examined whether participants met aerobic recommendations [38] and only four small-scale trials focused on balance or muscle strength as any outcomes and all were pilot studies [29, 31, 33, 34]. No program fully addressed the WHO PA older adults' guidelines, which highlight the need to incorporate balance exercises (≥3 times a week) and muscle strengthening (≥2 times a week) along with the aerobic recommendations of a minimum of 150 minutes per week of moderate-to-vigorous PA. Therefore, there is a noticeable mismatch between the aims and outcome measures of interventions and those recommended as ideal for older adults. Additionally, this review of the literature suggests that the initial levels of participation (i.e., intervention reach) was mostly low, which was also noted in another systematic review of workplace health promotion interventions [44].

High heterogeneity makes the pooled estimates difficult to interpret as it is unclear whether population characteristics, intervention implementation or methodological quality issues underlie the heterogeneous results across studies. For example, two cluster RCTs reported changes at 6 months in predicted maximal oxygen uptake; the Japanese study reported a significant medium effect size based on a sub-maximal stationary bike test [36] and the Dutch study [25] reported a small non-significant effect on the 2km walking test. The difference between these two measures could explain the high heterogeneity as well as differences

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related to the completion rate; high in the Japanese study (85% and 93%) and low in the Dutch study (56% and 66%) in the control and intervention groups, respectively.

Multiple risk factor interventions incorporated screening for cardiovascular risk factors, but did not address other age-related health problems, such as falls, which are preventable by appropriate exercise. Such screening could be carried out either through a single self-report question (e.g., history of falls) or through assessment of physiological fall risk (e.g., poor balance, impaired leg strength). Risk of falls and related injuries is an important safety issue because the consequences of falls and injuries are greater in older employees compared to their younger counterparts [6]. Further, despite emerging evidence that resistance training is as beneficial as aerobic training for the prevention and management of cardio-metabolic risks [45, 46], none of the multiple risk factor interventions explicitly included muscles strength training to reduce CVD/metabolic risk. Muscles strength training was promoted to employees with osteoarthritis, in accordance with clinical guidelines [31], but only one intervention specifically targeted muscle strength to generally healthy older employees [34], supporting the recent claim that strength training has been a neglected aspect of PA promotion [47]. From the mid-forties onwards, each decade adults lose 14% to 16% of their isokinetic leg strength [48], which is a greater decline than that observed in cardiovascular capacity (~10% per decade) [49]. Considering that employees aged 50 years and over are likely to stay at work for another decade or more, neglecting this domain is unwarranted. In fact, tests of older university employees (mean age 63 years; n=258) in Brazil found that 55.8% and 6.3% of them were classified in the “sarcopenia” and “severe sarcopenia” groups, respectively. In the same study, the prevalence of frailty was 9% and 63% were classified as “pre-frail” [50].

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The included PA programs varied substantially in terms of the PA modality, delivery methods and duration. Interestingly, compared with self-managed programs carried out in participants' discretionary time, on-site supervised sessions during the workday [25, 26, 29, 33, 34] tended to demonstrate high retention rates (71% to 100%) and good compliance. However, these programs were usually shorter in duration (less than 15 weeks). On-site pragmatic approaches, such as treadmill workstations, short breaks for balance and strength training or web-based interventions, can be easily integrated into the workday. However, the effectiveness of treadmill workstations was disappointing [32], and the collective evidence for other approaches was inconsistent [24, 27, 34]. These interventions should be further tested by high-quality replication studies before translational research can be conducted.

A concerning finding is the poor quality of the RCTs we identified. The study quality could be improved by appropriate randomisation, intention to treat analysis, and the use of both objective and self-report measures of PA. For example, most multiple risk factor interventions included objective measures of CVD risks (i.e., blood tests or anthropometrics), but failed to use objective fitness tests or activity trackers (e.g., pedometer, accelerometer). Further, to improve the directness of evidence (i.e., the selection of outcome measures should better reflect the fitness dimension(s) that the proposed PA is likely to affect), such as measuring balance and strength outcomes in interventions involving yoga or tai-chi or aerobic fitness in Nordic walking interventions.

Research demonstrates increases in leisure-time PA after transition to retirement compared to during full employment [51, 52]. However, a systematic review has found disparities by socioeconomic status (SES); employees with low SES tend to be less active after retirement whereas those with high SES tend to be more active after retirement [52]. Focus groups with

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older retirees revealed that retirees from lower SES appeared to place lower value on the importance of leisure-time PA [53]. Therefore, improving PA whilst in the workforce may be of even greater importance to older people with low SES. In this review we identified three studies that specifically targeted low paid employees in manual occupations (i.e., factory employees) in Japan [36] and the UK [30, 38]. Only the study from Japan, which included social and environmental support in addition to behavioural counselling, demonstrated significant positive effects. However, this was a non-randomised study with a high risk of bias for almost all bias domains. Further work is therefore needed to ascertain what types of interventions can most effectively improve PA outcomes in older employees with low SES.

4.1 Strengths and limitations

This review applied guideline-informed, rigorous methods to synthesise and evaluate the current evidence on workplace PA interventions for older employees. Compared with the 2016 systematic review by Poscia and colleagues [13], we have expanded upon the list of workplace health promotion interventions delivered to older employees by 11 studies [28-32, 34-36, 38-40] by carrying out a more sensitive and comprehensive literature search. However, some limitations need to be acknowledged: first, we limited our search to electronic databases which can result in missing unpublished workplace interventions. Second, studies tended to apply different measures or used different units for similar measures. Despite our best attempt to harmonise measures (e.g., converting kcal to minutes of at least moderate-intensity PA), comparability across studies is still limited and may explain the high heterogeneity; for example measuring postural control by balance platform as opposed to a Functional Reach Test. In addition, the current evidence base is limited by the quality of existing studies. Based on the limitations we have identified in the literature we

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have mapped out research gaps and suggestions for the conduct and reporting of future interventions targeting older employees (Table 4).

Table 4: Recommendations for the conduct and reporting of future workplace interventions targeting older adults

Aspect	Recommendation
Reach and recruitment	1. Employers' endorsed invitations to the target population (e.g., age or year prior to retirement), to enable calculation of the reach of workplace interventions and may boost recruitment
Population	2. Expand the current target population (ie. mostly university and healthcare staff) to include a broader range of occupations, including manual jobs and workers from low socioeconomic status (SES) background. This is important for reducing gaps between high and low SES workers in lifestyle behaviours after retirement.
Intervention content	3. Incorporate strength, balance, and flexibility training along with aerobic activities. Multi-dimensional interventions addressing the broader physical activity recommendations for older adults are recommended. 4. Moving interventions from a "pilot" stage to larger trials and scalable programs; 5. Intervention target health risk should include screening for falls and injury risk, fragility and sarcopenia 6. Improve the reporting on the nature of physical activity interventions. For example, for interventions that involve a referral to a gym or other facility, describe precisely what exactly these referrals included: advice only, a written program, or a gym membership 7. Improve reporting of the implementation fidelity of interventions
Study methodology	8. Cluster RCTs should include several clusters (workplaces) and should adjust for cluster effect; 9. Quasi-experimental design should assess and adjust for confounders. 10. Interventions that incorporate multi-dimensional activities (e.g., yoga, taichi, dance) should assess the fitness outcomes expected to be improve by these activities (leg strength, balance, gait speeds). Harmonising measures across studies. 11. Improving the conduct / reporting of RCTs in terms of methods of randomisation, concealment of allocation, blinding the assessors and analysts, reporting results for all randomised participants first before any ad-hoc analyses and including objective measures of physical activity
Duration and ustainability	12. Consider intervention beyond 6 months and evaluation of maintenance beyond a year

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Declaration

Ethics approval and consent to participation

Not applicable for systematic review

Availability of Data and Materials

Not applicable

Competing interests

No competing interests to declare.

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Author contributions

DM conceptualised the research and drafted the first manuscript. DM and JS conducted the systematic search and selection. FS and DD conducted the quality assessment. AT and DM conduced the meta-analysis. DM and SM conducted the data extraction to tables. All authors (DM, FS, JS, AT, SM, KG, DD) participated in the interpretation of results, commented on first draft and gave final approval of the manuscript.

Consent to publication

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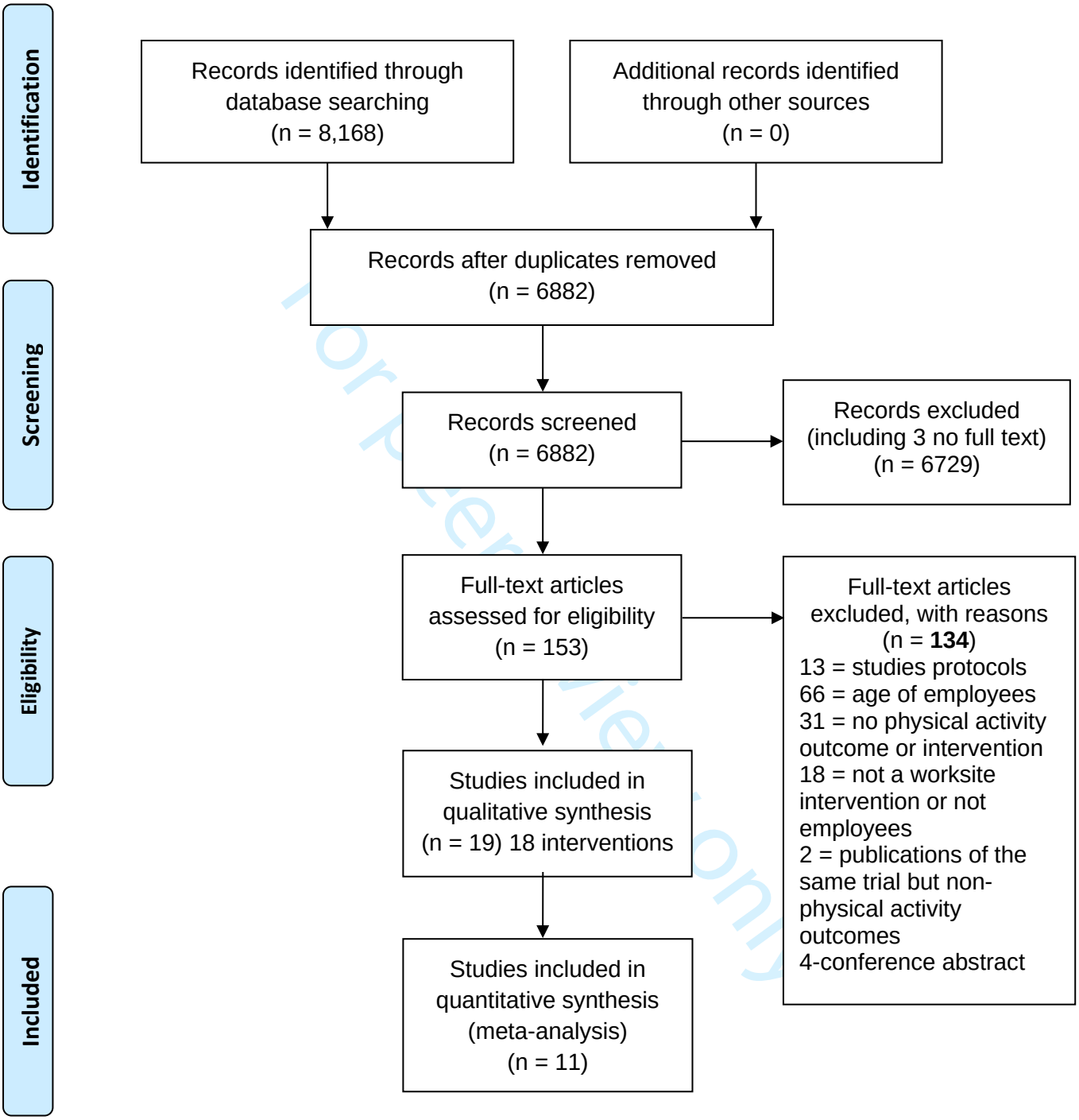
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Figure 1: PRISMA flow diagram of selected workplace physical activity intervention studies

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Appendix Table 1 Characteristic of worksite physical activity (PA) interventions delivered to older employees (organised by publication dates)

Study	Recruitment / reach	Intervention content / theoretical guidance	Intervention deliverers	Duration	Process data
Bassey, E. J., et al, 1987 [27]	Initial discussions for the support of the management; After a publicity launch in the factory newspaper, a personal letter of invitation was sent to all workers through factory's internal mail system; 35% attended introductory meetings and ultimately 23% agreed to participate	"The New Aerobics" (1970) Cooper's prescription for people aged ≥50yrs, which isa progressive walking program instructed at workplace but carried out at employees' leisure time. 1st week, walking 20mins$_{*}^{-1}$d and at a pace of 5.2km$_{*}^{-1}$hr preferably 5 days per week. Subjects can increase duration of walks and frequency. In 2nd phase the intensity can increase based on heart rate response to exercise at 4.8km$_{*}^{-1}$hr. The target dose was 40mins at 7km$_{*}^{-1}$hr 5 days per week. Individual discussion at beginning (how to progress) and half way through. The progression was individually discussion on courses and timing at start and mid program. Participants were asked to keep daily log-card of duration and distance walked. Encouraging articles in the factory newspaper were published monthly.	Unsupervised program – no mentioning of the profession of the person conducted the meeting to inform the grade of effort	12 weeks program divided to 6 weeks each phase	59 out of 108 completed the intervention (54%). Log cards: only 1/3 nearly achieved the speed and distance prescribed for the last 2 weeks, 1/3 provided evidence on logs for extra walking but substantially below prescription, and 1/3 failed to return log card. The most compliant were the most active before program started
Sharpe, P. A. et al 1992 [31]	Advertising at work, direct contact with departmental representatives, telephone calls from the research staff and provision of additional dates and sites to enhance the accessibility of screening; 250 employees did health screening but the % from total employees is unknown	Health promotion activities after health risk screening (smoking, blood pressure, level of blood lipids and body weight) and brief communication of results by nurse. Participants could choose to have face-to-face with meetings with health counsellors, to join walking groups at work and/or exercise on their own at home or at university facilities. No other promotional activities were reported.	health counsellors at work provided face-to-face with counselling based on participants choice	12 months	60 out of 95 experimental participants (63%) completed the program. Behaviour change self-report: 48% of Control and 53% of have done anything to increase fitness. started program on their own- 63% and 73% of C and Int. took worksite program 13% of C and 11% of Int; used worksite facilities: 11% of C 16% Int.
Arao, T. et al 2007 [34]	Two companies were recruited, all together 5 sites. Medical check-ups were conducted and 192 employees were identified to have CVD risk factor(s) of those 177 (92%) were eligible and consented and ultimately 155 (87%) were allocated as a cluster to LiSM (n=92) and Control (n=85) groups.	Theoretically driven (TTM and SCT) Life Style Modification Program for Physical Activity and Nutrition (LiSM-PAN). Structured counselling for goals setting for each behaviour and monthly meeting. Log book for steps a day (10,000 as a goal); increasing incrementally; if reached take another sport/exercise class. Environment support at worksite: 1) setting up a walking course; 2) installing exercise facilities; 3) displaying caloric content table of the lunch menu at cafeteria; 4) cafeteria manager advised on healthy cooking for better diet. Social support at home: printed material on healthy diet and cooking; participant and spouse going together for exercise	Research staff were trained in structured counselling;	6 months	95.2% completion. 97.1% attended monthly counselling 86.7% self-monitored the walking steps 54.7% controlling their dietary targeted activities

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Study	Recruitment / Reach	Intervention content / theoretical guidance	Intervention deliverers	Duration	Process data
Granacher et al. 2011 [30]	Several companies in Basel were approached and 2 large companied (~ 500 employees) agreed (randomization at cluster level). Informational session was held explaining the intervention and eligibility criteria, 34 employees consented and participated.	Three times a day five days a week of training in: Static balance exercises – in morning at arrival to office Dynamic balance exercises – after lunch break Strength enhancing exercises before leaving office New exercise routines were assigned each week to be repeat for the whole week before progressing.	Expert in balance and strength training saw each participant every week	8 weeks of training followed by 8 weeks without training data collected at 8 and 16 weeks	All 17 intervention participants completed the program (100%). Program compliance was excellent with participants of the intervention group completing 99 % of the scheduled training sessions (~ 119 out of 120 training sessions).
Hughes et al. 2011 [22]	Worksite advertisement Research participants recruited via announcements on staff list-serves, targeted e-mails, staffed recruitment tables at events in highly trafficked buildings, and flyers posted throughout the university. Out of 647 enquiries 66% deemed eligible and 423(65%) enrolled in the study.	COACH –shaped by the Enhance Wellness Program for seniors in the community and guided by TTM. One face- to- face meeting with coach to review the participant's health-risk profile (baseline assessment), discuss the behaviors the participant was interested in adopting, and discuss the availability of health behavior change programs on or off campus, assist in creating plan to achieve goals. Biweekly contact by email/telephone for the first 6 months to re-evaluate plan, goals and actions. In the remaining 7 to 12 months the contact was monthly. Health-risk profile assessments were at 6 and12 months. RealAge- A Web page contain health risk test and generate health profile based on family history, diet and nutrition, PA, tobacco and alcohol use, satisfaction with family relationships, and use of preventive health care. Participants used the Web site to select behaviors to work on and create plans to meet behavioral goals. The RealAge site tracked each time a participant used it, and the site forwarded this information to the study team at regular intervals, enabling us to document use.	COACH -MPH graduate trained in behavioural change coaching RealAge – research staff	12 months	Retention: 91% in COACH and 84% RealAge. COAC: 150 participants in Coach spent 9.25hr/participant – there were 688 emails 77% by the coach and 48 telephones FU, RealAge-135 participants 59% completed all health risk assessment and 53% engaged in at least one of the web link.
Palumbo et al. 2012 [23]	One medical centre targeting nurses currently employed full time or part time in staff nurse position that involved lifting patients. About 70 nurses place their interests and 14, on a first come first served basis, were randomized.	Tai Chi workplace wellness program, guided by the Revised Health Promotion Model by Pender (1996). Classes on site were carried once a week (including 15 minutes breathing exercise, 30 mins tai chi practice 5mins visualization cool down). Additional practice on their own 10mins 4 times a week	Experienced tai chi instructors (22 years of teaching)	15 weeks	Five out of the 7 intervention participants (71%) completed the tai-chi program. Fidelity- two classes were cancelled attendance 82% range:69%-92%

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Study	Recruitment & Reach	Intervention content / theoretical guidance	Intervention deliverers	Duration	Process data
Strijk et al. 2011 , 2012 [20,21]	All workers of two academic hospitals were invited to participate between April and October 2009; Among 3756 employees invited to participate, 1101 (29%) agreed to participate and 730 (19%) allocated into two groups	3 visits to Personal Vitality Coach (PVC) at worksite: at start of intervention, followed by 2 consecutive visits at 4-6 weeks and 10-12 weeks. Meeting 30 minutes each. Weekly yoga 45mins and workout sessions (intensity 65%-90% of the age-predicted maximum heart rate) for 45mins. Plus 45 min unsupervised aerobic exercise. Fruit & Vegie at the session program name 'Vital@work'	Certified yoga instructor and exercise physiologist	6 month interventio n period and additional 6 months sustainabili ty testing	293 out of 367 (80%) intervention participants completed the program. PVC visits 100%. Dose delivered – Yoga sessions 72.3% (Amsterdam - 89.3%; Leiden - 58.3%) Workout sessions 96.3% (Amsterdam – 95.1%; Leiden – 97.4%) High satisfaction score
Cook, et al. 2015 [24]	A recruitment flyer was emailed by company officials to all employees 50 years plus; interested employees were provided information on study and a link to the online baseline survey (received $25 if completed); out of 290 employees, 279 (96%) completed the baseline survey.	HealthyPast50, an automated web-based program containing multimedia health promotion information and guidance based on Social Cognitive Theory. Modules: Facts about Healthy Ageing, My Health Profile – assessment module with questionnaires and recommendations, Stress and Mood Management, Healthy Eating, Active Lifestyle, and Tobacco Free. Encouragement of self-assessment (My Health Profile) before accessing modules. the content of the Active Lifestyle module is oriented toward older users, including the use of the Physical Activity Readiness Questionnaire (PAR-Q), assessing one's readiness for physical activity, the relationship of physical activity to mobility and energy in later years, and the need to moderate the intensity of exercise as one ages	No human contact for the program; project staff can be reached by telephone for queries	3 months; program progress by monthly email and reminder to access	Formative evaluation of the program including focus groups and testing materials. Process evaluation: mean # log-in:4.33 (SD 4.28, range 0-28), the mean minutes in the program was 102.26 minutes (SD 148.32), and the mean number of pages viewed was 11.04 (SD 20.08, range 0-120).
Abbas et al. 2015 [34]	Targeted recruitment of low-paid unskilled or semi-skilled employees age >40 on social care. A single individualised invitation was sent with simultaneous promotion of the programme within standard employer–staff communication routes. Out of 3264 target group (low SES employees), 665 (20% response rate) attended health screening. Of them, 579 were eligible and 427 (74%) returned for the follow-up assessment	Health screening of all major cardiovascular risks was offered to employees of low SES in deprived area. Based on screening results participants were stratified to either "high" , "moderate" or "low" CVD risk. The intervention included referrals to either GPs (if pharmacological treatment were needed), Weight management (in the community, exercise training, smoking cessation and alcohol reduction counselling either face to face or by telephone. Employers ensured that time of assessments and time spent on health promotion will be accounted towards the daily working hours. These services were provided in the community.	Occupational health nursing staff. The nurses were trained to provide smoking cessation and alcohol awareness delivery at the national standard.	9 months interval from initial assessment to follow-up	The attendance to health screen was 20%, much lower than the general population that offered similar program (health screen). There was record of the referral number given to each domain of intervention (exercise 38%, smoking 5%, alcohol 1%and weight management 20%). No account on the attendance to service referred.

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Study	Recruitment & Reach	Intervention content / theoretical guidance	Intervention deliverers	Duration	Process data
Low et al. 2015 [25]	A promotional flyer to all female at age range & invited to call the centre if interested in reduction of CVD risk; once reached 60 employees who had ≥1 risk of CVD no further recruitment effort was made	Multi-components risk reduction educational sessions in groups on: weight loss/nutrition, stress management, exercise training, and smoking cessation accompanied by access to an on-site gymnasium, and organized group walks. Intervention groups received weekly communication via phone or e-mail, incorporating goal-setting and suggestions for overcoming obstacles.	Not specified	6-months for classes, gym availability organized walks and weekly communication Follow-up survey at 12 months	Attendance to 1st classes was highest for weight loss (42% & 47% control). Stress reduction (30% & 21%) and physical activity 25% to 26%) followed by very poor attendance to consecutive classes. 3 attended group walking; use of Gym 0-20 days over 6 months . Follow-up 16/57 (28%)
Chen et al. 2016 [33]	Approached six worksites small- and medium scale enterprises (SMEs). Three sites showed interest and approved. Among 1245 workers in the three sites, 438 (35.2%) met study criteria of being full-time employed and aged over 50 years. These were invited. Of those, 264 (60.3%) showed interest and finally 108 (25%) agreed to participate.	Phase I: on-site lectures and workshops on healthy living specifically impact of metabolic disorders and individualized self-management goal setting. Biweekly meetings face-to-face with occupational health nurse (OHN) to share experience, give mutual support and encouragement, learn about managing own health, and cultivate leadership. Phase II: on-site team led meetings; discussing barriers reinforcing personal goals; recording food consumption and exercise. Logs were reviewed by OHN and awards/prizes were given for excellent improvement	Occupational Health Nurse	Total 24 weeks Phase 1 was 4-week phase II 20- weeks.	111 from initial 136 intervention participants (80%) completed the program Fidelity - Participants found it difficult to records daily PA and diet so no review of records was made; Working overtime reduce the compliance with planned activities and goals; activities forced to cut short due lack of support from management at low level
Naug et al. 2016 [35]	Contacted with employer; Information session meetings were arranged to build rapport with the employers and drivers; after that, drivers were invited to an information presentation and were given an information package.	Educational sessions of 60mins delivered fortnightly in groups at worksite, interactive and fun. Content were derived from the "Life! Taking action on diabetes" program which was based on health psychology theories with a focus on self-regulation. Targeting physiological change due to inactivity, harm of sitting, healthy nutrition. Participants were provided pedometers and asked to record steps each day for 6 weeks.	Two postgraduate students with expertise in adults' education, human physiology & nutrition	6 weeks	21 out of 33 that started program (64%) completed the program. Qualitative after intervention : drivers said " I walk around the bus during breaks" "I, use the back seats of the bus to do push-ups"; "park my car far from depot to walk more"

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Study	Recruitment & Reach	Intervention content / theoretical guidance	Intervention deliverers	Duration	Process data
Kocur et al. 2017 [26]	Workplace advertisement Postmenopausal female who replied to the advertisement regarding the study (n=91) and net the inclusion criteria (n=44) but no data on % from the total number of the target population.	Nordic walking training program techniques and training 3 times a week for one hour including 10-min warm up phase at start 5-min cool down at end. Intensity progress 40%-70% of heart-rate reserve	Two instructors specialised in Nordic walking regulated the pace	12 weeks	No report on program fidelity or attendance.
Chopp-Hurley et al. 2017 [28]	Workplace advertisement through e-mail list serves, webpage at McMaster workplace wellness committee Potential participants had to have clinical osteoarthritis either of the hip or the knee; no number was provided for those that were screened or refused.	Program of static lower extremity strengthening exercise; designed initially for Knee Osteoathritis to avoid KAM (knee adduction moment); The program includes Yoga poses such as squats and lunges; A perceived intensity of 5-7 from a total 10-point scale was recommended. On-site 4 timea a week sessions in the University sport and recreation facility were provided before work (7-8 AM). Participants were asked to attend at least 3 sessions/week. Participants in intervention received weekly feedback on their performance of exercises. Incentive of $50 was given to maintain participation in both groups and control participants were offered the program at the end of trial	Not specified	12 weeks	From initial 24 participants (12;12) 3 intervention participants did not complete FU (25%) as opposed to 1 in the control (12%). Exercise group attended on average 1.2 sessions per week; median was 0.5 classes per week, lower quartile was 0 a week and upper quartile was 3.9 per week thus estimate that 33% adhered to 3 sessions per week
Bergman et al. 2018 [29] (2015 Protocol)	13 different companies in 17 sites were involved (number of employees age 40-67 not given). Information session were given to office workers- the information included who is eligible (i.e., at least 4 days/wk in the office with sedentary work tasks with a sit-to-stand workstation, don't have a travel day during their work). Information on study strict procedures and measurement only 100 assessed and 80 enrolled	Health consultation about 30-40 minutes about their screening results and sitting time at work and diet. Folder with written information about PA recommendations for moderate and vigorouse aerobic activity and strength, how to increase light PA and Non-exercise PA such as climbing stairs, walk and talk meetings. After this session the treadmill adapted to desk use was installed next to the existing workstation. Physiotherapist adjust the workstation and instruct the worker how to use the treadmill in the first week – any pace 15min twice a week. Thereafter they were recommended to use it for 1hr a day preferably longer. Booster email from research team remined the hazard of sitting, reminder to use treadmill at least 1hour a day. These emails were sent at 6,20,31 and 50th week over the program.	Nurses conducted the health consultation and boosters emails by research team	13 months	Of the 80 randomised (40;40) 34 continued their participation for 13 months. In the control 92% continued for 13 months. No data was collected on usage of the treadmill during work Threat for contamination between group was acknowledged

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Study	Recruitment & Reach	Intervention content / theoretical guidance	Intervention deliverers	Duration	Process data
Scapellato et al. 2018 [36]	Targeted strategy; screening for eligibility (at least 1 CVD risk: blood pressure, weight, waist circumference lipids, impaired fasting glucose) 1892 workers on voluntary basis agreed to be screened; 1574 did not meet the inclusion criteria 318 recruited (50% from 632 eligible). Final recruitment 226 (35% from eligible).	As part of the Occupational Health and Safety approach, a workplace intervention was offered to promote healthy lifestyle for healthcare workers in the hospital who are at risk; intervention included individualized counselling meeting a month after baseline assessment to discuss results of PA and diet, and diverted to health counselor to identify PA and diet goals and strategies to achieve them. Helpful printed material were given such as food pyramids and a brochure of exercise. After 3 months participants received further motivational support via phone counselling to remind their goals and assist to change lifestyle habits.	Hospital Employees Occupational Health practitioners, health promotion counsellors and hospital dietitians	6 months	92 out of 318 eligible recruited (29%) did not receive the intervention. Of the 226 that received intervention, only 169 remained for 6-month assessment. No information on % phone calls given at 3-month. Reason to lost to follow up were attributed to “no time” & “too many work commitments”; only 40% who completed the study favoured adherence to dietary recommendations
Qi et al. 2019	Worksite advertisement; posters, person-to-person invitations using card-sized flyers, promotion of the study by colleagues and an e-mail request from the University Administration promoting the research projects that office workers may be interested to partake in. No data on number of potential eligible (55+ in the Uni) or how many were interested	Based on evidence that taichi or resistance training alone can improve physical function and psychological wellbeing of older adults. The intervention offered 3 session per week from a choice of 8 different timetables to allow flexibility: Monday, Wednsday and Fridays (2 session) 7-9 AM midday and afternoon 3-6PM. Session took 45 minutes. Ten Taichi movements were practiced with theraband tied to palms to a relaxing/ calming music. Theraband degree of effort were progressive. Every fortnight a Rating of Perceived Exertion was completed. When RPE was below 5 the degree the difficulty was increased	Taichi instructor, which is the first author of the paper, delivered the intervention	12 weeks with mid-term (6-weeks) assessment	At 6 week 9 participants withdrew (22%) and no further withdrawals happened at 12 weeks. Reason to lost to follow up were: back issue, knee issue, time conflict flue and no reason given. Data on compliance was not reported; there were 36 sessions offered but no data on mean session attend. .
Edman et al 2019	Worksite advertisement; e-mail invitations (3) to get health coaching for free. Flyers were posted in various locations in the main hospital including the cafeteria and in satellite office locations.	Health coaching on cardiovascular health delivered face-to-face or by phone or in combination. A cyclie is considered 12 weeks during which 6 consultations 2-30min each must be completed. Sessions are client centred behavioural changes strategies and include provision of education and individualized support to make change. Sessions start with assessment, changes from previous session, goal setting (fortnightly), elaborate on topic the client chose (the SMART framework model)	A certified Health Coach with 4 years’ experience with a PhD degree in Nutrition (the first author)	5.5 months with cycle is 12 weeks with option to continue.	74% (40/50 participants) completed 12 weeks. Participants averaged 7.8 sessions (range 6–12) over 5.5 months (range 3.0–10.0). Coaching were in-person for 57% and 17% receive only telephonic coaching and 26% received combination of both. .

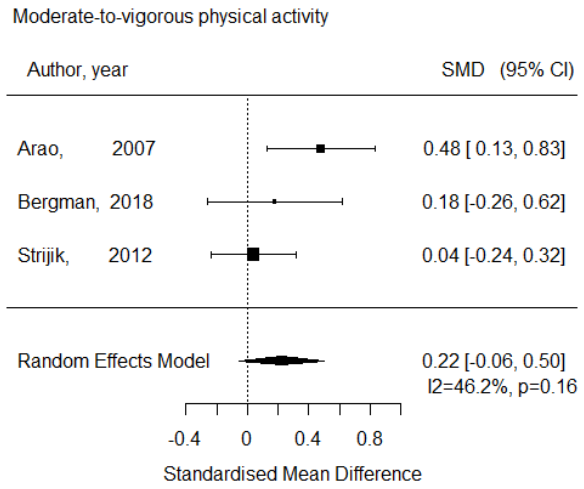
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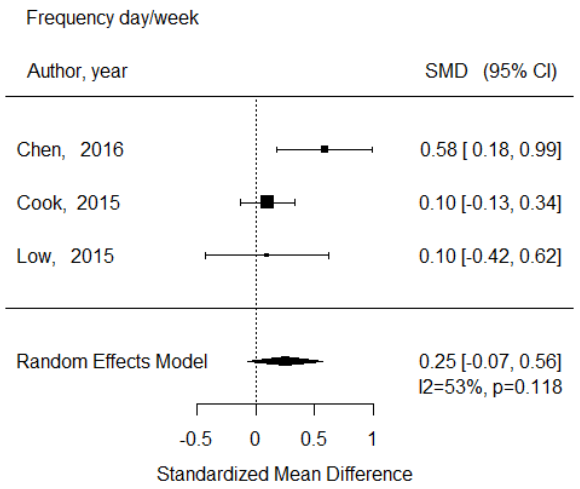
Supplementary Table 2: Risk of bias assessment for RCTs and quasi-experiments

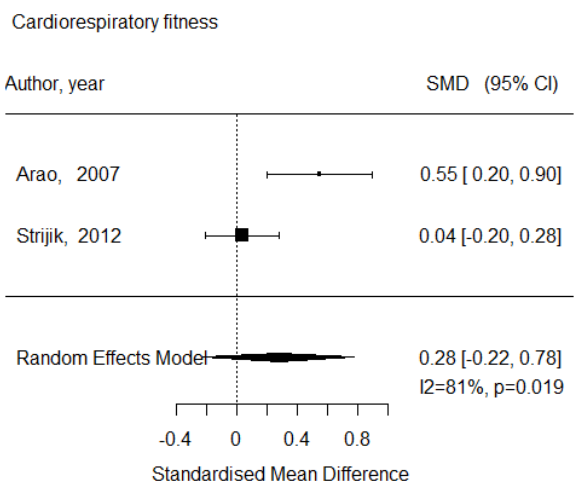
RCTs	Risk of bias (High, low, unclear)					
Source	Randomisation sequence Allocation	Concealment	Performance bias	Ascertainment bias [a]	Incomplete outcome data	selective reporting
Bassey et al. 1987	Unclear	Unclear	High	Low	High	Low
Sharpe et al. 1992	High	Unclear	High	High	High	Low
Granacher et al. 2011	High	Unclear	High	Low	Unclear	Low
Hughes et al. 2011	Low	Unclear	High	High	High	Low
Strijk et al. 2012	Low	Low	High	Low	High	High
Palumbo et al. 2012	Unclear	Unclear	High	low	High	High
Cook et al. 2015	Low	Low	High	High	High	Low
Low 2015	Unclear	Unclear	High	high	Unclear	High
Kocur et al. 2017	Unclear	Low	High	Low	High	Low
Chopp-Hurley et al. 2017	Low	Low	High	Low	Low	Low
Bergman et al. 2018	Low	unclear	High	Low	Low	Low
Qi et al. 2019	unclear	Low	High	low	High	Low
Non-RCT: Confounding						
Chen et al., 2016	High	Low	High	High	High	Low
Arao, 2007	High	Low	High	Low	High	Low

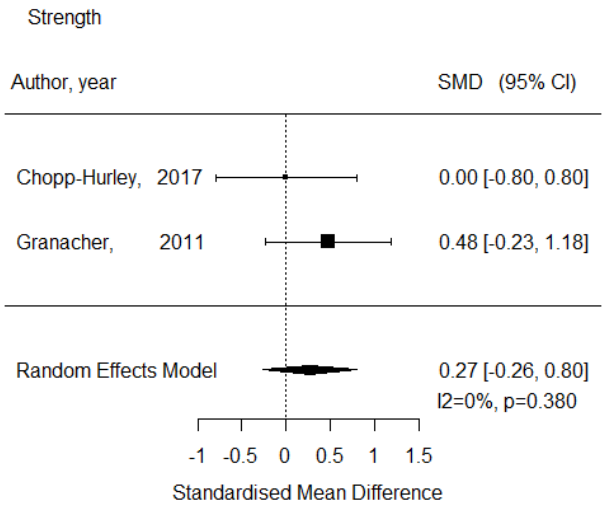
[a] Studies that had included objective physical activity outcomes and subjective outcomes were classified as low

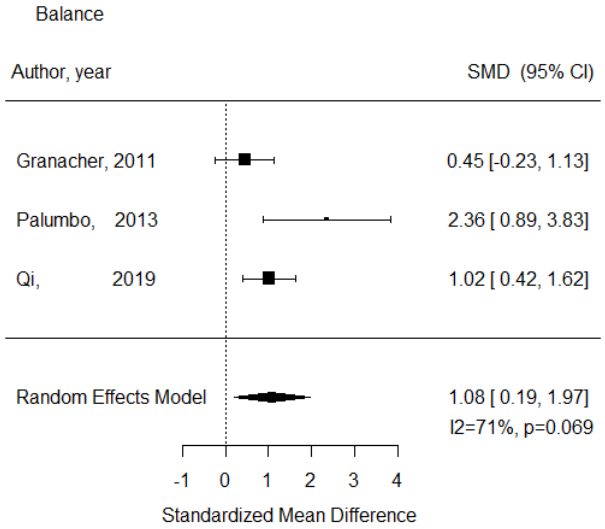
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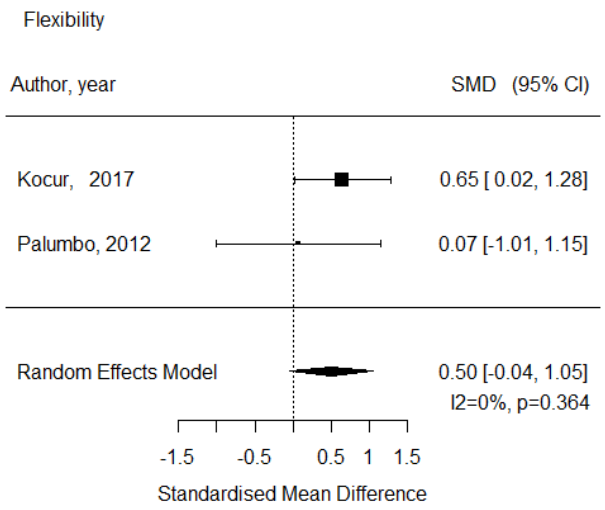












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PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
TITLE			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	1
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	2
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	4-5
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	5
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	6
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	6
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	6-7
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	NA
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	6-7
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	7
Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	6
Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	7-8
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	8
Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I^2) for each meta-analysis.	8 Table 3

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PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	8, GRADE Table 3
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	NA
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	8 -9 & Figure 1
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	Table 1 & Supp T1
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	19, & Supp T2
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	18 Table 2
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	19-21 Table 3
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	Table 3
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	NA
DISCUSSION			
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	23
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	25-26
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	26 & Table 4
FUNDING			
Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	NA

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From: Moher D, Liberati A, Tetzlaff J, Altman DG, The PRISMA Group (2009). Preferred Reporting Items for Systematic Reviews and Meta-Analyses: The PRISMA Statement. PLoS Med 6(7): e1000097.



PRISMA 2009 Checklist

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Supporting active ageing before retirement: a systematic review and meta-analysis of workplace physical activity interventions targeting older employees

Dafna Merom[1], Fiona Stanaway[2], Klaus Gebel[3,2] Joanna J. Sweeting[4], Anne Tiedemann[5], Shirin J. Mumu[1], Ding Ding[2]

1. School of Health Science, Western Sydney University, Penrith, New South Wales, Australia
2. Sydney School of Public Health, University of Sydney, Camperdown, New South Wales, Australia
3. Australian Centre for Public and Population Health Research, School of Public Health, Faculty of Health, University of Technology Sydney, Ultimo, New South Wales, Australia
4. Cardio Genomics Program, Centenary Institute Newtown New South Wales Australia
5. Sydney School of Public Health, Institute of Musculoskeletal Health, The University of Sydney, Camperdown, New South Wales, Australia

Corresponding Author:

Prof Dafna Merom

School ofHealth Sciences

Campbelltown Campus 24.2.33

The University of Western Sydney,

Locked Bag 1797, Penrith NSW 2751, Australia

Tel : +61-2- 4620 3747 **E** d.merom@westernsydney.edu.au

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Abstract

Objective: With the growing representation of older adults in the workforce the health and fitness of older employees are critical to support Active Ageing policies. This systematic review aimed to characterise and evaluate the effects on physical activity (PA) and fitness outcomes of workplace PA interventions targeting older employees.

Design: We searched Medline, PreMedline, PsycInfo, CINAHL, and the Cochrane Controlled Register of Trials (CENTRAL) for articles published from inception to 17th of February 2020. Eligible studies were of any experimental design, included employees aged ≥50 years, had PA as an intervention component and reported PA-related outcomes.

Results: Titles and abstracts of 8168 records were screened and 18 unique interventions were included (3,309 participants). Twelve studies were RCTs. Seven interventions targeted multiple risk factors (n=1,640), involving screening for cardiovascular disease risk factors, but had a non-specific description of the PA intervention. Four interventions targeted nutrition and PA (n=1,127), and seven (n=235) focused only on PA. Interventions overwhelmingly targeted aerobic PA, compared to only four interventions targeting strength and/or balance (n=106). No studies involved screening for falls/injury risk and only two interventions targeted employees of low socioeconomic status. Computation of effect sizes (ES) was only possible in a maximum of three RCTs per outcome. ESs were medium for PA behaviour (ES=0.25 95% CI: -0.07 to 0.56), muscle strength (ES=0.27, 95%CI: -0.26- 0.80), cardiorespiratory fitness (ES=0.28, 95%CI: -22 to 0.78), flexibility (ES=0.50, 95% CI: -0.04-1.05) and balance (ES=0.74, 95% CI: -0.21 to 1. 69). GRADE criteria-rated quality of evidence was 'low' due to high risk of bias, imprecision and inconsistency.

Conclusions: The lack of high-quality effective workplace PA interventions contrasts the importance and urgency to improve the health and fitness in this population. Future

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interventions should incorporate strength and balance training and screening of falls/injury risk in multi risk factors approaches.

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Key words: older employees, Physical activity, exercise, workplace, interventions, systematic review, meta-analyses

Strengths and Limitations of this study

- This systematic review was registered in PROSPERO.
- We followed PRISMA reporting guidelines, and the Cochrane Collaboration and the GRADE approach to rate the quality of evidence.
- We used five databases from inception and backward citation tracking of all included studies.
- Pooled effect sizes that were not expressed in similar units, although they measured the same concept, may have artificially increased the heterogeneity of the effect.

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1. INTRODUCTION

Population ageing poses individual, social, economic and political challenges and is predicted to accelerate even further in the 21st century [1]. The World Health Organization's (WHO) "Active Ageing" concept is a leading global policy strategy for successful ageing [2], particularly influencing retirement policies towards maintaining engagement with paid employment. In the developed world, strategies have been implemented to promote economic activities among aged workers, including incentives for late retirement, penalties for early retirement, and an increase in or abolishment of the age of mandatory retirement [3]. Older retirement age means that more employees beyond the age of 55 years will remain in the workforce [4].

While a number of mental and social abilities can improve with increasing age, and such changes should be viewed as strengths and be respected [4], age-related physiological decline requires special considerations regarding occupational health, safety and productivity [5]. In-depth reviews on ageing at work [6, 7] identified a number of physiological, sensory and psychological differences between older workers and their younger counterparts, including reduced muscle strength, endurance, trunk flexibility, structural and sensorineural degeneration which affect balance, vision and hearing, aerobic capacity, heat tolerance, increased anthropometric risks, psychological exhaustion, anxiety and depression. Further, older employees suffer from greater prevalence of comorbidities, musculoskeletal problems, and sickness-related absence and severe injuries [6, 7].

It is well established that regular physical activity (PA) is effective in attenuating age-related physiological declines in all body systems, and in preventing and managing many age-related chronic and musculoskeletal conditions [8-10]. Adherence to older adults' comprehensive PA

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guidelines including balance exercises, muscle strengthening and moderate-intensity aerobic activity, is likely to benefit older employees in terms of health as well as work productivity and safety. Hence, recently physical employment standards for older employees have been suggested [7]. The workplace has long been considered a good setting for health promotion and many PA interventions have been conducted in this setting, with the first systematic review of their effectiveness published in 1998 [11]. However, we have not identified any systematic reviews of workplace PA interventions that specifically targeted older employees. PA interventions targeting older adults should additionally include activities to promote muscle mass, strength, and balance, which are important for the prevention of sarcopenia, functional disabilities, falls and injuries [12]. A recent systematic review by Poscia et al. synthesised the evidence on health promotion interventions targeting older employees [13]. However, this review examined a variety of health promotion initiatives and have missed interventions that focused on PA, by only using 'fitness' and 'capacity' as search terms. Given that PA guidelines for older adults include a range of physical outcomes relevant to health beyond cardiorespiratory fitness or capacity, such as muscle strength, balance and flexibility [14, 15], it is important to include broader terms when reviewing health promotion programs. Furthermore, the main outcomes included in the review by Poscia and colleagues [13] were health status, wellbeing and work productivity, not PA measures. Thus, the effectiveness of workplace interventions on PA and fitness outcomes of older employees is uncertain.

The objectives of this systematic review were to: i) identify and characterise workplace PA interventions delivered to employees aged ≥50 years; ii) assess the methodological quality of the studies; iii) assess the effect of interventions on PA behaviour and fitness outcomes highlighted in PA guidelines for older adults.

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2. METHODS

This review followed guidance published by the Centre for Reviews and Dissemination, the Cochrane Collaboration [16], and PRISMA guidelines [17] (see Supplementary file:PRISMA checklist). The protocol was submitted to the International Prospective Register of Systematic Reviews (PROSPERO) on the 21st of December 2017 and final registration received on 29th of January 2018; registration number CRD42018084863 available at https://www.crd.york.ac.uk/prospero/display_record.php?RecordID=84863)

2.1 Inclusion and exclusion criteria

We included studies that involved any PA interventions (i.e, intervention that will encourage people to move large skeletal muscles) and were delivered at workplaces to employees aged ≥50 years. Studies that included a wider age range, such as whole workplace intervention, were eligible for inclusion if the mean age of participants was at least 50 years. This decision was based on a systematic review of workplace health promotion interventions by Posica et al who included studies that explicitly targeted older employees and found that the mean age of these workers ranged from 51 to 55 years [13]. Interventions could include those related to aerobic PA (e.g., walking), stretching, balance, muscle strength, yoga, tai chi, Pilates, gym workout, sport or any other form of PA/exercise. Studies that targeted other health risks in addition to physical inactivity (hereafter, multiple risk factors interventions) were only included if PA/exercise was one of the intervention components and a PA-related outcome was reported. Studies with any experimental design (pre-post with or without a comparison group, nonrandomised trials, RCTs and cluster RCTs) were included because of the small number of randomised controlled trials (RCT) anticipated [13].

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Studies were excluded if the PA intervention was not delivered in the workplace (e.g., community, health practice), or if the interventions targeted the whole workplace with a wider age range and did not include specific PA-related outcomes.

2.2 Search strategies

We systematically searched five databases from inception to 23rd of December 2017. We then updated our search on the 13th of June 2019 to include all 2018 and again on the 17th of February 2020. Included databases were: Medline, PreMedline, PsycInfo, CINAHL, and the Cochrane Controlled Register of Trials (CENTRAL). Specific search strategies were developed for each database, using a combination of search terms and subject headings where applicable. All electronic searches included MeSH terms and keywords for workplace (e.g. occupations, worksite), PA (e.g., walking, postural balance, muscle strength) and intervention (e.g., health promotion, program, RCT) and were limited to humans, and age (PsycInfo: 40+ all others: 45+) (see Supplement file: search strategy). One reviewer (JS) retrieved the entries from the database using EndNote (Version X9.3.3) and screened the titles and abstracts to identify potential papers for inclusion as well as running backward citation tracking of potential studies selected for full text review. Two reviewers (JS, DM) independently conducted the final selection of papers based on full text. Disagreements were discussed and resolved by consensus.

2.3 Data extraction

Three data extraction formats were used: A table summarising the included studies was created using the subheadings: source, study design, population characteristics, study duration, type of intervention, comparison group, and outcomes. A second table describes characteristics of each intervention in terms of recruitment, content, intervention deliverer

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and process outcomes based on data extracted by DM and SM. A third table summarises the quality of the included papers and another Table summarises the effects sizes for each outcome along with the level of evidence of this effect.

2.4 Quality assessments and analysis

The Cochrane risk of bias tool for RCTs and additional domains from the ROBINS-I tool for non-randomised studies with a comparison group were used to assess the quality of individual studies [18]. The domains included were: randomisation sequence generation, allocation concealment, blinding of participants and personnel, blinding of outcome assessment, incomplete outcome data and selective outcome reporting. The first two domains were only assessed in RCTs. In non-randomised studies with a control group the additional domains of confounding and selection bias were assessed. Each domain was assessed independently by two authors (DD, FS) who assigned a rating of either 'low', 'high' or 'unclear' risk of bias. Disagreement was resolved by consensus or, where needed, by consultation with a third reviewer (DM). We used Comprehensive Meta-Analysis software (Version 2, Biostat, Englewood, New Jersey, USA) to conduct random effects meta-analyses for each outcome accounting for heterogeneity between studies. The standardised mean difference (SMD) (Hedges' g) was calculated for each meta-analysis, standardised by postscore SD (or its estimate) and calculated using the pre- and post- means and SD or, when unavailable, the mean change score and SD. Effect sizes were categorised as small (≤0.20), medium (> 0.20-0.79) or large (≥0.8) [19]. For trials with moderate-to-vigorous intensity PA as the outcome, we used data from accelerometers when possible. For studies reporting energy expenditure as the outcome, we converted the value to minutes of moderate-intensity PA using the population mean body weight and 4 Metabolic Equivalents for Task (METs) for moderate-intensity PA. Statistical heterogeneity was determined by the I^2 and χ^2 tests to

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measure the proportion of variability between studies and tau-square (T^2) was reported to indicate by how much the true effect sizes vary. We followed the recommendation of the Cochrane Collaboration which states that tests for funnel plot asymmetry should be used only when there are at least 10 studies included in the meta-analysis [16]. Since we had no more than 3 studies for each outcome, we did not assess publication bias. Finally, we used the Grading of Recommendations Assessment, Development and Evaluation (GRADE) approach to rate the overall quality of evidence for each outcome from 'very low' to 'high' [20, 21], including a narrative summary of findings for outcomes without meta-analysis [21]. GRADE ratings are based on the domains of risk of bias in included studies, indirectness of evidence, imprecision, inconsistency and likelihood of publication bias.

Patient and Public Involvement

Neither patients nor the public were involved in the design, conduct, reporting or dissemination of this research.

3. RESULTS

3.1. Study selections and characteristics

Our search resulted in 8,168 records which were screened by title and abstract. Of these, 153 were further assessed for eligibility based on full text (Figure 1). Nineteen studies met the inclusion criteria, however, two publications referred to the same intervention [22, 23], resulting in 18 unique interventions.

Insert Figure 1

The characteristics of the studies are summarised in Table 1. Ten studies were RCTs [24-33], and two were cluster RCTs - one involving only two worksites [34] and the other several university units [35]. There were also two non-randomised studies [36, 37] and four pre-and post-evaluations with no comparison group [38-41].

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3.2 Participants

In total, 3,309 participants (of those 2085 participated in RCTs) were included in the 18 worksite PA interventions: six in the USA [24, 26-28, 35, 41], two in the UK [30, 38], two in Australia [33, 39] and one each in Taiwan [37], Japan [36], Poland [29], the Netherlands [22, 25], Switzerland [34], Canada [31], Sweden [32], and Italy [40].

Five interventions were delivered in academic institutions [24, 29, 31, 33, 35], four in hospitals or medical centres [25, 28, 40, 41], and four were delivered to factory employees [30, 36-38]. One intervention was delivered to transport workers (drivers) [39], and four were delivered to office workers [27, 32-34]. Most studies (n=12) targeted both genders, with a high percentage of male participants (>60%) in transport [39] and IT workplaces [27], and fewer male participants (5% - 25%) in academic / administrative [24] and medical occupations [25, 31, 41]. Three interventions targeted females only [26, 28, 29] and one males only [36].

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Table 1: Characteristics of the studies that met the inclusion criteria in worksite older employees' physical activity (PA) interventions arranged by design and year

	Design Sample	Population	Duration	Intervention Description	Comparison	Physical activity related outcomes
Bassey et al. 1983 [30]	RCT n=108	UK, factory floor workers (blue collar) from light industrial company, age 55-60, 51% males	12 weeks	PA focus; prescribed aerobic (walking) program with a goal to increase cardiorespiratory fitness	No intervention	1) Physical condition - heart rate (HR) at walking 4.8km/hr 2) Sustained $HR_{4.8km/h}^{-1}$ for at least 7 minutes 3) Daily total minutes walked at $HR_{4.8km/h}^{-1}$ 4) Step count
Sharpe et al. 1992 [35]	Cluster RCT n=250	USA, employees from 'support and academic staff" at the University of Michigan, age 50-69, 53% males	12 months	Multiple risk factor program; after health risk screening a face-to-face meeting with health promoter; optional: walking groups at work/ or use facilities.	Health screening, but no exposure to health promotion program	1) Self-report frequency item with a 5-point response format -which was not explained or referenced. 2) Self–reported action taken to improve fitness in the past year 3) Precursors of exercise behaviours (self-efficacy to exercise 3 times a week and intention to exercise vigorously
Hughes et al. 2011 [24]	RCT (3 arms) n=423	USA, older employees from "support and academic" staff at the University of Illinois, Chicago Age ≥ 55 years 18% males	12 months	Multiple risk factor program to; 1) Enhance Wellness one-on-one coaching program (COACH) 2) a web-based health promotion program (RealAge)	Printed health promotion materials, programs and services offered by university or by community organisations	1) Self -report minutes of moderate-intensity and minutes in vigorous-intensity Using the Behavioral Risk Factor Surveillance System (typical week -7 questions) 2) The Rapid Assessment of Physical Activity (RAPA) for meeting PA recommendations based on 9 items (aerobic muscle training and flexibility, response: yes=1, no=0 combined to total score)
Strijk et al. 2011 [22] & 2012 [25]	RCT n=730	Netherlands, workers from two academic hospitals Age ≥ 45 years Mean age 52.3±4.9 25% males	6 months	PA & nutrition: Vital@work Yoga session 60 minutes and workout classes 30 minutes plus home based aerobic session	General healthy lifestyle written information – diet, physical activity and relaxation	1) The frequency, duration and intensity of participation in commuting, household, occupation and leisure-time PA using the SQUASH [a] questionnaire: 2) Minutes spent on sport-from SQUASH 3) Accelerometer minutes in MVPA on sub-sample (GTM1 Actigraph) 4) Aerobic capacity (VO_2 max) was estimated using the UKK [b] 2 km walk test.

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	Design Sample	Population	Duration	Intervention Description	Comparison	Physical activity related outcomes
Palumbo et al. 2012 [26]	RCT n=14	USA, female nurses from an academic medical centre in Vermont Age ≥ 49 years 0% males	15 weeks	PA focus: Once a week Tai-chi guided classes at work;45 minutes and unguided home-based practice 10-mins 4 times a week	No intervention	1) Flexibility of trunk "Sit-and-Reach" test 2) Strength: isometric knee extensor test with dynamometer 3) Balance-Functional Reach test (cm)
Cook et al. 2015 [27]	RCT n=278	USA, IT employees from two major global IT companies in California and Boston Age – 50 to 68 years 67% males	12 weeks	Multiple risk factor web-based educational program "HealthyPast50" with no human contact.	No intervention Wait-listed	1) Exercise habits: based on Godin LTPA score measuring frequency per week exercising at 3 levels (mild, moderate strenuous) and the number multiplied by intensity factor –reported frequency at each level and in total 2) Exercise self-efficacy score (8-items) 4-point scale 3) Exercise intention
Low et al. 2015 [28]	RCT n=62	USA, female employees of a busy community hospital, North Carolina, 40-65 years old; mean age 52 ± 6.3. 0% males	6 months	Multiple risk factor program to reduce CVD [c] risk as in control, plus weekly motivational sessions by mail or telephone for goal setting	Risk reduction educational classes; Free access to gym on site and walking group	1) Frequency – days per week of exercising 2) Minutes per session of exercising 3) Level of intensity: no exercise, leisurely, moderate, or vigorous exercise (reported as %) 4) Readiness to change exercise
Granacher et al. 2011 [34]	Cluster RCT n=32	Switzerland, sedentary office workers from two large companies in Basel, Age ≥50 37% males	8 weeks	PA focus; progressive balance and strength training at the office using exercise charts to perform 3 times a day; each session lasts 8 minutes	No intervention	1) Balance (static) postural control using balance platform, standing on one leg 30 seconds eyes open–displacement of COP[d] 2) Gait variability and speed on special treadmill 3) Force jumping height measured on a force platform 4) Maximal isometric & isokinetic torque (60° / and rate of torque development (RTD) of the plantar flexor using force platform

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	Design Sample	Population	Duration	Intervention Description	Comparison	Physical activity related outcomes
Kocur et al. 2017 [29]	RCT n=44	Poland, female administrative and academic office workers from 4 higher education institutions in Poznań Age 50-60 years 0% males	12 weeks	PA focus - Nordic walking training program 3 times a week for 1 hour each	No intervention	1) Perceived Pain Threshold of upper body (PPT, kg/cm^2) (i.e., the minimum force that can be applied which induced the feeling of pain) using an electronic pressure algometry 2) Flexibility (shoulders) using back scratch: measuring how close the hands can be brought together behind the back
Chopp-Hurley et al. 2017 [31]	RCT n=24	Canada, Hamilton University employees (McMaster) with clinical osteoarthritis Age > 50 years 21% men	12 weeks	PA focus – Specific Osteoarthritis exercises: static leg strengthening (e.g., yoga poses) 3-4 times a week 7-8 AM	No intervention	1) Hip and knee strengths measured by dynamometer the peak extension and flexion torque out of 5 trials were recorded normalised to body weight 2) 6-minutes walking test 3) Lower limb strength: chair sit to stand during 30 seconds 4) Timed –up-and go test 5) 40meter fast paced (time to complete)
Bergman et al. 2018 [32]	RCT n=80	Sweden, Umea office workers (13 companies). Overweight and obese Age 40-67 years (Mean age 51yrs) 45% males	13 months	PA focus – treadmill workstation recommended to walk 1hr a day at moderate-intensity, but no jogging	Working as usual at their sit-and stand working desk	1) Daily walking time at weekdays and weekend measured by ActiPAL accelerometer 2) Number of steps ActiPAL 3) Time spent in moderate to vigorous PA by Actigraph accelerometer 4) PA bouts of more than 10mins on weekdays and weekend
Qi et al. 2019 [33]	RCT n=40	Queensland, Australia University insufficiently active office workers Age 55-77 (mean 59.6) 22% males	12 weeks	PA focus Taichi movements (10) performed with resistance band (Theraband) 3x per week 45min per session on-site. Employee could choose from 8 different time slots	Active control: Taichi only without the theraband in different room to intervention. Similar number of sessions were offered	1) Lower limb strength using chair sit to stand during 30 seconds 2) Balance Functional Reach test (cm) 3) Hand grip strength 4) 2-minute walk test (work capacity)

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	Design Sample	Population	Duration	Intervention Description	Comparison	Physical activity related outcomes
Chen et al. 2016 [37]	Quasi-experiment (individual allocation) n=108	Taiwan, workers from small- and medium-scale enterprises, Age ≥ 50 years 45% males	6 months	Multiple risk factors; onsite educational workshops during first 4 weeks + meetings with OHN[e] to set goals, plans and on-site group support	Only educational workshops in the first 4 weeks	1) Frequency: number of times per week doing physical activity using Taiwan Longitudinal Study on Aging - no information on the PA domains asked (e.g., leisure, work etc) 2) Sedentary time (hrs/d)
Arao et al. 2007 [36]	Quasi-experiment (cluster allocation) n=197	Japan, employees working in 5 sites of two factories in Tokyo who had at least 1 CVD[c] risk Age 40-59 years (mean age=55) 0% females	6 months	PA & nutrition (LiSM[f]-PAN) multi-components program; individual counselling 5x10-min around goals; social and environmental support	Feedback from the medical check-ups plus recommendations on diet and physical activity including printed materials on exercise, healthy diet and cooking	1) Self-report energy expenditure (kcal/week) derived from the leisure time exercise was assessed by the Kuopio Ischaemic Heart Disease Risk Factor Study 2) Maximum oxygen uptake 3) VO_2max (ml/kg/min) from a sub-maximum bicycle test (Astrad) 4) Stage of change for exercise
Abbas et al. 2015 [38]	Pre-post n=665	UK, low-paid local government employees from socially and economically deprived areas in NE England, Age ≥40 years, mean age 50.5±6.4 37% males	9 months	Multiple risk factor health screening staging risk level and referrals to exercise, weight management, smoking cessation, promotion of mental health and alcohol reduction	Not applicable	1) Participation in aerobic exercise dichotomised to not meeting recommendation (less than 5 times a week of less than 30-minutes session) 2) Referred to exercise question on doing exercise outside work (report on % before and after)
Naug et al. 2016 [39]	Pre-post pilot trial n=33	Australia, bus drivers from two depots of South East Queensland Age: 50-68 years 64% males	6 weeks	PA & nutrition, Group educational sessions; harm of sitting, healthy eating & exercise; PA - pedometer to track steps	Not applicable	1) Exercise levels – no report on questionnaire type; researcher driven classification: a) none (no exercise), b) moderate level (e.g. 30-40min walking twice a week or tennis once a week, c) intense (e.g., gym 4-7 times a week or cycling 5 days/week), sedentary behaviour (hrs/week)

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	Design Sample	Population	Duration	Intervention Description	Comparison	Physical activity related outcomes
Scapellato et al. 2018 [40]	Pre-post n=167	Padua, Italy healthcare workers at risk of CVD Mean age 50±7.3 31.7% males	6 months	PA & nutrition, brochure on exercise and motivational counselling on site at baseline and mid-term by phone	Not applicable	1) PA MET[h] based on a 4-day diary about the type of activity, frequency (day per week) and duration (minutes)
Edman et al. 2019 [41]	Pre-post pilot trial n=54	Philadelphia, USA healthcare hospital workers volunteered with at least one CVD risk Mean age 53.3±10 5% males	6 months 12 weeks cycles with option to continue	Multiple risk factor program - six sessions of Health Coaching f2f or by phone including goal setting, education, motivational strategies	Not applicable	1) Self-reported change in number of sessions per week exercising aerobically for at least 20 minutes 2) Self-reported number of times per week doing muscle strengthening exercise for at least 20 minutes

[a] SQUASH –Short Questionnaire to ASsess Health- enhancing physical activity; [b] UKK- The Finnish Urho Kaleva Kekkonen walking test; [c] CVD – cardiovascular disease; [d] COP – Centre of Pressure; [e] OHN – Occupational health nurse; [f] LiSM – Lifestyle Modification; [g] GHQ – General Health Questionnaire; [h] MET- Metabolic Equivalent

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3. 3 Characteristics of PA interventions

3.3.1 PA types and targeted fitness area

Table 2 summarises the characteristics of the interventions (more details are provided in Supplementary Table 1). Interventions overwhelmingly focused on aerobic PA and five studies targeted muscle strength, balance and flexibility; one study delivered a combination of progressive balance and muscle strengthening [34], two studies focused on tai chi [26, 33] and two studies focused on yoga classes [23, 31].

Seven studies addressed multiple cardiovascular risk factors [24, 27, 28, 35, 37, 38, 41], four targeted PA and nutrition [25, 36, 39, 40] and seven focused only on PA [26, 29-34]. Nearly all multiple risk factor interventions used a generic description of the PA intervention such as 'exercise on their own' [35], 'exercise training' [28], referrals to exercise classes [38], or choosing PA goals [24, 27]. In the studies by Hughes et al. [24] and Low et al. [28] the interventions also offered on-site self-managed activities such as walking groups or using workplace exercise facilities.

In addition, in these multiple risk factor interventions, some forms of 'risk assessment', such as a nurses' check-up for CVD risks, were applied before the targeted behaviour was selected by or for participants [38] or self-appraisal of risk using questionnaires [24, 27], or assessment carried out during the first educational session [37, 41]. No study assessed risk of falls or injuries.

The PA and nutrition interventions focused on aerobic regimens such as steps accumulation [25, 36, 39], counselling to shape PA goals [40] or an aerobic workout on site [25]. The study by Strijk et al. [25] also offered yoga sessions, an activity that is recognised as multi-

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dimensional and which enhances muscle strength, balance and mobility, but these fitness outcomes were not assessed [42].

The PA-only interventions either focused on aerobic regimens, such as use of a treadmill workstation [32], self-managed aerobic walking [30] or Nordic walking [29]. Although Nordic walking is primarily aerobic activity, it can also increase upper body strength and flexibility (Table 2), we therefore consider the study multi-dimensional [29]. Two studies offered tai-chi classes, which is also considered multi-dimensional PA (tai-chi) [43]. One study invited older office employees to novel tai chi combined with theraband exercises, adding progressive strength training to tai-chi [33]. The other study offered tai-chi classes to nurses as part of a wellness program [26]. The study by Granacher et al. offered progressive strength and balance exercises performed in the office. At the beginning of each week the trainer introduced the series of exercises to be performed three times a day, five days a week [34].

3.3.2 Delivery modes

Three delivery modes were noted (Table 2). In eight studies (mostly PA only interventions) the PA intervention was delivered on site during working hours [25, 26, 28, 29, 32-34], or on site but before working hours [31], in six studies participants were counselled in the workplace in face-to-face meetings [35-37], or by telephone calls (COACH-arm) [24], or a combination of both [40, 41], but the PA sessions took place in employees' discretionary time. In four studies the PA was self-managed with little or no supervision [27, 30, 38, 39]. The study by Hughes et al. [24] had two intervention arms; one involved personal contact with students trained in behaviour change coaching ("COACH"), while the other arm was

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self-managed through a web-based intervention ("RealAge") with no other contact [24]. The study by Cook et al. [27] was also a web-based self-managed intervention (HealthyPast50).

3.3.3 Intervention duration and compliance

Eight interventions were short-term, ranging from 6 to 15 weeks [26, 27, 29-31, 33, 34, 39]. Seven interventions lasted between 5 and 9 months [25, 28, 36-38, 40, 41], and three interventions lasted 12 to 13 months [24, 32, 35]. All studies provided data related to intervention retention rates (i.e., withdrawals) with the exception of one pre-post study [38]. Seven studies did not report intervention compliance (e.g. attendance at classes, recorded logs). Studies of short duration also reported higher compliance (>75%) compared to studies with medium or long duration. Four studies reported on program implementation fidelity [24, 26, 38, 40].

3.3.4 Intervention reach

Program reach (i.e., proportion of study population recruited to the trials) varied substantially. When recruitment methods involved a targeted strategy based on prior 'screening for eligibility criteria', a strategy reported in five studies [30, 36-38, 40], intervention reach was highly variable. In the Japanese study, 92% of the targeted eligible employees were recruited [36], while two interventions in the UK [30, 38], the Taiwanese study [37] and the Italian study [40] using the same strategy achieved much lower recruitment rates (ranging from 20% to 35%). In the ten studies that used workplace advertisement as the method of recruitment [22, 24, 26-29, 31, 33, 35, 41], only two studies reported on the reach of the program: Strijk et al. invited all employees listed in the targeted age range (n=3756) and indicated that 27% of those agreed to participate [25]. Cook et al. [27] offered a financial incentive ($25) for completing the baseline survey, which 96% of the employees aged >50 years did, resulting in

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276 participants [27]. The rest of the studies either did not report or could not estimate reach because key information, such as the number of eligible potential participants exposed to the advertisement, was missing.

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Table 2: Summary of the characteristics of workplace physical activity interventions delivered to older employees (18 studies)

	Multiple risk factors including physical inactivity							Physical activity & diet				Physical activity only						
Intervention characteristics	Abbas 2015[38]	Chen 2016[37]	Cook 2015[27]	Hughes 2011[24]	Low 2015[28]	Sharp 1992[35]	Edman 2019[41]	Arao 2007[36]	Strijk 2011-12[22,23]	Naug 2016[39]	Scapel-lato 2018[40]	Bassey 1983[30]	Grana-cher 2011[34]	Palum-bo 2012[26]	Kocur 2017[29]	Chop Hurley 2017[31]	Berg-man 2018[32]	Qi 2019[33]
Aerobic workout, walking, steps accumulation				X	X	X		X	X	X		X	X				X	
Nordic walking															X			
Balance exercise													X					
Muscle strength													X			X		X
Flexibility																		
Multi-dimensional PA									Yoga					Tai-chi		Yoga poses		Tai-Chi+
Generic description 'exercise goals', referral to class	X	X	X	X	X	X	X				X			X				
Delivery mode [a]	SM	IC	SM	IC/SM	OS	IC	IC	IC	OS	SM	IC	SM	OS	OS	OS	OS	OS	OS
Duration [b]	M	M	S	L	M	L	M	M	M	S	M	S	S	S	S	S	L	S
Recruitment method [c]	TS	TS	WA	WA	WA	WA	WA	TS	WA	I	TS	TS	I	WA	WA	WA	I	WA
Process outcomes [d]	R 64% C - no report F	R 96% C-no report Q	R 80% C 72%	R_{arm1} 91% R_{arm2} 51% C_{arm1} 97% C_{arm2} 57% F	R 28% C Means ses/wk very poor	R 63% C 73%	R 70% C Mean ses/per trial 7.5 out of 12	R 95% C 55%	R 80% C 73%	R 64% C- no report Q	R 53% C – no report F	R 54% C 33%	R 100% C 99%	R 71% C 82% F	R 91% C- no report	R 75% C Means ses/wk 1.2 (poor)	R 85% C – no report	R 75% C- no report
PA-related outcomes [e]	NA	+	+	-	-	-	NA	+	NA	-	NA	+	$+_1$	+	$+_1$	$+_1$	+	+

[a] Delivery mode: OS-on site PA sessions, SM - self-managed program, IC -individualised counselling
[b] Duration: S -short ≤ 15 weeks, M - medium 5-9 months, L - long >9 months
[c] Recruitment: TS-targeted strategy (e.g., screening), WA – worksite advertisement emails, posters, word of mouth inviting participation, I –information sessions and volunteers to take part
[d] Process outcome: R - retention of intervention participants, C - compliance % from ideal target, F - fidelity of intervention delivery, Q - qualitative comments
[e] NA = if no comparison group + = significant between groups positive effect/s in favour of interventions, $+_1$ = significant between groups positive effect in favour of intervention only in one PA-related outcome "-" = no significant between group effect on PA measure

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3.4 Study quality

Included RCTs and non-randomised studies were mainly of poor quality (Supplementary Table 2 and Supp Figure 1). In 5 of the RCTs the randomisation was poorly described and in 7 of the RCTs the concealment of allocation was poorly described. Only five out of 12 RCTs were deemed at a 'low risk' of selection bias [25, 27, 31, 32]. Performance bias is expected due to inability to blind participants, as would be expected in trials involving behaviour modification. Four of the 12 RCTs relied only on self-reported PA-related outcomes [24, 27, 28, 35], resulting in a high risk of ascertainment bias. Eight trials included objective measures of PA or fitness outcomes; of those, four involved less than 40 participants [26, 29, 31, 34]. An additional major concern was the risk of selection bias due to attrition given that most trials (78%) had incomplete outcome data, and intention-to-treat analysis was done in only five RCTs [25, 27, 31, 32]. The two included studies that were not RCTs had a high risk of bias in three, respectively four out of the five domains.

3. 5 Effectiveness of interventions

Two studies assessed PA behaviour, but were not included in meta-analysis as we could not extract SMD from the mixed model with interaction, both reported no-between groups difference [24, 35]. Table 3 summarises the effect of the interventions on PA behaviour and fitness outcomes, including pooled effect sizes which are also presented in Supplementary Figure 2. Overall, no pooled outcomes showed statistically significant differences between intervention and control groups.

Three studies that examined PA/exercise frequency [27, 28, 37] had a pooled SMD of 0.25 (95% CI -0.07 to 0.56) and moderate heterogeneity (I^2=53%, p=0.19).. Time spent in moderate-to-vigorous PA was computed from the studies using accelerometers [25, 32] and

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the study by Arao et al. [36] which used self-report. All these studies presented results at 6-month follow-up. The pooled SMD was 0.22 (95%CI -0.05 to 0.50) with acceptable heterogeneity (I^2=46%, p=0.16). .

Aerobic fitness using VO_2 max was assessed in two studies [25, 36] with a pooled SMD of 0.28 (95% CI -0.22 to 0.78) and high heterogeneity (I^2=82%, p=0.02); in the study by Strijk et al. [25] a small proportion of the sample took the test, introducing performance bias. In the study by Arao et al [36], the participants were not randomised, introducing potential selection bias and confounding.

Two small-scale studies (n=57) reported the effect of the intervention on muscle strength (i.e., plantar strength and knee flexor) with an SMD of 0.27 (95%CI -0.26 to 0.80) [31, 34]. Another study included isometric knee extensor strength as an outcome, but did not provide data, suggesting selective reporting [26]. Three small-scale studies (n=46) reported on static balance (measured by postural control [34] and functional reach [26, 33]) with a non-significant medium pooled effect size (SMD=0.74, 95% CI: -0.21 to 1.69) and high heterogeneity (I^2=74.3%, p=0.069). Two studies (n=58) examined the effect of the intervention on flexibility in trials of tai-chi [26] and Nordic walking [29] and found a moderate effect size (0.50) without heterogeneity, but that was also statistically non-significant (95% CI -0.04 to 1.05).

GRADE ratings of the certainty or quality of evidence for each outcome were low, mostly due to the high risk of bias of included studies, high heterogeneity (inconsistency), and imprecision (all pooled CIs included 0 and were statistically non-significant).

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Table 3: Pooled effect sizes for PA behaviour and fitness outcomes with GRADE rating of evidence in trials with comparison group

PA outcomes References	Pooled effect size Standardised Mean Difference	Numbers (studies)	Quality of evidence (GRADE)[a]	Comments related to GRADE and SMD
Frequency PA /exercise (per week) [27,28,37]	SMD=0.25 95% CI: -0.07 to 0.56 Heterogeneity=54%, *p*=0.114 T^2 = 0.0419	N=448 (3 studies)	Very Low ⊗OOO Serious methodological limitations; serious indirectness; inconsistency with studies not included	All interventions were multiple risk factors and were self-managed; the exact nature of the exercise intervention and the goals set for people were unclear, limiting attribution of increases to the intervention. The RCT by Sharpe et al.[35] was not included as it was not possible to extract SMD from the mixed model.
Moderate-to-vigorous physical activity (MVPA) [25,32,36]	SMD=0.22, 95% CI: -0.05 to 0.50 Heterogeneity=46.2%, *p*=0.16 T^2 =0.079	N=404 (3 studies)	Low ⊗⊗OO Two RCTs used accelerometer data and one self-report non-randomised trial and therefore reduced overall quality	Additional study by Hughes et al. [24] was not included as it was not possible to extract SMD from mixed model with interactions term. The study reported no between-groups significant effect.
Aerobic fitness, VO_2 max [25,36]	SMD=0.28 95% CI: -0.22 to 0.78 Heterogeneity=82%, *p*=0.019 T^2 =0.1069	N=389 (2 studies)	Low ⊗⊗OO Serious methodological limitations; inconsistency	Strijik et al. [25] had high loss for this measure and high risk of performance & ascertainment bias; Arao et al. [36] was a non-randomised controlled trial with high risk of bias due to confounding.
Balance [26,33,34]	SMD=0.74 95% CI: -0.21 to 1.69- Heterogeneity=74%, *p*=0.020 T^2 =0.4980	N=86 (3 studies)	Low ⊗⊗OO Serious methodological limitations; serious imprecision; inconsistency	Granacher et al.[33] used balance platforms which has greater precision and also measures horizontal and vertical planes. Palumbo et al. [26] and Qi et al.[33] only measured horizontal displacement (functional reach).
Muscle strength [31,34]	SMD=0.27 95% CI: -0.26 to 0.80 Heterogeneity=0%, *p*=0.380 T^2 =0.000	N=57 (2 studies)	Low ⊗⊗OO Serious methodological limitations; serious imprecision	The trial by Palumbo et al [26] was not included despite listing isometric knee extension because the results were not presented (selective reporting). Chopp-Hurley et al. [31] recruited older employees with osteoarthritis; Granacher et al. [34] did not provide information to judge the trial quality;
Flexibility [26, 29]	SMD =0.50 95% CI: -0.05 to 1.05 Heterogeneity=0%, *p*=0.366 T^2 =0.000	N=58 (2 studies)	Low ⊗⊗OO Serious methodological limitations; imprecision	In both RCTs [by Plumbo et al [26] and Kocur et al [29] the randomisation process was unclear; small sample sizes and high risk of performance bias and selective reporting

a= High quality: Further research is very unlikely to change our confidence in the estimate of effect
Moderate quality: Further research is likely to have an important impact on our confidence in the estimate of effect and may change the estimate. Low quality: Further research is very likely to have an important impact on our confidence in the estimate of effect and is likely to change the estimate. Very low quality: We are very uncertain about the estimate

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4. DISCUSSION

To our knowledge this is the first systematic review and meta-analysis of workplace PA interventions specifically designed for older employees. Although some interventions demonstrated a significant effect on one fitness outcome, collectively we found no definitive evidence on effectiveness as pooled effects remained non-significant. The methodological quality of the studies was generally low. Only one pre-post study examined whether participants met aerobic recommendations [38] and only four small-scale trials focused on balance or muscle strength as any outcomes and all were pilot studies [29, 31, 33, 34]. No program fully addressed the WHO PA older adults' guidelines, which highlight the need to incorporate balance exercises (≥3 times a week) and muscle strengthening (≥2 times a week) along with the aerobic recommendations of a minimum of 150 minutes per week of moderate-to-vigorous PA. Therefore, there is a noticeable mismatch between the aims and outcome measures of interventions and those recommended as ideal for older adults. Additionally, this review of the literature suggests that the initial levels of participation (i.e., intervention reach) was mostly low, which was also noted in another systematic review of workplace health promotion interventions [44].

High heterogeneity makes the pooled estimates difficult to interpret as it is unclear whether population characteristics, intervention implementation or methodological quality issues underlie the heterogeneous results across studies. For example, two cluster RCTs reported changes at 6 months in predicted maximal oxygen uptake; the Japanese study reported a significant medium effect size based on a sub-maximal stationary bike test [36] and the Dutch study [25] reported a small non-significant effect on the 2km walking test. The difference between these two measures could explain the high heterogeneity as well as differences

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related to the completion rate; high in the Japanese study (85% and 93%) and low in the Dutch study (56% and 66%) in the control and intervention groups, respectively.

Multiple risk factor interventions incorporated screening for cardiovascular risk factors, but did not address other age-related health problems, such as falls, which are preventable by appropriate exercise [10, 14]. Such screening could be carried out either through a single self-report question (e.g., history of falls) or through assessment of physiological fall risk (e.g., poor balance, impaired leg strength). Risk of falls and related injuries is an important safety issue because the consequences of falls and injuries are greater in older employees compared to their younger counterparts [6]. Further, despite emerging evidence that resistance training is as beneficial as aerobic training for the prevention and management of cardio-metabolic risks [45, 46], none of the multiple risk factor interventions explicitly included muscles strength training to reduce CVD/metabolic risk. Muscles strength training was promoted to employees with osteoarthritis, in accordance with clinical guidelines [31], but only one intervention specifically targeted muscle strength to generally healthy older employees [34], supporting the recent claim that strength training has been a neglected aspect of PA promotion [47]. From the mid-forties onwards, each decade adults lose 14% to 16% of their isokinetic leg strength [48], which is a greater decline than that observed in cardiovascular capacity (~10% per decade) [49]. Considering that employees aged 50 years and over are likely to stay at work for another decade or more, neglecting this domain is unwarranted. In fact, tests of older university employees (mean age 63 years; n=258) in Brazil found that 55.8% and 6.3% of them were classified in the “sarcopenia” and “severe sarcopenia” groups, respectively. In the same study, the prevalence of frailty was 9% and 63% were classified as “pre-frail” [50].

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The included PA programs varied substantially in terms of the PA modality, delivery methods and duration. Interestingly, compared with self-managed programs carried out in participants' discretionary time, on-site supervised sessions during the workday [25, 26, 29, 33, 34] tended to demonstrate high retention rates (71% to 100%) and good compliance. However, these programs were usually shorter in duration (less than 15 weeks). On-site pragmatic approaches, such as treadmill workstations, short breaks for balance and strength training or web-based interventions, can be easily integrated into the workday. However, the effectiveness of treadmill workstations was disappointing [32], and the collective evidence for other approaches was inconsistent [24, 27, 34]. These interventions should be further tested by high-quality replication studies before translational research can be conducted.

Multiple risk factor interventions mostly had poor description of the PA component of the intervention which reduced their reproducibility. For example, referrals to exercise lacked specifications of the type of exercise programs that were available to clients, or goal settings lacked descriptions of how goals were tailored to individuals, how they were monitored and by whom. In addition, most studies rarely reported the fidelity of the interventions. Hence it is recommended to use a template for reporting of PA interventions, such as TIDieR [51], in order to allow better replication of sensible effective PA interventions.

It is worth noting that about 40% of the interventions were conducted in universities or academic hospitals/clinics. This could possibly be due to ease of doing research in these settings coupled with environments that are conducive to health promotion messages, such as health-care providers or academics. Hence, the generalisability of intervention implementations and the findings of these studies is limited.

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A concerning finding is the poor quality of the RCTs we identified. The study quality could be improved by appropriate randomisation, intention to treat analysis, and the use of both objective and self-report measures of PA. For example, most multiple risk factor interventions included objective measures of CVD risks (i.e., blood tests or anthropometrics), but failed to use objective fitness tests or activity trackers (e.g., pedometer, accelerometer). Further, to improve the directness of evidence (i.e., the selection of outcome measures should better reflect the fitness dimension(s) that the proposed PA is likely to affect), such as measuring balance and strength outcomes in interventions involving yoga or tai-chi or aerobic fitness in Nordic walking interventions.

Research demonstrates increases in leisure-time PA after transition to retirement compared to during full employment [52, 53]. However, a systematic review has found disparities by socioeconomic status (SES); employees with low SES tend to be less active after retirement whereas those with high SES tend to be more active after retirement [53]. Focus groups with older retirees revealed that retirees from lower SES appeared to place lower value on the importance of leisure-time PA [54]. Therefore, improving PA whilst in the workforce may be of even greater importance to older people with low SES. In this review we identified three studies that specifically targeted low paid employees in manual occupations (i.e., factory employees) in Japan [36] and the UK [30, 38]. Only the study from Japan, which included social and environmental support in addition to behavioural counselling, demonstrated significant positive effects. However, this was a non-randomised study with a high risk of bias for almost all bias domains. Further work is therefore needed to ascertain what types of interventions can most effectively improve PA outcomes in older employees with low SES.

4.1 Strengths and limitations

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This review applied guideline-informed, rigorous methods to synthesise and evaluate the current evidence on workplace PA interventions for older employees. Compared with the 2016 systematic review by Poscia and colleagues [13], we have expanded upon the list of workplace health promotion interventions delivered to older employees by 11 studies [28-32, 34-36, 38-40] by carrying out a more sensitive and comprehensive literature search. However, some limitations need to be acknowledged: first, we limited our search to electronic databases which can result in missing unpublished workplace interventions. Second, studies tended to apply different measures or used different units for similar measures. Despite our best attempt to harmonise measures (e.g., converting kcal to minutes of at least moderate-intensity PA), comparability across studies is still limited and may explain the high heterogeneity; for example measuring postural control by balance platform as opposed to a Functional Reach Test. In addition, the current evidence base is limited by the number of studies included for each PA outcome and the quality of existing studies. Based on these limitations we have mapped out research gaps and suggestions for the conduct and reporting of future interventions targeting older employees (Table 4).

Table 4: Recommendations for the conduct and reporting of future workplace interventions targeting older adults

Aspect	Recommendation
Reach and recruitment	1. To enable calculation of the reach of workplace interventions we suggest that as a first step employers will endorse the intervention and provide the list of all employees or at least the number of the potential target population (e.g., age or year prior to retirement). It may also boost recruitment when the management of the workplace endorses such intervention.
Population	2. Expand the current target population (ie. mostly university and healthcare staff) to include a broader range of occupations, including manual jobs and workers from low socioeconomic status (SES) background. This is important for increased generalisability and reducing gaps between high and low SES workers in lifestyle behaviours after retirement.
Intervention content	3. Incorporate strength, balance, and flexibility training along with aerobic activities.. 4. Moving interventions from a "pilot" stage to larger trials and scalable programs; 5. Intervention targets should include screening for falls and injury risk, frailty and sarcopenia

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	6. Improve the reporting on the nature of PA interventions. For example, for interventions that involve a referral to a gym or other facility, describe precisely what exactly these referrals included: advice only, a written program, or a gym membership 7. Improve reporting of the intervention using a standard checklist (TIDieR) including implementation fidelity of interventions.
Study methodology	8. Cluster RCTs should include several clusters (workplaces) and should adjust for cluster effect; 9. Studies should assess confounders and adjust for their presence. 10. Interventions that incorporate multi-dimensional activities (e.g., yoga, taichi, dance) should assess the fitness outcomes specific to be improved by these activities (leg strength, balance, gait speeds). Standardising measures across studies will increase the of studies included in future meta-analyses. 11. Improving the conduct / reporting of RCTs in terms of methods of randomisation, concealment of allocation, blinding the assessors and analysts, reporting results for all randomised participants first before any ad-hoc analyses and including objective measures of physical activity
Duration and sustainability	12. Consider intervention beyond 6 months and evaluation of maintenance beyond a year

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Declaration

Ethics approval and consent to participation

Not applicable for systematic review

Availability of Data and Materials

Not applicable

Competing interests

No competing interests to declare.

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Data availability

All data relevant to the study are included in the article or uploaded as supplementary information.

Author contributions

DM conceptualised the research and drafted the first manuscript. DM and JS conducted the systematic search and selection. FS and DD conducted the quality assessment. AT and DM conduced the meta-analysis. DM and SM conducted the data extraction to tables. All authors (DM, FS, JS, AT, SM, KG, DD) participated in the interpretation of results, commented on first draft and gave final approval of the manuscript.

Consent to publication

Not applicable

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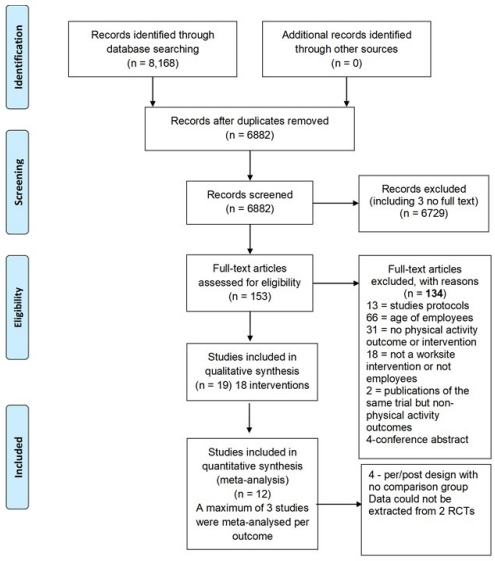
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Figure 1: PRISMA flow diagram of selected workplace physical activity intervention studies

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338x190mm (96 x 96 DPI)

PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
TITLE			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	1
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	2
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	4-5
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	5
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	6
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	6
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	6-7
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	Supplementary file
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	6-7
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	7
Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	6
Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	7-8
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	8
Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I^2) for each meta-analysis.	8 Table 3

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PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	8, GRADE Table 3
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	NA
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	8 -9 & Figure 1
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	Table 1 & Supp T1
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	19, & Supp T2
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	18 Table 2
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	19-21 Table 3
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	Table 3
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	NA
DISCUSSION			
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	23
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	25-26
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	26 & Table 4
FUNDING			
Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	NA

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PRISMA 2009 Checklist

From: Moher D, Liberati A, Tetzlaff J, Altman DG, The PRISMA Group (2009). Preferred Reporting Items for Systematic Reviews and Meta-Analyses: The PRISMA Statement. PLoS Med 6(7): e1000097. doi:10.1371/journal.pmed1000097

For more information, visit: **www.prisma-statement.org**.

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Physical Activity Worksite Interventions Delivered To Older Employees: A Systematic Review

Medline Search Strategy – limit final results by age (45 and up), humans, not reviews, English

Work Terms	Physical Activity	Intervention
MeSH terms		
Workplace	Exercise (exp)	Health Education
Occupations	Walking	Inservice Training (exp)
Employment (exp)	Postural balance	Occupational Health Services
	Physical fitness (exp)	Occupational Health
	Physical exertion	Health Promotion (exp)
	Exercise Therapy (exp) (inc. resistance training, Muscle Stretching Exercises)	"Physical education and training"
	Quality of Life	
	Life style (exp) (inc. Healthy Lifestyle)	
	Health Behavior	
	Yoga	
	Sports (exp)	
Keywords		
Worksite.mp	Aerobics.mp	Intervention.mp
Work environment.mp	Physical activity.mp	Program.mp
Job site.mp	Wellness.mp	Prevent*.mp
Employee*.mp	Fitness.mp	Promot*.mp
Worker*.mp	Pilates.mp	
Economically active.mp	Exercise.mp	
	Tai chi.mp	

CINAHL Search Strategy – limit final results by age (45 and up), humans, not reviews, English

Work Terms	Physical Activity	Intervention

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MeSH terms?		
Work Environment	Exercise (exp)	Health Education
Occupations	Walking	
Employment (exp)	Postural balance	Occupational Health Services
Work Environment	Physical fitness (exp)	Occupational Health
	Exertion	Health Promotion (exp)
	Therapeutic Exercise	"Physical education and training"
	Quality of Life	
	Life style (exp)	
	Resistance Training	
	Yoga	
	Sports (exp)	
	Group exercise	
Keywords		
Worksite.mp	Aerobic Exercises	Intervention.mp
	Physical activity	Program.mp
Job site.mp	Wellness	Prevent*.mp
Employee*.mp	Psychological wellbeing	Promot*.mp
Worker*.mp	Pilates	Inservice training.mp
Economically active.mp	Exercise.mp	
Workplace.mp	Tai chi.mp	

PsycInfo Search Strategy – limit final results by age (40 and up), humans, English

Work Terms	**Physical Activity**	**Intervention**
MeSH terms		
	Exercise (exp)	Health Education
Occupations	Walking	Inservice Training (exp)
Employment (exp)	Physical activity	
	Physical fitness (exp)	Occupational Health

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	Movement Therapy	Health Promotion (exp)
	Lifestyle changes	Workplace Intervention
	Quality of Life	
	Aerobic exercise	
	Lifestyle (exp)	
	Health Behavior	
	Yoga	
	Sports (exp)	
Keywords		
Worksite.mp	Aerobics.mp	Intervention.mp
Work environment.mp	Physical activity.mp	Program.mp
Job site.mp	Wellness.mp	Prevent*.mp
Employee*.mp	Fitness.mp	Promot*.mp
Worker*.mp	Pilates.mp	
Economically active.mp	Exercise.mp	
Workplace.mp	Tai chi.mp	
Employment.mp	Resistance training.mp	

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▼ **Search History** (47)

	# ▲	Searches	Results	Type	Actions	Annotation
	1	Workplace/	23992	Advanced	Display Results | More	
	2	occupations/	23208	Advanced	Display Results | More	
	3	exp Employment/	88952	Advanced	Display Results | More	
	4	worksite.mp.	2972	Advanced	Display Results | More	
	5	work environment.mp.	9401	Advanced	Display Results | More	
	6	job site.mp.	77	Advanced	Display Results | More	
	7	employee*.mp.	64566	Advanced	Display Results | More	
	8	worker*.mp.	201742	Advanced	Display Results | More	
	9	economically active.mp.	606	Advanced	Display Results | More	
	10	exp exercise/	203707	Advanced	Display Results | More	
	11	walking/	34042	Advanced	Display Results | More	
	12	postural balance/	24102	Advanced	Display Results | More	
	13	exp physical fitness/	31436	Advanced	Display Results | More	
	14	physical exertion/	56619	Advanced	Display Results | More	
	15	exp exercise therapy/	52973	Advanced	Display Results | More	
	16	quality of life/	204059	Advanced	Display Results | More	
	17	exp life style/	96894	Advanced	Display Results | More	
	18	health behavior/	51575	Advanced	Display Results | More	
	19	yoga/	2968	Advanced	Display Results | More	
	20	exp sports/	188295	Advanced	Display Results | More	
	21	aerobics.mp.	645	Advanced	Display Results | More	
	22	physical activity.mp.	118123	Advanced	Display Results | More	
	23	wellness.mp.	10986	Advanced	Display Results | More	
	24	fitness.mp.	96101	Advanced	Display Results | More	
	25	pilates.mp.	542	Advanced	Display Results | More	
	26	exercise.mp.	366702	Advanced	Display Results | More	
	27	tai chi.mp.	1893	Advanced	Display Results | More	
	28	health education/	61303	Advanced	Display Results | More	
	29	exp inservice training/	29418	Advanced	Display Results | More	
	30	Occupational Health Services/	10609	Advanced	Display Results | More	
	31	occupational health/	34329	Advanced	Display Results | More	
	32	exp health promotion/	78692	Advanced	Display Results | More	
	33	physical education.mp. and training/ [mp=title, abstract, original title, name of substance word, subject heading word, floating	0	Advanced	Save | More	

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Supplementary Table 1 Characteristic of worksite physical activity (PA) interventions delivered to older employees (organised by publication dates)

Study	Recruitment / reach	Intervention content / theoretical guidance	Intervention deliverers	Duration	Process data
Bassey, E. J., et al, 1987 [27]	Initial discussions for the support of the management; After a publicity launch in the factory newspaper, a personal letter of invitation was sent to all workers through factory's internal mail system; 35% attended introductory meetings and ultimately 23% agreed to participate	"The New Aerobics" (1970) Cooper's prescription for people aged ≥50yrs, which isa progressive walking program instructed at workplace but carried out at employees' leisure time. 1st week, walking 20mins$_{*}^{-1}$d and at a pace of 5.2km$_{*}^{-1}$hr preferably 5 days per week. Subjects can increase duration of walks and frequency. In 2nd phase the intensity can increase based on heart rate response to exercise at 4.8km$_{*}^{-1}$hr. The target dose was 40mins at 7km$_{*}^{-1}$hr 5 days per week. Individual discussion at beginning (how to progress) and half way through. The progression was individually discussion on courses and timing at start and mid program. Participants were asked to keep daily log-card of duration and distance walked. Encouraging articles in the factory newspaper were published monthly.	Unsupervised program – no mentioning of the profession of the person conducted the meeting to inform the grade of effort	12 weeks program divided to 6 weeks each phase	59 out of 108 completed the intervention (54%). Log cards: only 1/3 nearly achieved the speed and distance prescribed for the last 2 weeks, 1/3 provided evidence on logs for extra walking but substantially below prescription, and 1/3 failed to return log card. The most compliant were the most active before program started
Sharpe, P. A. et al 1992 [35]	Advertising at work, direct contact with departmental representatives, telephone calls from the research staff and provision of additional dates and sites to enhance the accessibility of screening; 250 employees did health screening but the % from total employees is unknown	Health promotion activities after health risk screening (smoking, blood pressure, level of blood lipids and body weight) and brief communication of results by nurse. Participants could choose to have face-to-face with meetings with health counsellors, to join walking groups at work and/or exercise on their own at home or at university facilities. No other promotional activities were reported.	health counsellors at work provided face-to-face with counselling based on participants choice	12 months	60 out of 95 experimental participants (63%) completed the program. Behaviour change self-report: 48% of Control and 53% of have done anything to increase fitness. started program on their own- 63% and 73% of C and Int. took worksite program 13% of C and 11% of Int; used worksite facilities: 11% of C 16% Int.
Arao, T. et al 2007 [36]	Two companies were recruited, all together 5 sites. Medical check-ups were conducted and 192 employees were identified to have CVD risk factor(s) of those 177 (92%) were eligible and consented and ultimately 155 (87%) were allocated as a cluster to LiSM (n=92) and Control (n=85) groups.	Theoretically driven (Transtheoretical model and Social Cognitive Theory) Life Style Modification Program for Physical Activity and Nutrition (LiSM-PAN). Structured counselling for goals setting for each behaviour and monthly meeting. Log book for steps a day (10,000 as a goal); increasing incrementally; if reached take another sport/exercise class. Environment support at worksite: 1) setting up a walking course; 2) installing exercise facilities; 3) displaying caloric content table of the lunch menu at cafeteria; 4) cafeteria manager advised on healthy cooking for better diet. Social support at home: printed material on healthy diet and cooking; participant and spouse going together for exercise	Research staff were trained in structured counselling;	6 months	95.2% completion. 97.1% attended monthly counselling 86.7% self-monitored the walking steps 54.7% controlling their dietary targeted activities

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Study	Recruitment / Reach	Intervention content / theoretical guidance	Intervention deliverers	Duration	Process data
Granacher et al. 2011 [34]	Several companies in Basel were approached and 2 large companied (~ 500 employees) agreed (randomization at cluster level). Informational session was held explaining the intervention and eligibility criteria, 34 employees consented and participated.	Three times a day five days a week of training in: Static balance exercises – in morning at arrival to office Dynamic balance exercises – after lunch break Strength enhancing exercises before leaving office New exercise routines were assigned each week to be repeat for the whole week before progressing.	Expert in balance and strength training saw each participant every week	8 weeks of training followed by 8 weeks without training data collected at 8 and 16 weeks	All 17 intervention participants completed the program (100%). Program compliance was excellent with participants of the intervention group completing 99% of the scheduled training sessions (~ 119 out of 120 training sessions).
Hughes et al. 2011 [24]	Worksite advertisement Research participants recruited via announcements on staff list-serves, targeted e-mails, staffed recruitment tables at events in highly trafficked buildings, and flyers posted throughout the university. Out of 647 enquiries 66% deemed eligible and 423(65%) enrolled in the study.	COACH –shaped by the Enhance Wellness Program for seniors in the community and guided by the Transtheoretical Model. One face- to- face meeting with coach to review the participant's health-risk profile (baseline assessment), discuss the behaviors the participant was interested in adopting, and discuss the availability of health behavior change programs on or off campus, assist in creating plan to achieve goals. Biweekly contact by email/telephone for the first 6 months to re-evaluate plan, goals and actions. In the remaining 7 to 12 months the contact was monthly. Health-risk profile assessments were at 6 and12 months. RealAge- A Web page contain health risk test and generate health profile based on family history, diet and nutrition, PA, tobacco and alcohol use, satisfaction with family relationships, and use of preventive health care. Participants used the Web site to select behaviors to work on and create plans to meet behavioral goals. The RealAge site tracked each time a participant used it, and the site forwarded this information to the study team at regular intervals, enabling us to document use.	COACH -MPH graduate trained in behavioural change coaching RealAge – research staff	12 months	Retention: 91% in COACH and 84% RealAge. Fidelity data: COAC: Coach spent 9.25hr per participant – there were 688 emails 77% by the coach and 48 telephones FU, RealAge- 59% completed all health risk assessment and 53% engaged in at least one of the web link.
Palumbo et al. 2012 [26]	One medical centre targeting nurses currently employed full time or part time in staff nurse position that involved lifting patients. About 70 nurses place their interests and 14, on a first come first served basis, were randomized.	Tai Chi workplace wellness program, guided by the Revised Health Promotion Model by Pender (1996). Classes on site were carried once a week (including 15 minutes breathing exercise, 30 mins tai chi practice 5mins visualization cool down). Additional practice on their own 10mins 4 times a week	Experienced tai chi instructors (22 years of teaching)	15 weeks	Five out of the 7 intervention participants (71%) completed the tai-chi program. Attendance 82% range:69%-92% Fidelity- two classes were cancelled

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Study	Recruitment & Reach	Intervention content / theoretical guidance	Intervention deliverers	Duration	Process data
Strijk et al. 2011 , 2012 [22,23]	All workers of two academic hospitals were invited to participate between April and October 2009; Among 3756 employees invited to participate, 1101 (29%) agreed to participate and 730 (19%) allocated into two groups	3 visits to Personal Vitality Coach (PVC) at worksite: at start of intervention, followed by 2 consecutive visits at 4-6 weeks and 10-12 weeks. Meeting 30 minutes each. Weekly yoga 45mins and workout sessions (intensity 65%-90% of the age-predicted maximum heart rate) for 45mins. Plus 45 min unsupervised aerobic exercise. Fruit & Vegetable at the session program name 'Vital@work'	Certified yoga instructor and exercise physiologist	6 month interventio n period and additional 6 months sustainabili ty testing	293 out of 367 (80%) intervention participants completed the program. PVC visits 100%. Dose delivered – Yoga sessions 72.3% (Amsterdam - 89.3%; Leiden - 58.3%) Workout sessions 96.3% (Amsterdam – 95.1%; Leiden – 97.4%) High satisfaction score
Cook, et al. 2015 [27]	A recruitment flyer was emailed by company officials to all employees 50 years plus; interested employees were provided information on study and a link to the online baseline survey (received $25 if completed); out of 290 employees, 279 (96%) completed the baseline survey.	HealthyPast50, an automated web-based program containing multimedia health promotion information and guidance based on Social Cognitive Theory. Modules: Facts about Healthy Ageing, My Health Profile – assessment module with questionnaires and recommendations, Stress and Mood Management, Healthy Eating, Active Lifestyle, and Tobacco Free. Encouragement of self-assessment (My Health Profile) before accessing modules. the content of the Active Lifestyle module is oriented toward older users, including the use of the Physical Activity Readiness Questionnaire (PAR-Q), assessing one's readiness for physical activity, the relationship of physical activity to mobility and energy in later years, and the need to moderate the intensity of exercise as one ages	No human contact for the program; project staff can be reached by telephone for queries	3 months; program progress by monthly email and reminder to access	Formative evaluation of the program including focus groups and testing materials. Process evaluation: mean # log-in:4.33 (SD 4.28, range 0-28), the mean minutes in the program was 102.26 minutes (SD 148.32), and the mean number of pages viewed was 11.04 (SD 20.08, range 0-120).
Abbas et al. 2015 [38]	Targeted recruitment of low-paid unskilled or semi-skilled employees age >40 on social care. A single individualised invitation was sent with simultaneous promotion of the programme within standard employer–staff communication routes. Out of 3264 target group (low SES employees), 665 (20% response rate) attended health screening. Of them, 579 were eligible and 427 (74%) returned for the follow-up assessment	Health screening of all major cardiovascular risks was offered to employees of low SES in deprived area. Based on screening results participants were stratified to either "high" , "moderate" or "low" CVD risk. The intervention included referrals to either GPs (if pharmacological treatment were needed), Weight management (in the community, exercise training, smoking cessation and alcohol reduction counselling either face to face or by telephone. Employers ensured that time of assessments and time spent on health promotion will be accounted towards the daily working hours. These services were provided in the community.	Occupational health nursing staff. The nurses were trained to provide smoking cessation and alcohol awareness delivery at the national standard.	9 months interval from initial assessment to follow-up	The attendance to health screen was 20%, much lower than the general population that offered similar program (health screen). Fidelity=There was a record of the referral number given to each domain of intervention (exercise 38%, smoking 5%, alcohol 1%and weight management 20%). No account on the attendance to service referred.

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Study	Recruitment & Reach	Intervention content / theoretical guidance	Intervention deliverers	Duration	Process data
Low et al. 2015 [28]	A promotional flyer to all female at age range & invited to call the centre if interested in reduction of CVD risk; once reached 60 employees who had ≥1 risk of CVD no further recruitment effort was made	Multi-components risk reduction educational sessions in groups on: weight loss/nutrition, stress management, exercise training, and smoking cessation accompanied by access to an on-site gymnasium, and organized group walks. Intervention groups received weekly communication via phone or e-mail, incorporating goal-setting and suggestions for overcoming obstacles.	Not specified	6-months for classes, gym availability organized walks and weekly communication Follow-up survey at 12 months	Attendance to 1st classes was highest for weight loss (42% & 47% control). Stress reduction (30% & 21%) and physical activity 25% to 26%) followed by very poor attendance to consecutive classes. 3 attended group walking; use of Gym 0-20 days over 6 months . Follow-up 16/57 (28%)
Chen et al. 2016 [37]	Approached six worksites small- and medium scale enterprises (SMEs). Three sites showed interest and approved. Among 1245 workers in the three sites, 438 (35.2%) met study criteria of being full-time employed and aged over 50 years. These were invited. Of those, 264 (60.3%) showed interest and finally 108 (25%) agreed to participate.	Phase I: on-site lectures and workshops on healthy living specifically impact of metabolic disorders and individualized self-management goal setting. Biweekly meetings face-to-face with occupational health nurse (OHN) to share experience, give mutual support and encouragement, learn about managing own health, and cultivate leadership. Phase II: on-site team led meetings; discussing barriers reinforcing personal goals; recording food consumption and exercise. Logs were reviewed by OHN and awards/prizes were given for excellent improvement	Occupational Health Nurse	Total 24 weeks Phase 1 was 4-week phase II 20- weeks.	111 from initial 136 intervention participants (80%) completed the program Qualitative Evaluation: Participants found it difficult to records daily PA and diet so no review of records was made; Working overtime reduce the compliance with planned activities and goals; activities forced to cut short due lack of support from management at low level
Naug et al. 2016 [39]	Contacted with employer; Information session meetings were arranged to build rapport with the employers and drivers; after that, drivers were invited to an information presentation and were given an information package.	Educational sessions of 60mins delivered fortnightly in groups at worksite, interactive and fun. Content were derived from the "Life! Taking action on diabetes" program which was based on health psychology theories with a focus on self-regulation. Targeting physiological change due to inactivity, harm of sitting, healthy nutrition. Participants were provided pedometers and asked to record steps each day for 6 weeks.	Two postgraduate students with expertise in adults' education, human physiology & nutrition	6 weeks	21 out of 33 that started program (64%) completed the program. Qualitative after intervention : drivers said " I walk around the bus during breaks" "I, use the back seats of the bus to do push-ups"; "park my car far from depot to walk more"

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Study	Recruitment & Reach	Intervention content / theoretical guidance	Intervention deliverers	Duration	Process data
Kocur et al. 2017 [29]	Workplace advertisement Postmenopausal female who replied to the advertisement regarding the study (n=91) and net the inclusion criteria (n=44) but no data on % from the total number of the target population.	Nordic walking training program techniques and training 3 times a week for one hour including 10-min warm up phase at start 5-min cool down at end. Intensity progress 40%-70% of heart-rate reserve	Two instructors specialised in Nordic walking regulated the pace	12 weeks	No report on program attendance or fidelity.
Chopp-Hurley et al. 2017 [31]	Workplace advertisement through e-mail list serves, webpage at McMaster workplace wellness committee Potential participants had to have clinical osteoarthritis either of the hip or the knee; no number was provided for those that were screened or refused.	Program of static lower extremity strengthening exercise; designed initially for Knee Osteoathritis to avoid KAM (knee adduction moment); The program includes Yoga poses such as squats and lunges; A perceived intensity of 5-7 from a total 10-point scale was recommended. On-site 4 timea a week sessions in the University sport and recreation facility were provided before work (7-8 AM). Participants were asked to attend at least 3 sessions/week. Participants in intervention received weekly feedback on their performance of exercises. Incentive of $50 was given to maintain participation in both groups and control participants were offered the program at the end of trial	Not specified	12 weeks	From initial 24 participants (12;12) 3 intervention participants did not complete FU (25%) as opposed to 1 in the control (12%). Exercise group attended on average 1.2 sessions per week; median was 0.5 classes per week, lower quartile was 0 a week and upper quartile was 3.9 per week thus estimate that 33% adhered to 3 sessions per week.
Bergman et al. 2018 [32]	13 different companies in 17 sites were involved (number of employees age 40-67 not given). Information session were given to office workers- the information included who is eligible (i.e., at least 4 days/wk in the office with sedentary work tasks with a sit-to-stand workstation, don't have a travel day during their work). Information on study strict procedures and measurement only 100 assessed and 80 enrolled	Health consultation about 30-40 minutes about their screening results and sitting time at work and diet. Folder with written information about PA recommendations for moderate and vigorouse aerobic activity and strength, how to increase light PA and Non-exercise PA such as climbing stairs, walk and talk meetings. After this session the treadmill adapted to desk use was installed next to the existing workstation. Physiotherapist adjust the workstation and instruct the worker how to use the treadmill in the first week – any pace 15min twice a week. Thereafter they were recommended to use it for 1hr a day preferably longer. Booster email from research team remined the hazard of sitting, reminder to use treadmill at least 1hour a day. These emails were sent at 6,20,31 and 50th week over the program.	Nurses conducted the health consultation and boosters emails by research team	13 months	Of the 80 randomised (40;40) 34 continued their participation for 13 months. In the control 92% continued for 13 months. No data was collected on usage of the treadmill during work Threat for contamination between group was acknowledged

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Study	Recruitment & Reach	Intervention content / theoretical guidance	Intervention deliverers	Duration	Process data
Scapellato et al. 2018 [40]	Targeted strategy; screening for eligibility (at least 1 CVD risk: blood pressure, weight, waist circumference lipids, impaired fasting glucose) 1892 workers on voluntary basis agreed to be screened; 1574 did not meet the inclusion criteria 318 recruited (50% from 632 eligible). Final recruitment 226 (35% from eligible).	As part of the Occupational Health and Safety approach, a workplace intervention was offered to promote healthy lifestyle for healthcare workers in the hospital who are at risk; intervention included individualized counselling meeting a month after baseline assessment to discuss results of PA and diet, and diverted to health counselor to identify PA and diet goals and strategies to achieve them. Helpful printed material were given such as food pyramids and a brochure of exercise. After 3 months participants received further motivational support via phone counselling to remind their goals and assist to change lifestyle habits.	Hospital Employees Occupational Health practitioners, health promotion counsellors and hospital dietitians	6 months	Fidelity=92 out of 318 eligible recruited (29%) did not receive the intervention. Of the 226 that received intervention, only 169 remained for 6-month assessment. No information on % phone calls given at 3-month. Reason to lost to follow up were attributed to “no time” & “too many work commitments”; only 40% who completed the study favoured adherence to dietary recommendations
Qi et al. 2019 [33]	Worksite advertisement; posters, person-to-person invitations using card-sized flyers, promotion of the study by colleagues and an e-mail request from the University Administration promoting the research projects that office workers may be interested to partake in. No data on number of potential eligible (55+ in the Uni) or how many were interested	Based on evidence that taichi or resistance training alone can improve physical function and psychological wellbeing of older adults. The intervention offered 3 session per week from a choice of 8 different timetables to allow flexibility: Monday, Wednsday and Fridays (2 session) 7-9 AM midday and afternoon 3-6PM. Session took 45 minutes. Ten Taichi movements were practiced with theraband tied to palms to a relaxing/ calming music. Theraband degree of effort were progressive. Every fortnight a Rating of Perceived Exertion was completed. When RPE was below 5 the degree the difficulty was increased	Taichi instructor, which is the first author of the paper, delivered the intervention	12 weeks with mid-term (6-weeks) assessment	At 6 week 9 participants withdrew (22%) and no further withdrawals happened at 12 weeks. Reason to lost to follow up were: back issue, knee issue, time conflict flue and no reason given. Data on compliance was not reported; there were 36 sessions offered but no data on mean session attend.
Edman et al 2019 [41]	Worksite advertisement; e-mail invitations (3) to get health coaching for free. Flyers were posted in various locations in the main hospital including the cafeteria and in satellite office locations.	Health coaching on cardiovascular health delivered face-to-face or by phone or in combination. A cyclie is considered 12 weeks during which 6 consultations 2-30min each must be completed. Sessions are client centred behavioural changes strategies and include provision of education and individualized support to make change. Sessions start with assessment, changes from previous session, goal setting (fortnightly), elaborate on topic the client chose (the SMART framework model)	A certified Health Coach with 4 years’ experience with a PhD degree in Nutrition (the first author)	5.5 months with cycle is 12 weeks with option to continue.	74% (40/50 participants) completed 12 weeks. Participants averaged 7.8 sessions (range 6–12) over 5.5 months (range 3.0–10.0). Coaching were in-person for 57% and 17% receive only telephonic coaching and 26% received combination of both.

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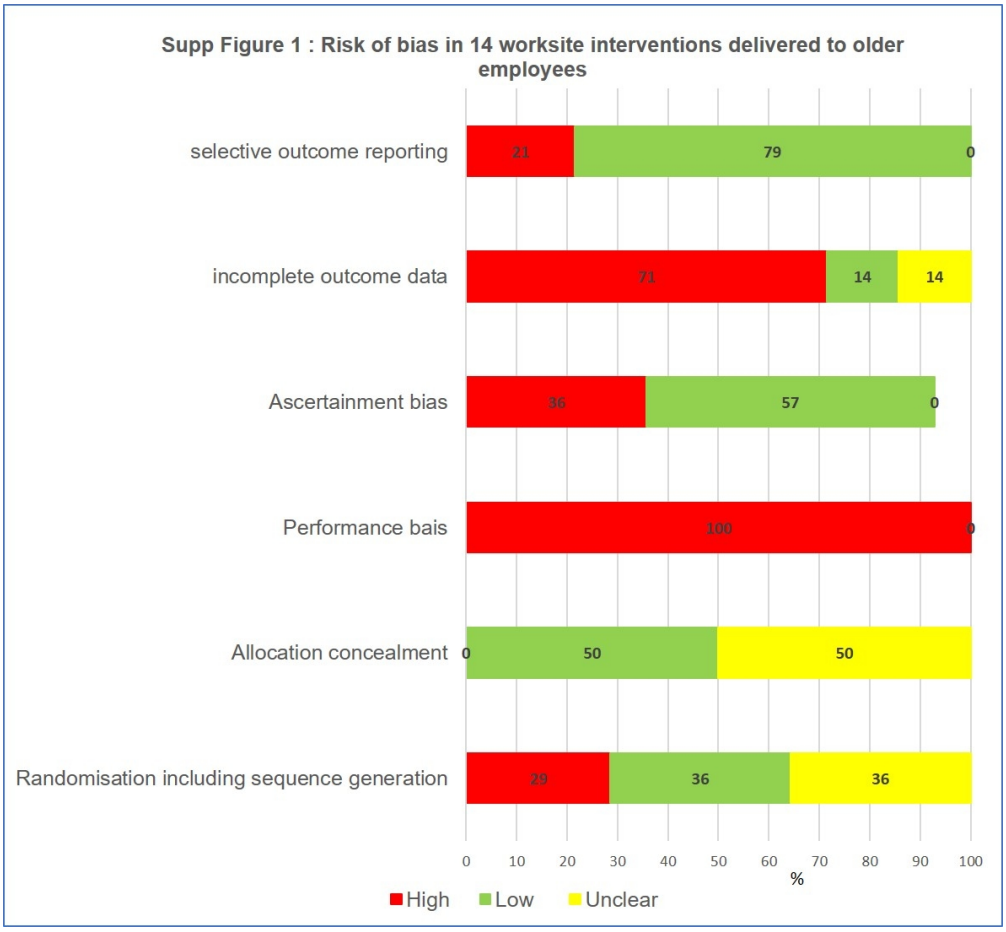
Supplementary Table 2: Risk of bias assessment for RCTs and quasi-experiments

RCTs	Risk of bias (High, low, unclear)					
Source	Randomisation sequence Allocation	Concealment	Performance bias	Ascertainment bias [a]	Incomplete outcome data	selective reporting
Bassey et al. 1987	Unclear	Unclear	High	Low	High	Low
Sharpe et al. 1992	High	Unclear	High	High	High	Low
Granacher et al. 2011	High	Unclear	High	Low	Unclear	Low
Hughes et al. 2011	Low	Unclear	High	High	High	Low
Strijk et al. 2012	Low	Low	High	Low	High	High
Palumbo et al. 2012	Unclear	Unclear	High	low	High	High
Cook et al. 2015	Low	Low	High	High	High	Low
Low 2015	Unclear	Unclear	High	high	Unclear	High
Kocur et al. 2017	Unclear	Low	High	Low	High	Low
Chopp-Hurley et al. 2017	Low	Low	High	Low	Low	Low
Bergman et al. 2018	Low	unclear	High	Low	Low	Low
Qi et al. 2019	unclear	Low	High	low	High	Low
Non-RCT: Confounding						
Chen et al., 2016	High	Low	High	High	High	Low
Arao, 2007	High	Low	High	Low	High	Low

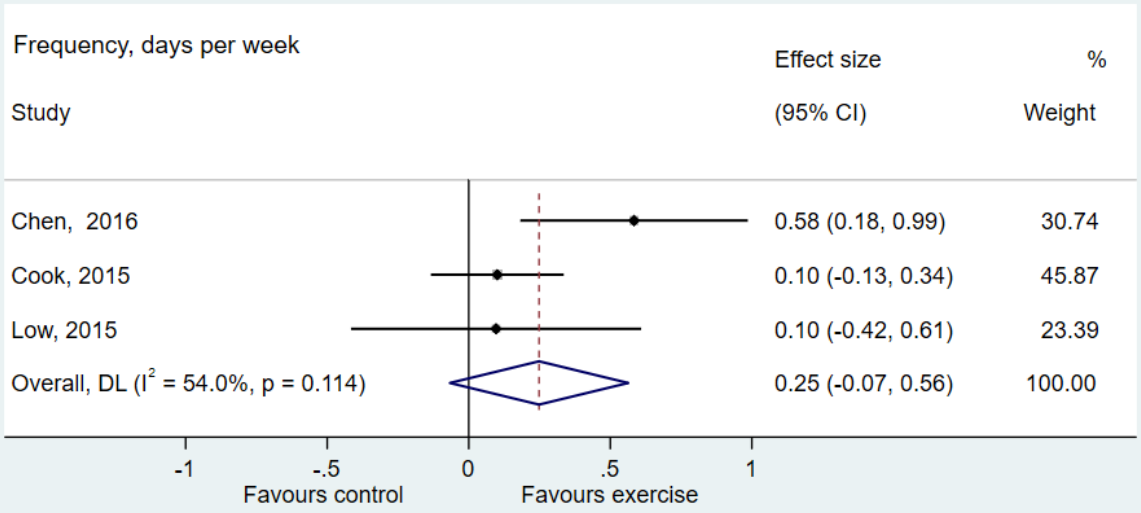
[a] Studies that had included objective physical activity outcomes and subjective outcomes were classified as low

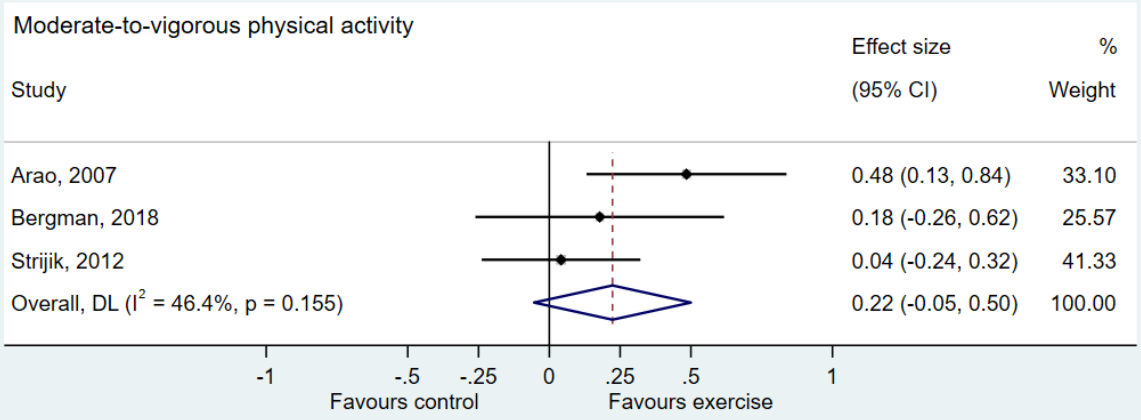
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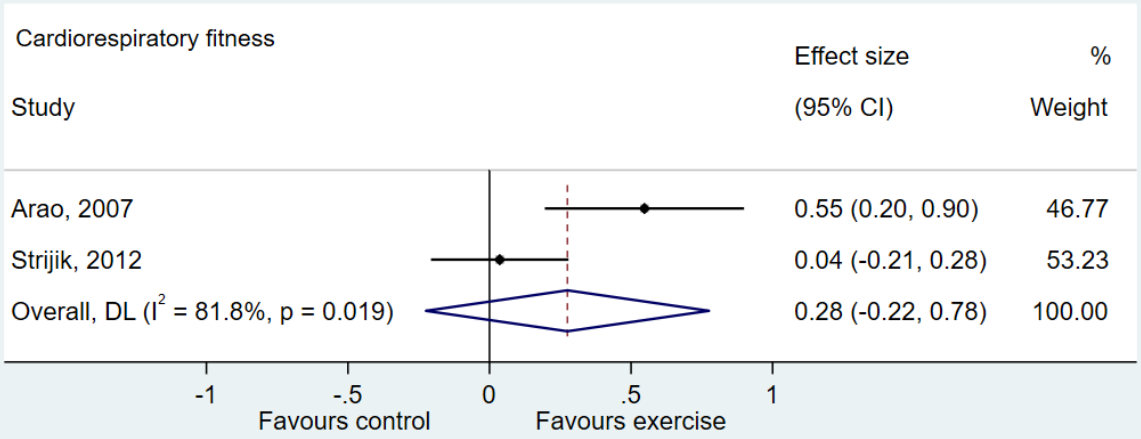
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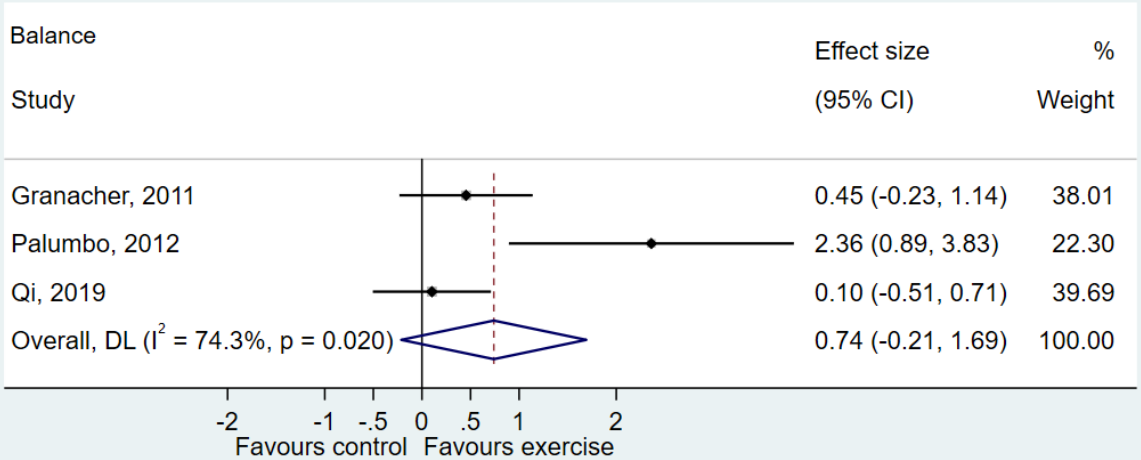


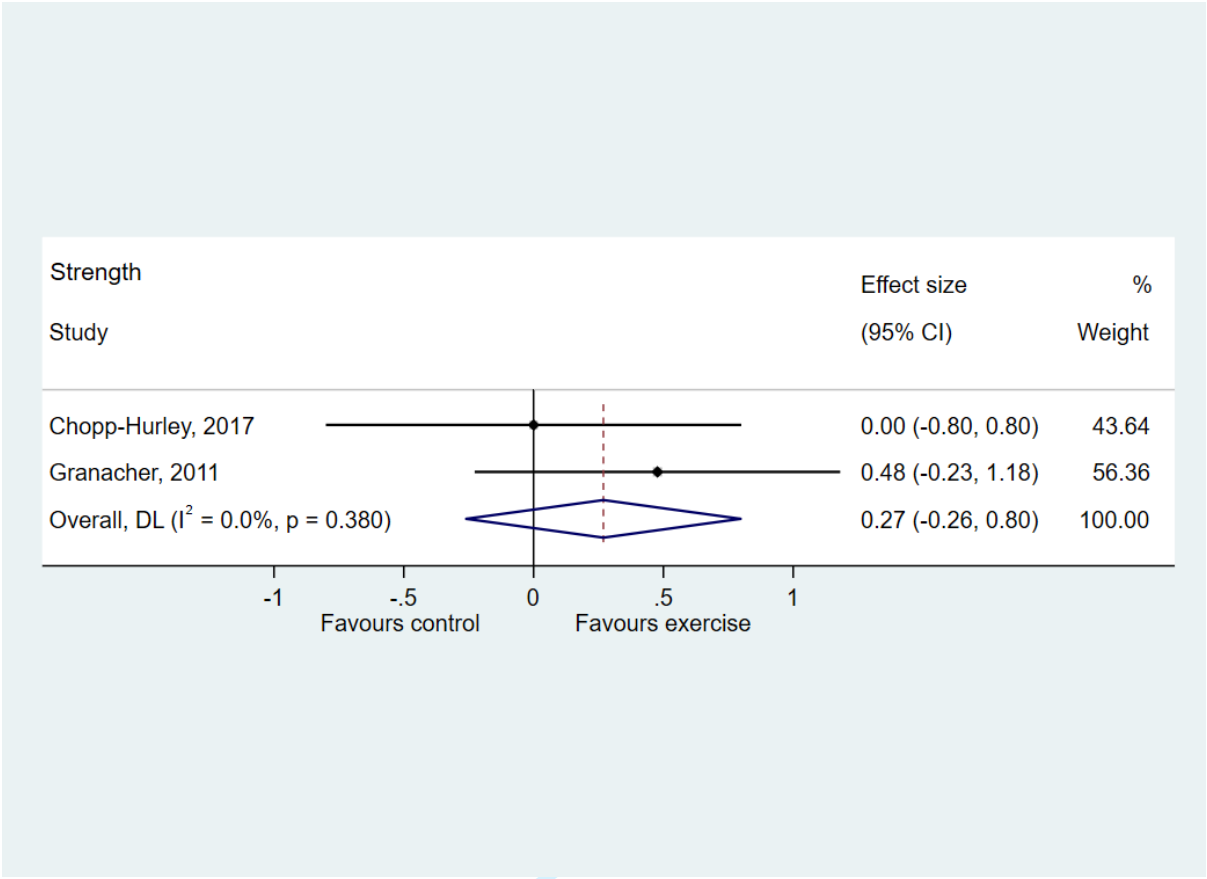
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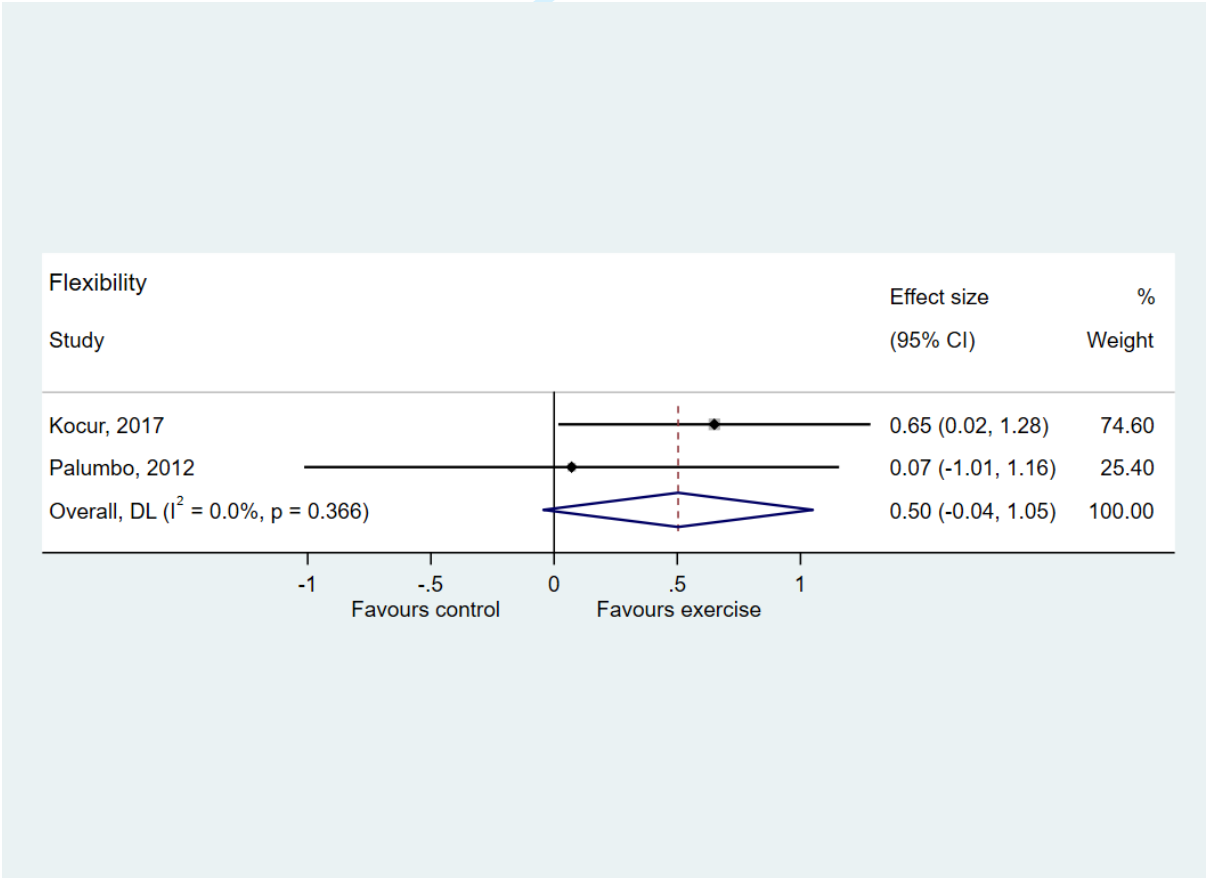

Frequency, days per week
Study
Effect size (95% CI)
%
Weight
Chen, 2016
0.58 (0.18, 0.99)
30.74
Cook, 2015
0.10 (-0.13, 0.34)
45.87
Low, 2015
0.10 (-0.42, 0.61)
23.39
Overall, DL (I² = 54.0%, p = 0.114)
0.25 (-0.07, 0.56)
100.00
-1
-.5
0
.5
1
Favours control
Favours exercise


Moderate-to-vigorous physical activity
Study
Effect size (95% CI)
%
Weight
Arao, 2007
0.48 (0.13, 0.84)
33.10
Bergman, 2018
0.18 (-0.26, 0.62)
25.57
Strijik, 2012
0.04 (-0.24, 0.32)
41.33
Overall, DL (I² = 46.4%, p = 0.155)
0.22 (-0.05, 0.50)
100.00
-1
-.5
-.25
0
.25
.5
1
Favours control
Favours exercise


Cardiorespiratory fitness
Study
Effect size
(95% CI)
%
Weight
Arao, 2007
0.55 (0.20, 0.90)
46.77
Strijik, 2012
0.04 (-0.21, 0.28)
53.23
Overall, DL (I² = 81.8%, p = 0.019)
0.28 (-0.22, 0.78)
100.00
-1
-.5
0
.5
1
Favours control
Favours exercise


Balance
Study
Effect size
(95% CI)
%
Weight
Granacher, 2011
0.45 (-0.23, 1.14)
38.01
Palumbo, 2012
2.36 (0.89, 3.83)
22.30
Qi, 2019
0.10 (-0.51, 0.71)
39.69
Overall, DL (I² = 74.3%, p = 0.020)
0.74 (-0.21, 1.69)
100.00
-2
-1
-.5
0
.5
1
2
Favours control
Favours exercise


Strength
Study
Effect size (95% CI)
% Weight
Chopp-Hurley, 2017
0.00 (-0.80, 0.80)
43.64
Granacher, 2011
0.48 (-0.23, 1.18)
56.36
Overall, DL (I^2 = 0.0%, p = 0.380)
0.27 (-0.26, 0.80)
100.00
-1
-.5
0
.5
1
Favours control
Favours exercise


Flexibility
Study
Effect size (95% CI)
% Weight
Kocur, 2017
0.65 (0.02, 1.28)
74.60
Palumbo, 2012
0.07 (-1.01, 1.16)
25.40
Overall, DL (I^2 = 0.0%, p = 0.366)
0.50 (-0.04, 1.05)
100.00
-1
-.5
0
.5
1
Favours control
Favours exercise

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Supporting active ageing before retirement: a systematic review and meta-analysis of workplace physical activity interventions targeting older employees

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Supporting active ageing before retirement: a systematic review and meta-analysis of workplace physical activity interventions targeting older employees

Dafna Merom[1], Fiona Stanaway[2], Klaus Gebel[3,2] Joanna J. Sweeting[4], Anne Tiedemann[5], Shirin J. Mumu[1], Ding Ding[2]

1. School of Health Science, Western Sydney University, Penrith, New South Wales, Australia
2. Sydney School of Public Health, University of Sydney, Camperdown, New South Wales, Australia
3. Australian Centre for Public and Population Health Research, School of Public Health, Faculty of Health, University of Technology Sydney, Ultimo, New South Wales, Australia
4. Cardio Genomics Program, Centenary Institute Newtown New South Wales Australia
5. Sydney School of Public Health, Institute of Musculoskeletal Health, The University of Sydney, Camperdown, New South Wales, Australia

Corresponding Author:

Prof Dafna Merom

School ofHealth Sciences

Campbelltown Campus 24.2.33

The University of Western Sydney,

Locked Bag 1797, Penrith NSW 2751, Australia

Tel : +61-2- 4620 3747 **E** d.merom@westernsydney.edu.au

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Abstract

Objective: With the growing representation of older adults in the workforce the health and fitness of older employees are critical to support Active Ageing policies. This systematic review aimed to characterise and evaluate the effects on physical activity (PA) and fitness outcomes of workplace PA interventions targeting older employees.

Design: We searched Medline, PreMedline, PsycInfo, CINAHL, and the Cochrane Controlled Register of Trials (CENTRAL) for articles published from inception to 17th of February 2020. Eligible studies were of any experimental design, included employees aged ≥50 years, had PA as an intervention component and reported PA-related outcomes.

Results: Titles and abstracts of 8168 records were screened and 18 unique interventions were included (3,309 participants). Twelve studies were RCTs. Seven interventions targeted multiple risk factors (n=1,640), involving screening for cardiovascular disease risk factors, but had a non-specific description of the PA intervention. Four interventions targeted nutrition and PA (n=1,127), and seven (n=235) focused only on PA. Interventions overwhelmingly targeted aerobic PA, compared to only four interventions targeting strength and/or balance (n=106). No studies involved screening for falls/injury risk and only two interventions targeted employees of low socioeconomic status. Computation of effect sizes (ES) was only possible in a maximum of three RCTs per outcome. ESs were medium for PA behaviour (ES=0.25 95% CI: -0.07 to 0.56), muscle strength (ES=0.27, 95%CI: -0.26- 0.80), cardiorespiratory fitness (ES=0.28, 95%CI: -22 to 0.78), flexibility (ES=0.50, 95% CI: -0.04-1.05) and balance (ES=0.74, 95% CI: -0.21 to 1. 69). GRADE criteria-rated quality of evidence was 'low' due to high risk of bias, imprecision and inconsistency.

Conclusions: The lack of high-quality effective workplace PA interventions contrasts the importance and urgency to improve the health and fitness in this population. Future

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interventions should incorporate strength and balance training and screening of falls/injury risk in multi risk factors approaches.

PROSPERO; registration number CRD42018084863

https://www.crd.york.ac.uk/prospero/display_record.php?RecordID=84863)

Key words: older employees, Physical activity, exercise, workplace, interventions, systematic review, meta-analyses

Strengths and Limitations of this study

- This systematic review was registered in PROSPERO.
- We followed PRISMA reporting guidelines, and the Cochrane Collaboration and the GRADE approach to rate the quality of evidence.
- We used five databases from inception and backward citation tracking of all included studies.
- Pooled effect sizes that were not expressed in similar units, although they measured the same concept, may have artificially increased the heterogeneity of the effect.

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1. INTRODUCTION

Population ageing poses individual, social, economic and political challenges and is predicted to accelerate even further in the 21st century [1]. The World Health Organization's (WHO) "Active Ageing" concept is a leading global policy strategy for successful ageing [2], particularly influencing retirement policies towards maintaining engagement with paid employment. In the developed world, strategies have been implemented to promote economic activities among aged workers, including incentives for late retirement, penalties for early retirement, and an increase in or abolishment of the age of mandatory retirement [3]. Older retirement age means that more employees beyond the age of 55 years will remain in the workforce [4].

While a number of mental and social abilities can improve with increasing age, and such changes should be viewed as strengths and be respected [4], age-related physiological decline requires special considerations regarding occupational health, safety and productivity [5]. In-depth reviews on ageing at work [6, 7] identified a number of physiological, sensory and psychological differences between older workers and their younger counterparts, including reduced muscle strength, endurance, trunk flexibility, structural and sensorineural degeneration which affect balance, vision and hearing, aerobic capacity, heat tolerance, increased anthropometric risks, psychological exhaustion, anxiety and depression. Further, older employees suffer from greater prevalence of comorbidities, musculoskeletal problems, and sickness-related absence and severe injuries [6, 7].

It is well established that regular physical activity (PA) is effective in attenuating age-related physiological declines in all body systems, and in preventing and managing many age-related chronic and musculoskeletal conditions [8-10]. Adherence to older adults' comprehensive PA

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guidelines including balance exercises, muscle strengthening and moderate-intensity aerobic activity, is likely to benefit older employees in terms of health as well as work productivity and safety. Hence, recently physical employment standards for older employees have been suggested [7]. The workplace has long been considered a good setting for health promotion and many PA interventions have been conducted in this setting, with the first systematic review of their effectiveness published in 1998 [11]. However, we have not identified any systematic reviews of workplace PA interventions that specifically targeted older employees. PA interventions targeting older adults should additionally include activities to promote muscle mass, strength, and balance, which are important for the prevention of sarcopenia, functional disabilities, falls and injuries [12]. A recent systematic review by Poscia et al. synthesised the evidence on health promotion interventions targeting older employees [13]. However, this review examined a variety of health promotion initiatives and have missed interventions that focused on PA, by only using 'fitness' and 'capacity' as search terms. Given that PA guidelines for older adults include a range of physical outcomes relevant to health beyond cardiorespiratory fitness or capacity, such as muscle strength, balance and flexibility [14, 15], it is important to include broader terms when reviewing health promotion programs. Furthermore, the main outcomes included in the review by Poscia and colleagues [13] were health status, wellbeing and work productivity, not PA measures. Thus, the effectiveness of workplace interventions on PA and fitness outcomes of older employees is uncertain.

The objectives of this systematic review were to: i) identify and characterise workplace PA interventions delivered to employees aged ≥50 years; ii) assess the methodological quality of the studies; iii) assess the effect of interventions on PA behaviour and fitness outcomes highlighted in PA guidelines for older adults.

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2. METHODS

This review followed guidance published by the Centre for Reviews and Dissemination, the Cochrane Collaboration [16], and PRISMA guidelines [17] (see Supplementary file:PRISMA checklist). The protocol was submitted to the International Prospective Register of Systematic Reviews (PROSPERO) on the 21st of December 2017 and final registration received on 29th of January 2018; registration number CRD42018084863 available at https://www.crd.york.ac.uk/prospero/display_record.php?RecordID=84863)

2.1 Inclusion and exclusion criteria

We included studies that involved any PA interventions (i.e, intervention that will encourage people to move large skeletal muscles) and were delivered at workplaces to employees aged ≥50 years. Studies that included a wider age range, such as whole workplace intervention, were eligible for inclusion if the mean age of participants was at least 50 years. This decision was based on a systematic review of workplace health promotion interventions by Posica et al who included studies that explicitly targeted older employees and found that the mean age of these workers ranged from 51 to 55 years [13]. Interventions could include those related to aerobic PA (e.g., walking), stretching, balance, muscle strength, yoga, tai chi, Pilates, gym workout, sport or any other form of PA/exercise. Studies that targeted other health risks in addition to physical inactivity (hereafter, multiple risk factors interventions) were only included if PA/exercise was one of the intervention components and a PA-related outcome was reported. Studies with any experimental design (pre-post with or without a comparison group, nonrandomised trials, RCTs and cluster RCTs) were included because of the small number of randomised controlled trials (RCT) anticipated [13].

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Studies were excluded if the PA intervention was not delivered in the workplace (e.g., community, health practice), or if the interventions targeted the whole workplace with a wider age range and did not include specific PA-related outcomes.

2.2 Search strategies

We systematically searched five databases from inception to 23rd of December 2017. We then updated our search on the 13th of June 2019 to include all 2018 and again on the 17th of February 2020. Included databases were: Medline, PreMedline, PsycInfo, CINAHL, and the Cochrane Controlled Register of Trials (CENTRAL). Specific search strategies were developed for each database, using a combination of search terms and subject headings where applicable. All electronic searches included MeSH terms and keywords for workplace (e.g. occupations, worksite), PA (e.g., walking, postural balance, muscle strength) and intervention (e.g., health promotion, program, RCT) and were limited to humans, and age (PsycInfo: 40+ all others: 45+) (see Supplement file: search strategy). One reviewer (JS) retrieved the entries from the database using EndNote (Version X9.3.3) and screened the titles and abstracts to identify potential papers for inclusion as well as running backward citation tracking of potential studies selected for full text review. Two reviewers (JS, DM) independently conducted the final selection of papers based on full text. Disagreements were discussed and resolved by consensus.

2.3 Data extraction

Three data extraction formats were used: A table summarising the included studies was created using the subheadings: source, study design, population characteristics, study duration, type of intervention, comparison group, and outcomes. A second table describes characteristics of each intervention in terms of recruitment, content, intervention deliverer

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and process outcomes based on data extracted by DM and SM. A third table summarises the quality of the included papers and another Table summarises the effects sizes for each outcome along with the level of evidence of this effect.

2.4 Quality assessments and analysis

The Cochrane risk of bias tool for RCTs and additional domains from the ROBINS-I tool for non-randomised studies with a comparison group were used to assess the quality of individual studies [18]. The domains included were: randomisation sequence generation, allocation concealment, blinding of participants and personnel, blinding of outcome assessment, incomplete outcome data and selective outcome reporting. The first two domains were only assessed in RCTs. In non-randomised studies with a control group the additional domains of confounding and selection bias were assessed. Each domain was assessed independently by two authors (DD, FS) who assigned a rating of either 'low', 'high' or 'unclear' risk of bias. Disagreement was resolved by consensus or, where needed, by consultation with a third reviewer (DM). We used Comprehensive Meta-Analysis software (Version 2, Biostat, Englewood, New Jersey, USA) to conduct random effects meta-analyses for each outcome accounting for heterogeneity between studies. The standardised mean difference (SMD) (Hedges' g) was calculated for each meta-analysis, standardised by postscore SD (or its estimate) and calculated using the pre- and post- means and SD or, when unavailable, the mean change score and SD. Effect sizes were categorised as small (≤0.20), medium (> 0.20-0.79) or large (≥0.8) [19]. For trials with moderate-to-vigorous intensity PA as the outcome, we used data from accelerometers when possible. For studies reporting energy expenditure as the outcome, we converted the value to minutes of moderate-intensity PA using the population mean body weight and 4 Metabolic Equivalents for Task (METs) for moderate-intensity PA. Statistical heterogeneity was determined by the I^2 and χ^2 tests to

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measure the proportion of variability between studies and tau-square (T^2) was reported to indicate by how much the true effect sizes vary. We followed the recommendation of the Cochrane Collaboration which states that tests for funnel plot asymmetry should be used only when there are at least 10 studies included in the meta-analysis [16]. Since we had no more than 3 studies for each outcome, we did not assess publication bias. Finally, we used the Grading of Recommendations Assessment, Development and Evaluation (GRADE) approach to rate the overall quality of evidence for each outcome from 'very low' to 'high' [20, 21], including a narrative summary of findings for outcomes without meta-analysis [21]. GRADE ratings are based on the domains of risk of bias in included studies, indirectness of evidence, imprecision, inconsistency and likelihood of publication bias.

Patient and Public Involvement

Neither patients nor the public were involved in the design, conduct, reporting or dissemination of this research.

3. RESULTS

3.1. Study selections and characteristics

Our search resulted in 8,168 records which were screened by title and abstract. Of these, 153 were further assessed for eligibility based on full text (Figure 1). Nineteen studies met the inclusion criteria, however, two publications referred to the same intervention [22, 23], resulting in 18 unique interventions.

Insert Figure 1

The characteristics of the studies are summarised in Table 1. Ten studies were RCTs [24-33], and two were cluster RCTs - one involving only two worksites [34] and the other several

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university units [35]. There were also two non-randomised studies [36, 37] and four pre-and post-evaluations with no comparison group [38-41].

3.2 Participants

In total, 3,309 participants (of those 2085 participated in RCTs) were included in the 18 worksite PA interventions: six in the USA [24, 26-28, 35, 41], two in the UK [30, 38], two in Australia [33, 39] and one each in Taiwan [37], Japan [36], Poland [29], the Netherlands [22, 25], Switzerland [34], Canada [31], Sweden [32], and Italy [40].

Five interventions were delivered in academic institutions [24, 29, 31, 33, 35], four in hospitals or medical centres [25, 28, 40, 41], and four were delivered to factory employees [30, 36-38]. One intervention was delivered to transport workers (drivers) [39], and four were delivered to office workers [27, 32-34]. Most studies (n=12) targeted both genders, with a high percentage of male participants (>60%) in transport [39] and IT workplaces [27], and fewer male participants (5% - 25%) in academic / administrative [24] and medical occupations [25, 31, 41]. Three interventions targeted females only [26, 28, 29] and one males only [36].

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Table 1: Characteristics of the studies that met the inclusion criteria in worksite older employees' physical activity (PA) interventions arranged by design and year

	Design Sample	Population	Duration	Intervention Description	Comparison	Physical activity related outcomes
Bassey et al. 1983 [30]	RCT n=108	UK, factory floor workers (blue collar) from light industrial company, age 55-60, 51% males	12 weeks	PA focus; prescribed aerobic (walking) program with a goal to increase cardiorespiratory fitness	No intervention	1) Physical condition - heart rate (HR) at walking 4.8km/hr 2) Sustained $HR_{4.8km/h}^{-1}$ for at least 7 minutes 3) Daily total minutes walked at $HR_{4.8km/h}^{-1}$ 4) Step count
Sharpe et al. 1992 [35]	Cluster RCT n=250	USA, employees from 'support and academic staff" at the University of Michigan, age 50-69, 53% males	12 months	Multiple risk factor program; after health risk screening a face-to-face meeting with health promoter; optional: walking groups at work/ or use facilities.	Health screening, but no exposure to health promotion program	1) Self-report frequency item with a 5-point response format -which was not explained or referenced. 2) Self–reported action taken to improve fitness in the past year 3) Precursors of exercise behaviours (self-efficacy to exercise 3 times a week and intention to exercise vigorously
Hughes et al. 2011 [24]	RCT (3 arms) n=423	USA, older employees from "support and academic" staff at the University of Illinois, Chicago Age ≥ 55 years 18% males	12 months	Multiple risk factor program to; 1) Enhance Wellness one-on-one coaching program (COACH) 2) a web-based health promotion program (RealAge)	Printed health promotion materials, programs and services offered by university or by community organisations	1) Self -report minutes of moderate-intensity and minutes in vigorous-intensity Using the Behavioral Risk Factor Surveillance System (typical week -7 questions) 2) The Rapid Assessment of Physical Activity (RAPA) for meeting PA recommendations based on 9 items (aerobic muscle training and flexibility, response: yes=1, no=0 combined to total score)
Strijk et al. 2011 [22] & 2012 [25]	RCT n=730	Netherlands, workers from two academic hospitals Age ≥ 45 years Mean age 52.3±4.9 25% males	6 months	PA & nutrition: Vital@work Yoga session 60 minutes and workout classes 30 minutes plus home based aerobic session	General healthy lifestyle written information – diet, physical activity and relaxation	1) The frequency, duration and intensity of participation in commuting, household, occupation and leisure-time PA using the SQUASH [a] questionnaire: 2) Minutes spent on sport-from SQUASH 3) Accelerometer minutes in MVPA on sub-sample (GTM1 Actigraph) 4) Aerobic capacity (VO_2 max) was estimated using the UKK [b] 2 km walk test.

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	Design Sample	Population	Duration	Intervention Description	Comparison	Physical activity related outcomes
Palumbo et al. 2012 [26]	RCT n=14	USA, female nurses from an academic medical centre in Vermont Age ≥ 49 years 0% males	15 weeks	PA focus: Once a week Tai-chi guided classes at work;45 minutes and unguided home-based practice 10-mins 4 times a week	No intervention	1) Flexibility of trunk "Sit-and-Reach" test 2) Strength: isometric knee extensor test with dynamometer 3) Balance-Functional Reach test (cm)
Cook et al. 2015 [27]	RCT n=278	USA, IT employees from two major global IT companies in California and Boston Age – 50 to 68 years 67% males	12 weeks	Multiple risk factor web-based educational program "HealthyPast50" with no human contact.	No intervention Wait-listed	1) Exercise habits: based on Godin LTPA score measuring frequency per week exercising at 3 levels (mild, moderate strenuous) and the number multiplied by intensity factor –reported frequency at each level and in total 2) Exercise self-efficacy score (8-items) 4-point scale 3) Exercise intention
Low et al. 2015 [28]	RCT n=62	USA, female employees of a busy community hospital, North Carolina, 40-65 years old; mean age 52 ± 6.3. 0% males	6 months	Multiple risk factor program to reduce CVD [c] risk as in control, plus weekly motivational sessions by mail or telephone for goal setting	Risk reduction educational classes; Free access to gym on site and walking group	1) Frequency – days per week of exercising 2) Minutes per session of exercising 3) Level of intensity: no exercise, leisurely, moderate, or vigorous exercise (reported as %) 4) Readiness to change exercise
Granacher et al. 2011 [34]	Cluster RCT n=32	Switzerland, sedentary office workers from two large companies in Basel, Age ≥50 37% males	8 weeks	PA focus; progressive balance and strength training at the office using exercise charts to perform 3 times a day; each session lasts 8 minutes	No intervention	1) Balance (static) postural control using balance platform, standing on one leg 30 seconds eyes open–displacement of COP[d] 2) Gait variability and speed on special treadmill 3) Force jumping height measured on a force platform 4) Maximal isometric & isokinetic torque (60° / and rate of torque development (RTD) of the plantar flexor using force platform

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	Design Sample	Population	Duration	Intervention Description	Comparison	Physical activity related outcomes
Kocur et al. 2017 [29]	RCT n=44	Poland, female administrative and academic office workers from 4 higher education institutions in Poznań Age 50-60 years 0% males	12 weeks	PA focus - Nordic walking training program 3 times a week for 1 hour each	No intervention	1) Perceived Pain Threshold of upper body (PPT, kg/cm^2) (i.e., the minimum force that can be applied which induced the feeling of pain) using an electronic pressure algometry 2) Flexibility (shoulders) using back scratch: measuring how close the hands can be brought together behind the back
Chopp-Hurley et al. 2017 [31]	RCT n=24	Canada, Hamilton University employees (McMaster) with clinical osteoarthritis Age > 50 years 21% men	12 weeks	PA focus – Specific Osteoarthritis exercises: static leg strengthening (e.g., yoga poses) 3-4 times a week 7-8 AM	No intervention	1) Hip and knee strengths measured by dynamometer the peak extension and flexion torque out of 5 trials were recorded normalised to body weight 2) 6-minutes walking test 3) Lower limb strength: chair sit to stand during 30 seconds 4) Timed –up-and go test 5) 40meter fast paced (time to complete)
Bergman et al. 2018 [32]	RCT n=80	Sweden, Umea office workers (13 companies). Overweight and obese Age 40-67 years (Mean age 51yrs) 45% males	13 months	PA focus – treadmill workstation recommended to walk 1hr a day at moderate-intensity, but no jogging	Working as usual at their sit-and stand working desk	1) Daily walking time at weekdays and weekend measured by ActiPAL accelerometer 2) Number of steps ActiPAL 3) Time spent in moderate to vigorous PA by Actigraph accelerometer 4) PA bouts of more than 10mins on weekdays and weekend
Qi et al. 2019 [33]	RCT n=40	Queensland, Australia University insufficiently active office workers Age 55-77 (mean 59.6) 22% males	12 weeks	PA focus Taichi movements (10) performed with resistance band (Theraband) 3x per week 45min per session on-site. Employee could choose from 8 different time slots	Active control: Taichi only without the theraband in different room to intervention. Similar number of sessions were offered	1) Lower limb strength using chair sit to stand during 30 seconds 2) Balance Functional Reach test (cm) 3) Hand grip strength 4) 2-minute walk test (work capacity)

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	Design Sample	Population	Duration	Intervention Description	Comparison	Physical activity related outcomes
Chen et al. 2016 [37]	Quasi-experiment (individual allocation) n=108	Taiwan, workers from small- and medium-scale enterprises, Age ≥ 50 years 45% males	6 months	Multiple risk factors; onsite educational workshops during first 4 weeks + meetings with OHN[e] to set goals, plans and on-site group support	Only educational workshops in the first 4 weeks	1) Frequency: number of times per week doing physical activity using Taiwan Longitudinal Study on Aging - no information on the PA domains asked (e.g., leisure, work etc) 2) Sedentary time (hrs/d)
Arao et al. 2007 [36]	Quasi-experiment (cluster allocation) n=197	Japan, employees working in 5 sites of two factories in Tokyo who had at least 1 CVD[c] risk Age 40-59 years (mean age=55) 0% females	6 months	PA & nutrition (LiSM[f]-PAN) multi-components program; individual counselling 5x10-min around goals; social and environmental support	Feedback from the medical check-ups plus recommendations on diet and physical activity including printed materials on exercise, healthy diet and cooking	1) Self-report energy expenditure (kcal/week) derived from the leisure time exercise was assessed by the Kuopio Ischaemic Heart Disease Risk Factor Study 2) Maximum oxygen uptake 3) VO_2max (ml/kg/min) from a sub-maximum bicycle test (Astrad) 4) Stage of change for exercise
Abbas et al. 2015 [38]	Pre-post n=665	UK, low-paid local government employees from socially and economically deprived areas in NE England, Age ≥40 years, mean age 50.5±6.4 37% males	9 months	Multiple risk factor health screening staging risk level and referrals to exercise, weight management, smoking cessation, promotion of mental health and alcohol reduction	Not applicable	1) Participation in aerobic exercise dichotomised to not meeting recommendation (less than 5 times a week of less than 30-minutes session) 2) Referred to exercise question on doing exercise outside work (report on % before and after)
Naug et al. 2016 [39]	Pre-post pilot trial n=33	Australia, bus drivers from two depots of South East Queensland Age: 50-68 years 64% males	6 weeks	PA & nutrition, Group educational sessions; harm of sitting, healthy eating & exercise; PA - pedometer to track steps	Not applicable	1) Exercise levels – no report on questionnaire type; researcher driven classification: a) none (no exercise), b) moderate level (e.g. 30-40min walking twice a week or tennis once a week, c) intense (e.g., gym 4-7 times a week or cycling 5 days/week), sedentary behaviour (hrs/week)

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	Design Sample	Population	Duration	Intervention Description	Comparison	Physical activity related outcomes
Scapellato et al. 2018 [40]	Pre-post n=167	Padua, Italy healthcare workers at risk of CVD Mean age 50±7.3 31.7% males	6 months	PA & nutrition, brochure on exercise and motivational counselling on site at baseline and mid-term by phone	Not applicable	1) PA MET[h] based on a 4-day diary about the type of activity, frequency (day per week) and duration (minutes)
Edman et al. 2019 [41]	Pre-post pilot trial n=54	Philadelphia, USA healthcare hospital workers volunteered with at least one CVD risk Mean age 53.3±10 5% males	6 months 12 weeks cycles with option to continue	Multiple risk factor program - six sessions of Health Coaching f2f or by phone including goal setting, education, motivational strategies	Not applicable	1) Self-reported change in number of sessions per week exercising aerobically for at least 20 minutes 2) Self-reported number of times per week doing muscle strengthening exercise for at least 20 minutes

[a] SQUASH –Short Questionnaire to ASsess Health- enhancing physical activity; [b] UKK- The Finnish Urho Kaleva Kekkonen walking test; [c] CVD – cardiovascular disease; [d] COP – Centre of Pressure; [e] OHN – Occupational health nurse; [f] LiSM – Lifestyle Modification; [g] GHQ – General Health Questionnaire; [h] MET- Metabolic Equivalent

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3. 3 Characteristics of PA interventions

3.3.1 PA types and targeted fitness area

Table 2 summarises the characteristics of the interventions (more details are provided in Supplementary Table 1). Interventions overwhelmingly focused on aerobic PA and five studies targeted muscle strength, balance and flexibility; one study delivered a combination of progressive balance and muscle strengthening [34], two studies focused on tai chi [26, 33] and two studies focused on yoga classes [23, 31].

Seven studies addressed multiple cardiovascular risk factors [24, 27, 28, 35, 37, 38, 41], four targeted PA and nutrition [25, 36, 39, 40] and seven focused only on PA [26, 29-34]. Nearly all multiple risk factor interventions used a generic description of the PA intervention such as 'exercise on their own' [35], 'exercise training' [28], referrals to exercise classes [38], or choosing PA goals [24, 27]. In the studies by Hughes et al. [24] and Low et al. [28] the interventions also offered on-site self-managed activities such as walking groups or using workplace exercise facilities.

In addition, in these multiple risk factor interventions, some forms of 'risk assessment', such as a nurses' check-up for CVD risks, were applied before the targeted behaviour was selected by or for participants [38] or self-appraisal of risk using questionnaires [24, 27], or assessment carried out during the first educational session [37, 41]. No study assessed risk of falls or injuries.

The PA and nutrition interventions focused on aerobic regimens such as steps accumulation [25, 36, 39], counselling to shape PA goals [40] or an aerobic workout on site [25]. The study by Strijk et al. [25] also offered yoga sessions, an activity that is recognised as multi-

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dimensional and which enhances muscle strength, balance and mobility, but these fitness outcomes were not assessed [42].

The PA-only interventions either focused on aerobic regimens, such as use of a treadmill workstation [32], self-managed aerobic walking [30] or Nordic walking [29]. Although Nordic walking is primarily aerobic activity, it can also increase upper body strength and flexibility (Table 2), we therefore consider the study multi-dimensional [29]. Two studies offered tai-chi classes, which is also considered multi-dimensional PA (tai-chi) [43]. One study invited older office employees to novel tai chi combined with theraband exercises, adding progressive strength training to tai-chi [33]. The other study offered tai-chi classes to nurses as part of a wellness program [26]. The study by Granacher et al. offered progressive strength and balance exercises performed in the office. At the beginning of each week the trainer introduced the series of exercises to be performed three times a day, five days a week [34].

3.3.2 Delivery modes

Three delivery modes were noted (Table 2). In eight studies (mostly PA only interventions) the PA intervention was delivered on site during working hours [25, 26, 28, 29, 32-34], or on site but before working hours [31], in six studies participants were counselled in the workplace in face-to-face meetings [35-37], or by telephone calls (COACH-arm) [24], or a combination of both [40, 41], but the PA sessions took place in employees' discretionary time. In four studies the PA was self-managed with little or no supervision [27, 30, 38, 39]. The study by Hughes et al. [24] had two intervention arms; one involved personal contact with students trained in behaviour change coaching ("COACH"), while the other arm was

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self-managed through a web-based intervention ("RealAge") with no other contact [24]. The study by Cook et al. [27] was also a web-based self-managed intervention (HealthyPast50).

3.3.3 Intervention duration and compliance

Eight interventions were short-term, ranging from 6 to 15 weeks [26, 27, 29-31, 33, 34, 39]. Seven interventions lasted between 5 and 9 months [25, 28, 36-38, 40, 41], and three interventions lasted 12 to 13 months [24, 32, 35]. All studies provided data related to intervention retention rates (i.e., withdrawals) with the exception of one pre-post study [38]. Seven studies did not report intervention compliance (e.g. attendance at classes, recorded logs). Studies of short duration also reported higher compliance (>75%) compared to studies with medium or long duration. Four studies reported on program implementation fidelity [24, 26, 38, 40].

3.3.4 Intervention reach

Program reach (i.e., proportion of study population recruited to the trials) varied substantially. When recruitment methods involved a targeted strategy based on prior 'screening for eligibility criteria', a strategy reported in five studies [30, 36-38, 40], intervention reach was highly variable. In the Japanese study, 92% of the targeted eligible employees were recruited [36], while two interventions in the UK [30, 38], the Taiwanese study [37] and the Italian study [40] using the same strategy achieved much lower recruitment rates (ranging from 20% to 35%). In the ten studies that used workplace advertisement as the method of recruitment [22, 24, 26-29, 31, 33, 35, 41], only two studies reported on the reach of the program: Strijk et al. invited all employees listed in the targeted age range (n=3756) and indicated that 27% of those agreed to participate [25]. Cook et al. [27] offered a financial incentive ($25) for completing the baseline survey, which 96% of the employees aged >50 years did, resulting in

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276 participants [27]. The rest of the studies either did not report or could not estimate reach because key information, such as the number of eligible potential participants exposed to the advertisement, was missing.

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Table 2: Summary of the characteristics of workplace physical activity interventions delivered to older employees (18 studies)

	Multiple risk factors including physical inactivity							Physical activity & diet				Physical activity only						
Intervention characteristics	Abbas 2015[38]	Chen 2016[37]	Cook 2015[27]	Hughes 2011[24]	Low 2015[28]	Sharp 1992[35]	Edman 2019[41]	Arao 2007[36]	Strijk 2011-12[22,23]	Naug 2016[39]	Scapel-lato 2018[40]	Bassey 1983[30]	Grana-cher 2011[34]	Palum-bo 2012[26]	Kocur 2017[29]	Chop Hurley 2017[31]	Berg-man 2018[32]	Qi 2019[33]
Aerobic workout, walking, steps accumulation				X	X	X		X	X	X		X	X				X	
Nordic walking															X			
Balance exercise													X					
Muscle strength													X			X		X
Flexibility																		
Multi-dimensional PA									Yoga					Tai-chi		Yoga poses		Tai-Chi+
Generic description 'exercise goals', referral to class	X	X	X	X	X	X	X				X			X				
Delivery mode [a]	SM	IC	SM	IC/SM	OS	IC	IC	IC	OS	SM	IC	SM	OS	OS	OS	OS	OS	OS
Duration [b]	M	M	S	L	M	L	M	M	M	S	M	S	S	S	S	S	L	S
Recruitment method [c]	TS	TS	WA	WA	WA	WA	WA	TS	WA	I	TS	TS	I	WA	WA	WA	I	WA
Process outcomes [d]	R 64% C - no report F	R 96% C-no report Q	R 80% C 72%	R_{arm1} 91% R_{arm2} 51% C_{arm1} 97% C_{arm2} 57% F	R 28% C Means ses/wk very poor	R 63% C 73%	R 70% C Mean ses/per trial 7.5 out of 12	R 95% C 55%	R 80% C 73%	R 64% C- no report Q	R 53% C – no report F	R 54% C 33%	R 100% C 99%	R 71% C 82% F	R 91% C- no report	R 75% C Means ses/wk 1.2 (poor)	R 85% C – no report	R 75% C- no report
PA-related outcomes [e]	NA	+	+	-	-	-	NA	+	-	NA	NA	+	$+_1$	+	$+_1$	$+_1$	+	+
Included in meta-analysis	No	Yes	Yes	No	Yes	No	No	Yes	Yes	No	No	No	Yes	Yes	Yes	Yes	Yes	Yes

[a] Delivery mode: OS-on site PA sessions, SM - self-managed program, IC -individualised counselling
[b] Duration: S -short ≤ 15 weeks, M - medium 5-9 months, L - long >9 months
[c] Recruitment: TS-targeted strategy (e.g., screening), WA – worksite advertisement emails, posters, word of mouth inviting participation, I –information sessions and volunteers to take part
[d] Process outcome: R - retention of intervention participants, C - compliance % from ideal target, F - fidelity of intervention delivery, Q - qualitative comments
[e] NA = if no comparison group + = significant between groups positive effect/s in favour of interventions, $+_1$ = significant between groups positive effect in favour of intervention only in one PA-related outcome "-" = no significant between group effect on PA measure

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3.4 Study quality

Included RCTs and non-randomised studies were mainly of poor quality (Supplementary Table 2 and Supplementary Figure 1). In 5 of the RCTs the randomisation was poorly described and in 7 of the RCTs the concealment of allocation was poorly described. Only five out of 12 RCTs were deemed at a 'low risk' of selection bias [25, 27, 31, 32]. Performance bias is expected due to inability to blind participants, as would be expected in trials involving behaviour modification. Four of the 12 RCTs relied only on self-reported PA-related outcomes [24, 27, 28, 35], resulting in a high risk of ascertainment bias. Eight trials included objective measures of PA or fitness outcomes; of those, four involved less than 40 participants [26, 29, 31, 34]. An additional major concern was the risk of selection bias due to attrition given that most trials (78%) had incomplete outcome data, and intention-to-treat analysis was done in only five RCTs [25, 27, 31, 32]. The two included studies that were not RCTs had a high risk of bias in three, respectively four out of the five domains.

3. 5 Effectiveness of interventions

Two studies assessed PA behaviour, but were not included in meta-analysis as we could not extract SMD from the mixed model with interaction, both reported no-between groups difference [24, 35]. Another RCT assessed changes in daily walking on half of the sample but but reported the pre-post changes for the intervention and control group combined and therefore was not included in meta-analysis [30]. Table 3 summarises the effect of the interventions on PA behaviour and fitness outcomes, including pooled effect sizes which are also presented in Supplementary Figure 2. Overall, no pooled outcomes showed statistically significant differences between intervention and control groups.

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Three studies that examined PA/exercise frequency [27, 28, 37] had a pooled SMD of 0.25 (95% CI -0.07 to 0.56) and moderate heterogeneity (I^2=53%, p=0.19).. Time spent in moderate-to-vigorous PA was computed from the studies using accelerometers [25, 32] and the study by Arao et al. [36] which used self-report. All these studies presented results at 6-month follow-up. The pooled SMD was 0.22 (95%CI -0.05 to 0.50) with acceptable heterogeneity (I^2=46%, p=0.16). .

Aerobic fitness using VO_2 max was assessed in two studies [25, 36] with a pooled SMD of 0.28 (95% CI -0.22 to 0.78) and high heterogeneity (I^2=82%, p=0.02); in the study by Strijk et al. [25] a small proportion of the sample took the test, introducing performance bias. In the study by Arao et al [36], the participants were not randomised, introducing potential selection bias and confounding.

Two small-scale studies (n=57) reported the effect of the intervention on muscle strength (i.e., plantar strength and knee flexor) with an SMD of 0.27 (95%CI -0.26 to 0.80) [31, 34]. Another study included isometric knee extensor strength as an outcome, but did not provide data, suggesting selective reporting [26]. Three small-scale studies (n=46) reported on static balance (measured by postural control [34] and functional reach [26, 33]) with a non-significant medium pooled effect size (SMD=0.74, 95% CI: -0.21 to 1.69) and high heterogeneity (I^2=74.3%, p=0.069). Two studies (n=58) examined the effect of the intervention on flexibility in trials of tai-chi [26] and Nordic walking [29] and found a moderate effect size (0.50) without heterogeneity, but that was also statistically non-significant (95% CI -0.04 to 1.05).

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GRADE ratings of the certainty or quality of evidence for each outcome were low, mostly due to the high risk of bias of included studies, high heterogeneity (inconsistency), and imprecision (all pooled CIs included 0 and were statistically non-significant).

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Table 3: Pooled effect sizes for PA behaviour and fitness outcomes with GRADE rating of evidence in trials with comparison group

PA outcomes References	Pooled effect size Standardised Mean Difference	Numbers (studies)	Quality of evidence (GRADE)[a]	Comments related to GRADE and SMD
Frequency PA /exercise (per week) [27,28,37]	SMD=0.25 95% CI: -0.07 to 0.56 Heterogeneity=54%, *p*=0.114 T^2 = 0.0419	N=448 (3 studies)	Very Low ✇OOO Serious methodological limitations; serious indirectness; inconsistency with studies not included	All interventions were multiple risk factors and were self-managed; the exact nature of the exercise intervention and the goals set for people were unclear, limiting attribution of increases to the intervention. The RCT by Sharpe et al.[35] was not included as it was not possible to extract SMD from the mixed model.
Moderate-to-vigorous physical activity (MVPA) [25,32,36]	SMD=0.22, 95% CI: -0.05 to 0.50 Heterogeneity=46.2%, *p*=0.16 T^2 =0.079	N=404 (3 studies)	Low ✇✇OO Two RCTs used accelerometer data and one self-report non-randomised trial and therefore reduced overall quality	Additional study by Hughes et al. [24] was not included as it was not possible to extract SMD from mixed model with interactions term. The study reported no between-groups significant effect.
Aerobic fitness, VO_2 max [25,36]	SMD=0.28 95% CI: -0.22 to 0.78 Heterogeneity=82%, *p*=0.019 T^2 =0.1069	N=389 (2 studies)	Low ✇✇OO Serious methodological limitations; inconsistency	Strijik et al. [25] had high loss for this measure and high risk of performance & ascertainment bias; Arao et al. [36] was a non-randomised controlled trial with high risk of bias due to confounding.
Balance [26,33,34]	SMD=0.74 95% CI: -0.21 to 1.69- Heterogeneity=74%, *p*=0.020 T^2 =0.4980	N=86 (3 studies)	Low ✇✇OO Serious methodological limitations; serious imprecision; inconsistency	Granacher et al.[33] used balance platforms which has greater precision and also measures horizontal and vertical planes. Palumbo et al. [26] and Qi et al.[33] only measured horizontal displacement (functional reach).
Muscle strength [31,34]	SMD=0.27 95% CI: -0.26 to 0.80 Heterogeneity=0%, *p*=0.380 T^2 =0.000	N=57 (2 studies)	Low ✇✇OO Serious methodological limitations; serious imprecision	The trial by Palumbo et al [26] was not included despite listing isometric knee extension because the results were not presented (selective reporting). Chopp-Hurley et al. [31] recruited older employees with osteoarthritis; Granacher et al. [34] did not provide information to judge the trial quality;
Flexibility [26, 29]	SMD =0.50 95% CI: -0.05 to 1.05 Heterogeneity=0%, *p*=0.366 T^2 =0.000	N=58 (2 studies)	Low ✇✇OO Serious methodological limitations; imprecision	In both RCTs [by Plumbo et al [26] and Kocur et al [29] the randomisation process was unclear; small sample sizes and high risk of performance bias and selective reporting

a= High quality: Further research is very unlikely to change our confidence in the estimate of effect
Moderate quality: Further research is likely to have an important impact on our confidence in the estimate of effect and may change the estimate. Low quality: Further research is very likely to have an important impact on our confidence in the estimate of effect and is likely to change the estimate. Very low quality: We are very uncertain about the estimate

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4. DISCUSSION

To our knowledge this is the first systematic review and meta-analysis of workplace PA interventions specifically designed for older employees. Although some interventions demonstrated a significant effect on one fitness outcome, collectively we found no definitive evidence on effectiveness as pooled effects remained non-significant. The methodological quality of the studies was generally low. Only one pre-post study examined whether participants met aerobic recommendations [38] and only four small-scale trials focused on balance or muscle strength as any outcomes and all were pilot studies [29, 31, 33, 34]. No program fully addressed the WHO PA older adults' guidelines, which highlight the need to incorporate balance exercises (≥3 times a week) and muscle strengthening (≥2 times a week) along with the aerobic recommendations of a minimum of 150 minutes per week of moderate-to-vigorous PA. Therefore, there is a noticeable mismatch between the aims and outcome measures of interventions and those recommended as ideal for older adults. Additionally, this review of the literature suggests that the initial levels of participation (i.e., intervention reach) was mostly low, which was also noted in another systematic review of workplace health promotion interventions [44].

High heterogeneity makes the pooled estimates difficult to interpret as it is unclear whether population characteristics, intervention implementation or methodological quality issues underlie the heterogeneous results across studies. For example, two cluster RCTs reported changes at 6 months in predicted maximal oxygen uptake; the Japanese study reported a significant medium effect size based on a sub-maximal stationary bike test [36] and the Dutch study [25] reported a small non-significant effect on the 2km walking test. The difference between these two measures could explain the high heterogeneity as well as differences

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related to the completion rate; high in the Japanese study (85% and 93%) and low in the Dutch study (56% and 66%) in the control and intervention groups, respectively.

Multiple risk factor interventions incorporated screening for cardiovascular risk factors, but did not address other age-related health problems, such as falls, which are preventable by appropriate exercise [10, 14]. Such screening could be carried out either through a single self-report question (e.g., history of falls) or through assessment of physiological fall risk (e.g., poor balance, impaired leg strength). Risk of falls and related injuries is an important safety issue because the consequences of falls and injuries are greater in older employees compared to their younger counterparts [6]. Further, despite emerging evidence that resistance training is as beneficial as aerobic training for the prevention and management of cardio-metabolic risks [45, 46], none of the multiple risk factor interventions explicitly included muscles strength training to reduce CVD/metabolic risk. Muscles strength training was promoted to employees with osteoarthritis, in accordance with clinical guidelines [31], but only one intervention specifically targeted muscle strength to generally healthy older employees [34], supporting the recent claim that strength training has been a neglected aspect of PA promotion [47]. From the mid-forties onwards, each decade adults lose 14% to 16% of their isokinetic leg strength [48], which is a greater decline than that observed in cardiovascular capacity (~10% per decade) [49]. Considering that employees aged 50 years and over are likely to stay at work for another decade or more, neglecting this domain is unwarranted. In fact, tests of older university employees (mean age 63 years; n=258) in Brazil found that 55.8% and 6.3% of them were classified in the “sarcopenia” and “severe sarcopenia” groups, respectively. In the same study, the prevalence of frailty was 9% and 63% were classified as “pre-frail” [50].

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The included PA programs varied substantially in terms of the PA modality, delivery methods and duration. Interestingly, compared with self-managed programs carried out in participants' discretionary time, on-site supervised sessions during the workday [25, 26, 29, 33, 34] tended to demonstrate high retention rates (71% to 100%) and good compliance. However, these programs were usually shorter in duration (less than 15 weeks). On-site pragmatic approaches, such as treadmill workstations, short breaks for balance and strength training or web-based interventions, can be easily integrated into the workday. However, the effectiveness of treadmill workstations was disappointing [32], and the collective evidence for other approaches was inconsistent [24, 27, 34]. These interventions should be further tested by high-quality replication studies before translational research can be conducted.

Multiple risk factor interventions mostly had poor description of the PA component of the intervention which reduced their reproducibility. For example, referrals to exercise lacked specifications of the type of exercise programs that were available to clients, or goal settings lacked descriptions of how goals were tailored to individuals, how they were monitored and by whom. In addition, most studies rarely reported the fidelity of the interventions. Hence it is recommended to use a template for reporting of PA interventions, such as TIDieR [51], in order to allow better replication of sensible effective PA interventions.

It is worth noting that about 40% of the interventions were conducted in universities or academic hospitals/clinics. This could possibly be due to ease of doing research in these settings coupled with environments that are conducive to health promotion messages, such as health-care providers or academics. Hence, the generalisability of intervention implementations and the findings of these studies is limited.

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A concerning finding is the poor quality of the RCTs we identified. The study quality could be improved by appropriate randomisation, intention to treat analysis, and the use of both objective and self-report measures of PA. For example, most multiple risk factor interventions included objective measures of CVD risks (i.e., blood tests or anthropometrics), but failed to use objective fitness tests or activity trackers (e.g., pedometer, accelerometer). Further, to improve the directness of evidence (i.e., the selection of outcome measures should better reflect the fitness dimension(s) that the proposed PA is likely to affect), such as measuring balance and strength outcomes in interventions involving yoga or tai-chi or aerobic fitness in Nordic walking interventions.

Research demonstrates increases in leisure-time PA after transition to retirement compared to during full employment [52, 53]. However, a systematic review has found disparities by socioeconomic status (SES); employees with low SES tend to be less active after retirement whereas those with high SES tend to be more active after retirement [53]. Focus groups with older retirees revealed that retirees from lower SES appeared to place lower value on the importance of leisure-time PA [54]. Therefore, improving PA whilst in the workforce may be of even greater importance to older people with low SES. In this review we identified three studies that specifically targeted low paid employees in manual occupations (i.e., factory employees) in Japan [36] and the UK [30, 38]. Only the study from Japan, which included social and environmental support in addition to behavioural counselling, demonstrated significant positive effects. However, this was a non-randomised study with a high risk of bias for almost all bias domains. Further work is therefore needed to ascertain what types of interventions can most effectively improve PA outcomes in older employees with low SES.

4.1 Strengths and limitations

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This review applied guideline-informed, rigorous methods to synthesise and evaluate the current evidence on workplace PA interventions for older employees. Compared with the 2016 systematic review by Poscia and colleagues [13], we have expanded upon the list of workplace health promotion interventions delivered to older employees by 11 studies [28-32, 34-36, 38-40] by carrying out a more sensitive and comprehensive literature search. However, some limitations need to be acknowledged: first, we limited our search to electronic databases which can result in missing unpublished workplace interventions. Second, studies tended to apply different measures or used different units for similar measures. Despite our best attempt to harmonise measures (e.g., converting kcal to minutes of at least moderate-intensity PA), comparability across studies is still limited and may explain the high heterogeneity; for example measuring postural control by balance platform as opposed to a Functional Reach Test. In addition, the current evidence base is limited by the number of studies included for each PA outcome and the quality of existing studies. Based on these limitations we have mapped out research gaps and suggestions for the conduct and reporting of future interventions targeting older employees (Table 4).

Table 4: Recommendations for the conduct and reporting of future workplace interventions targeting older adults

Aspect	Recommendation
Reach and recruitment	1. To enable calculation of the reach of workplace interventions we suggest that as a first step employers will endorse the intervention and provide the list of all employees or at least the number of the potential target population (e.g., age or year prior to retirement). It may also boost recruitment when the management of the workplace endorses such intervention.
Population	2. Expand the current target population (ie. mostly university and healthcare staff) to include a broader range of occupations, including manual jobs and workers from low socioeconomic status (SES) background. This is important for increased generalisability and reducing gaps between high and low SES workers in lifestyle behaviours after retirement.
Intervention content	3. Incorporate strength, balance, and flexibility training along with aerobic activities.. 4. Moving interventions from a "pilot" stage to larger trials and scalable programs; 5. Intervention targets should include screening for falls and injury risk, frailty and sarcopenia

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	6. Improve the reporting on the nature of PA interventions. For example, for interventions that involve a referral to a gym or other facility, describe precisely what exactly these referrals included: advice only, a written program, or a gym membership 7. Improve reporting of the intervention using a standard checklist (TIDieR) including implementation fidelity of interventions.
Study methodology	8. Cluster RCTs should include several clusters (workplaces) and should adjust for cluster effect; 9. Studies should assess confounders and adjust for their presence. 10. Interventions that incorporate multi-dimensional activities (e.g., yoga, taichi, dance) should assess the fitness outcomes specific to be improved by these activities (leg strength, balance, gait speeds). Standardising measures across studies will increase the of studies included in future meta-analyses. 11. Improving the conduct / reporting of RCTs in terms of methods of randomisation, concealment of allocation, blinding the assessors and analysts, reporting results for all randomised participants first before any ad-hoc analyses and including objective measures of physical activity
Duration and sustainability	12. Consider intervention beyond 6 months and evaluation of maintenance beyond a year

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Declaration

Ethics approval and consent to participation

Not applicable for systematic review

Availability of Data and Materials

Not applicable

Competing interests

No competing interests to declare.

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Data availability

All data relevant to the study are included in the article or uploaded as supplementary information.

Author contributions

DM conceptualised the research and drafted the first manuscript. DM and JS conducted the systematic search and selection. FS and DD conducted the quality assessment. AT and DM conduced the meta-analysis. DM and SM conducted the data extraction to tables. All authors (DM, FS, JS, AT, SM, KG, DD) participated in the interpretation of results, commented on first draft and gave final approval of the manuscript.

Consent to publication

Not applicable

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Figure 1: PRISMA flow diagram of selected workplace physical activity intervention studies

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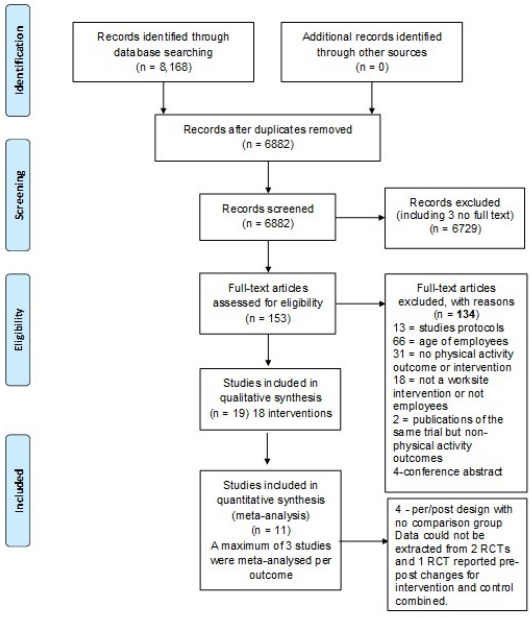


Figure 1: PRISMA flow diagram of selected workplace physical activity intervention studies

254x190mm (96 x 96 DPI)

Supplementary Table 1 Characteristic of worksite physical activity (PA) interventions delivered to older employees (organised by publication dates)

Study	Recruitment / reach	Intervention content / theoretical guidance	Intervention deliverers	Duration	Process data
Bassey, E. J., et al, 1987 [27]	Initial discussions for the support of the management; After a publicity launch in the factory newspaper, a personal letter of invitation was sent to all workers through factory's internal mail system; 35% attended introductory meetings and ultimately 23% agreed to participate	"The New Aerobics" (1970) Cooper's prescription for people aged ≥50yrs, which isa progressive walking program instructed at workplace but carried out at employees' leisure time. 1st week, walking 20mins$_{*}^{-1}$d and at a pace of 5.2km$_{*}^{-1}$hr preferably 5 days per week. Subjects can increase duration of walks and frequency. In 2nd phase the intensity can increase based on heart rate response to exercise at 4.8km$_{*}^{-1}$hr. The target dose was 40mins at 7km$_{*}^{-1}$hr 5 days per week. Individual discussion at beginning (how to progress) and half way through. The progression was individually discussion on courses and timing at start and mid program. Participants were asked to keep daily log-card of duration and distance walked. Encouraging articles in the factory newspaper were published monthly.	Unsupervised program – no mentioning of the profession of the person conducted the meeting to inform the grade of effort	12 weeks program divided to 6 weeks each phase	59 out of 108 completed the intervention (54%). Log cards: only 1/3 nearly achieved the speed and distance prescribed for the last 2 weeks, 1/3 provided evidence on logs for extra walking but substantially below prescription, and 1/3 failed to return log card. The most compliant were the most active before program started
Sharpe, P. A. et al 1992 [35]	Advertising at work, direct contact with departmental representatives, telephone calls from the research staff and provision of additional dates and sites to enhance the accessibility of screening; 250 employees did health screening but the % from total employees is unknown	Health promotion activities after health risk screening (smoking, blood pressure, level of blood lipids and body weight) and brief communication of results by nurse. Participants could choose to have face-to-face with meetings with health counsellors, to join walking groups at work and/or exercise on their own at home or at university facilities. No other promotional activities were reported.	health counsellors at work provided face-to-face with counselling based on participants choice	12 months	60 out of 95 experimental participants (63%) completed the program. Behaviour change self-report: 48% of Control and 53% of have done anything to increase fitness. started program on their own- 63% and 73% of C and Int. took worksite program 13% of C and 11% of Int; used worksite facilities: 11% of C 16% Int.
Arao, T. et al 2007 [36]	Two companies were recruited, all together 5 sites. Medical check-ups were conducted and 192 employees were identified to have CVD risk factor(s) of those 177 (92%) were eligible and consented and ultimately 155 (87%) were allocated as a cluster to LiSM (n=92) and Control (n=85) groups.	Theoretically driven (Transtheoretical model and Social Cognitive Theory) Life Style Modification Program for Physical Activity and Nutrition (LiSM-PAN). Structured counselling for goals setting for each behaviour and monthly meeting. Log book for steps a day (10,000 as a goal); increasing incrementally; if reached take another sport/exercise class. Environment support at worksite: 1) setting up a walking course; 2) installing exercise facilities; 3) displaying caloric content table of the lunch menu at cafeteria; 4) cafeteria manager advised on healthy cooking for better diet. Social support at home: printed material on healthy diet and cooking; participant and spouse going together for exercise	Research staff were trained in structured counselling;	6 months	95.2% completion. 97.1% attended monthly counselling 86.7% self-monitored the walking steps 54.7% controlling their dietary targeted activities

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Study	Recruitment / Reach	Intervention content / theoretical guidance	Intervention deliverers	Duration	Process data
Granacher et al. 2011 [34]	Several companies in Basel were approached and 2 large companied (~ 500 employees) agreed (randomization at cluster level). Informational session was held explaining the intervention and eligibility criteria, 34 employees consented and participated.	Three times a day five days a week of training in: Static balance exercises – in morning at arrival to office Dynamic balance exercises – after lunch break Strength enhancing exercises before leaving office New exercise routines were assigned each week to be repeat for the whole week before progressing.	Expert in balance and strength training saw each participant every week	8 weeks of training followed by 8 weeks without training data collected at 8 and 16 weeks	All 17 intervention participants completed the program (100%). Program compliance was excellent with participants of the intervention group completing 99% of the scheduled training sessions (~ 119 out of 120 training sessions).
Hughes et al. 2011 [24]	Worksite advertisement Research participants recruited via announcements on staff list-serves, targeted e-mails, staffed recruitment tables at events in highly trafficked buildings, and flyers posted throughout the university. Out of 647 enquiries 66% deemed eligible and 423(65%) enrolled in the study.	COACH –shaped by the Enhance Wellness Program for seniors in the community and guided by the Transtheoretical Model. One face- to- face meeting with coach to review the participant's health-risk profile (baseline assessment), discuss the behaviors the participant was interested in adopting, and discuss the availability of health behavior change programs on or off campus, assist in creating plan to achieve goals. Biweekly contact by email/telephone for the first 6 months to re-evaluate plan, goals and actions. In the remaining 7 to 12 months the contact was monthly. Health-risk profile assessments were at 6 and12 months. RealAge- A Web page contain health risk test and generate health profile based on family history, diet and nutrition, PA, tobacco and alcohol use, satisfaction with family relationships, and use of preventive health care. Participants used the Web site to select behaviors to work on and create plans to meet behavioral goals. The RealAge site tracked each time a participant used it, and the site forwarded this information to the study team at regular intervals, enabling us to document use.	COACH -MPH graduate trained in behavioural change coaching RealAge – research staff	12 months	Retention: 91% in COACH and 84% RealAge. Fidelity data: COAC: Coach spent 9.25hr per participant – there were 688 emails 77% by the coach and 48 telephones FU, RealAge- 59% completed all health risk assessment and 53% engaged in at least one of the web link.
Palumbo et al. 2012 [26]	One medical centre targeting nurses currently employed full time or part time in staff nurse position that involved lifting patients. About 70 nurses place their interests and 14, on a first come first served basis, were randomized.	Tai Chi workplace wellness program, guided by the Revised Health Promotion Model by Pender (1996). Classes on site were carried once a week (including 15 minutes breathing exercise, 30 mins tai chi practice 5mins visualization cool down). Additional practice on their own 10mins 4 times a week	Experienced tai chi instructors (22 years of teaching)	15 weeks	Five out of the 7 intervention participants (71%) completed the tai-chi program. Attendance 82% range:69%-92% Fidelity- two classes were cancelled

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Study	Recruitment & Reach	Intervention content / theoretical guidance	Intervention deliverers	Duration	Process data
Strijk et al. 2011 , 2012 [22,23]	All workers of two academic hospitals were invited to participate between April and October 2009; Among 3756 employees invited to participate, 1101 (29%) agreed to participate and 730 (19%) allocated into two groups	3 visits to Personal Vitality Coach (PVC) at worksite: at start of intervention, followed by 2 consecutive visits at 4-6 weeks and 10-12 weeks. Meeting 30 minutes each. Weekly yoga 45mins and workout sessions (intensity 65%-90% of the age-predicted maximum heart rate) for 45mins. Plus 45 min unsupervised aerobic exercise. Fruit & Vegetable at the session program name 'Vital@work'	Certified yoga instructor and exercise physiologist	6 month interventio n period and additional 6 months sustainabili ty testing	293 out of 367 (80%) intervention participants completed the program. PVC visits 100%. Dose delivered – Yoga sessions 72.3% (Amsterdam - 89.3%; Leiden - 58.3%) Workout sessions 96.3% (Amsterdam – 95.1%; Leiden – 97.4%) High satisfaction score
Cook, et al. 2015 [27]	A recruitment flyer was emailed by company officials to all employees 50 years plus; interested employees were provided information on study and a link to the online baseline survey (received $25 if completed); out of 290 employees, 279 (96%) completed the baseline survey.	HealthyPast50, an automated web-based program containing multimedia health promotion information and guidance based on Social Cognitive Theory. Modules: Facts about Healthy Ageing, My Health Profile – assessment module with questionnaires and recommendations, Stress and Mood Management, Healthy Eating, Active Lifestyle, and Tobacco Free. Encouragement of self-assessment (My Health Profile) before accessing modules. the content of the Active Lifestyle module is oriented toward older users, including the use of the Physical Activity Readiness Questionnaire (PAR-Q), assessing one's readiness for physical activity, the relationship of physical activity to mobility and energy in later years, and the need to moderate the intensity of exercise as one ages	No human contact for the program; project staff can be reached by telephone for queries	3 months; program progress by monthly email and reminder to access	Formative evaluation of the program including focus groups and testing materials. Process evaluation: mean # log-in:4.33 (SD 4.28, range 0-28), the mean minutes in the program was 102.26 minutes (SD 148.32), and the mean number of pages viewed was 11.04 (SD 20.08, range 0-120).
Abbas et al. 2015 [38]	Targeted recruitment of low-paid unskilled or semi-skilled employees age >40 on social care. A single individualised invitation was sent with simultaneous promotion of the programme within standard employer–staff communication routes. Out of 3264 target group (low SES employees), 665 (20% response rate) attended health screening. Of them, 579 were eligible and 427 (74%) returned for the follow-up assessment	Health screening of all major cardiovascular risks was offered to employees of low SES in deprived area. Based on screening results participants were stratified to either "high" , "moderate" or "low" CVD risk. The intervention included referrals to either GPs (if pharmacological treatment were needed), Weight management (in the community, exercise training, smoking cessation and alcohol reduction counselling either face to face or by telephone. Employers ensured that time of assessments and time spent on health promotion will be accounted towards the daily working hours. These services were provided in the community.	Occupational health nursing staff. The nurses were trained to provide smoking cessation and alcohol awareness delivery at the national standard.	9 months interval from initial assessment to follow-up	The attendance to health screen was 20%, much lower than the general population that offered similar program (health screen). Fidelity=There was a record of the referral number given to each domain of intervention (exercise 38%, smoking 5%, alcohol 1%and weight management 20%). No account on the attendance to service referred.

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Study	Recruitment & Reach	Intervention content / theoretical guidance	Intervention deliverers	Duration	Process data
Low et al. 2015 [28]	A promotional flyer to all female at age range & invited to call the centre if interested in reduction of CVD risk; once reached 60 employees who had ≥1 risk of CVD no further recruitment effort was made	Multi-components risk reduction educational sessions in groups on: weight loss/nutrition, stress management, exercise training, and smoking cessation accompanied by access to an on-site gymnasium, and organized group walks. Intervention groups received weekly communication via phone or e-mail, incorporating goal-setting and suggestions for overcoming obstacles.	Not specified	6-months for classes, gym availability organized walks and weekly communication Follow-up survey at 12 months	Attendance to 1st classes was highest for weight loss (42% & 47% control). Stress reduction (30% & 21%) and physical activity 25% to 26%) followed by very poor attendance to consecutive classes. 3 attended group walking; use of Gym 0-20 days over 6 months . Follow-up 16/57 (28%)
Chen et al. 2016 [37]	Approached six worksites small- and medium scale enterprises (SMEs). Three sites showed interest and approved. Among 1245 workers in the three sites, 438 (35.2%) met study criteria of being full-time employed and aged over 50 years. These were invited. Of those, 264 (60.3%) showed interest and finally 108 (25%) agreed to participate.	Phase I: on-site lectures and workshops on healthy living specifically impact of metabolic disorders and individualized self-management goal setting. Biweekly meetings face-to-face with occupational health nurse (OHN) to share experience, give mutual support and encouragement, learn about managing own health, and cultivate leadership. Phase II: on-site team led meetings; discussing barriers reinforcing personal goals; recording food consumption and exercise. Logs were reviewed by OHN and awards/prizes were given for excellent improvement	Occupational Health Nurse	Total 24 weeks Phase 1 was 4-week phase II 20- weeks.	111 from initial 136 intervention participants (80%) completed the program Qualitative Evaluation: Participants found it difficult to records daily PA and diet so no review of records was made; Working overtime reduce the compliance with planned activities and goals; activities forced to cut short due lack of support from management at low level
Naug et al. 2016 [39]	Contacted with employer; Information session meetings were arranged to build rapport with the employers and drivers; after that, drivers were invited to an information presentation and were given an information package.	Educational sessions of 60mins delivered fortnightly in groups at worksite, interactive and fun. Content were derived from the "Life! Taking action on diabetes" program which was based on health psychology theories with a focus on self-regulation. Targeting physiological change due to inactivity, harm of sitting, healthy nutrition. Participants were provided pedometers and asked to record steps each day for 6 weeks.	Two postgraduate students with expertise in adults' education, human physiology & nutrition	6 weeks	21 out of 33 that started program (64%) completed the program. Qualitative after intervention : drivers said " I walk around the bus during breaks" "I, use the back seats of the bus to do push-ups"; "park my car far from depot to walk more"

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Study	Recruitment & Reach	Intervention content / theoretical guidance	Intervention deliverers	Duration	Process data
Kocur et al. 2017 [29]	Workplace advertisement Postmenopausal female who replied to the advertisement regarding the study (n=91) and net the inclusion criteria (n=44) but no data on % from the total number of the target population.	Nordic walking training program techniques and training 3 times a week for one hour including 10-min warm up phase at start 5-min cool down at end. Intensity progress 40%-70% of heart-rate reserve	Two instructors specialised in Nordic walking regulated the pace	12 weeks	No report on program attendance or fidelity.
Chopp-Hurley et al. 2017 [31]	Workplace advertisement through e-mail list serves, webpage at McMaster workplace wellness committee Potential participants had to have clinical osteoarthritis either of the hip or the knee; no number was provided for those that were screened or refused.	Program of static lower extremity strengthening exercise; designed initially for Knee Osteoathritis to avoid KAM (knee adduction moment); The program includes Yoga poses such as squats and lunges; A perceived intensity of 5-7 from a total 10-point scale was recommended. On-site 4 timea a week sessions in the University sport and recreation facility were provided before work (7-8 AM). Participants were asked to attend at least 3 sessions/week. Participants in intervention received weekly feedback on their performance of exercises. Incentive of $50 was given to maintain participation in both groups and control participants were offered the program at the end of trial	Not specified	12 weeks	From initial 24 participants (12;12) 3 intervention participants did not complete FU (25%) as opposed to 1 in the control (12%). Exercise group attended on average 1.2 sessions per week; median was 0.5 classes per week, lower quartile was 0 a week and upper quartile was 3.9 per week thus estimate that 33% adhered to 3 sessions per week.
Bergman et al. 2018 [32]	13 different companies in 17 sites were involved (number of employees age 40-67 not given). Information session were given to office workers- the information included who is eligible (i.e., at least 4 days/wk in the office with sedentary work tasks with a sit-to-stand workstation, don't have a travel day during their work). Information on study strict procedures and measurement only 100 assessed and 80 enrolled	Health consultation about 30-40 minutes about their screening results and sitting time at work and diet. Folder with written information about PA recommendations for moderate and vigorouse aerobic activity and strength, how to increase light PA and Non-exercise PA such as climbing stairs, walk and talk meetings. After this session the treadmill adapted to desk use was installed next to the existing workstation. Physiotherapist adjust the workstation and instruct the worker how to use the treadmill in the first week – any pace 15min twice a week. Thereafter they were recommended to use it for 1hr a day preferably longer. Booster email from research team remined the hazard of sitting, reminder to use treadmill at least 1hour a day. These emails were sent at 6,20,31 and 50th week over the program.	Nurses conducted the health consultation and boosters emails by research team	13 months	Of the 80 randomised (40;40) 34 continued their participation for 13 months. In the control 92% continued for 13 months. No data was collected on usage of the treadmill during work Threat for contamination between group was acknowledged

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Study	Recruitment & Reach	Intervention content / theoretical guidance	Intervention deliverers	Duration	Process data
Scapellato et al. 2018 [40]	Targeted strategy; screening for eligibility (at least 1 CVD risk: blood pressure, weight, waist circumference lipids, impaired fasting glucose) 1892 workers on voluntary basis agreed to be screened; 1574 did not meet the inclusion criteria 318 recruited (50% from 632 eligible). Final recruitment 226 (35% from eligible).	As part of the Occupational Health and Safety approach, a workplace intervention was offered to promote healthy lifestyle for healthcare workers in the hospital who are at risk; intervention included individualized counselling meeting a month after baseline assessment to discuss results of PA and diet, and diverted to health counselor to identify PA and diet goals and strategies to achieve them. Helpful printed material were given such as food pyramids and a brochure of exercise. After 3 months participants received further motivational support via phone counselling to remind their goals and assist to change lifestyle habits.	Hospital Employees Occupational Health practitioners, health promotion counsellors and hospital dietitians	6 months	Fidelity=92 out of 318 eligible recruited (29%) did not receive the intervention. Of the 226 that received intervention, only 169 remained for 6-month assessment. No information on % phone calls given at 3-month. Reason to lost to follow up were attributed to “no time” & “too many work commitments”; only 40% who completed the study favoured adherence to dietary recommendations
Qi et al. 2019 [33]	Worksite advertisement; posters, person-to-person invitations using card-sized flyers, promotion of the study by colleagues and an e-mail request from the University Administration promoting the research projects that office workers may be interested to partake in. No data on number of potential eligible (55+ in the Uni) or how many were interested	Based on evidence that taichi or resistance training alone can improve physical function and psychological wellbeing of older adults. The intervention offered 3 session per week from a choice of 8 different timetables to allow flexibility: Monday, Wednsday and Fridays (2 session) 7-9 AM midday and afternoon 3-6PM. Session took 45 minutes. Ten Taichi movements were practiced with theraband tied to palms to a relaxing/ calming music. Theraband degree of effort were progressive. Every fortnight a Rating of Perceived Exertion was completed. When RPE was below 5 the degree the difficulty was increased	Taichi instructor, which is the first author of the paper, delivered the intervention	12 weeks with mid-term (6-weeks) assessment	At 6 week 9 participants withdrew (22%) and no further withdrawals happened at 12 weeks. Reason to lost to follow up were: back issue, knee issue, time conflict flue and no reason given. Data on compliance was not reported; there were 36 sessions offered but no data on mean session attend.
Edman et al 2019 [41]	Worksite advertisement; e-mail invitations (3) to get health coaching for free. Flyers were posted in various locations in the main hospital including the cafeteria and in satellite office locations.	Health coaching on cardiovascular health delivered face-to-face or by phone or in combination. A cyclie is considered 12 weeks during which 6 consultations 2-30min each must be completed. Sessions are client centred behavioural changes strategies and include provision of education and individualized support to make change. Sessions start with assessment, changes from previous session, goal setting (fortnightly), elaborate on topic the client chose (the SMART framework model)	A certified Health Coach with 4 years’ experience with a PhD degree in Nutrition (the first author)	5.5 months with cycle is 12 weeks with option to continue.	74% (40/50 participants) completed 12 weeks. Participants averaged 7.8 sessions (range 6–12) over 5.5 months (range 3.0–10.0). Coaching were in-person for 57% and 17% receive only telephonic coaching and 26% received combination of both.

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Supplementary Table 2: Risk of bias assessment for RCTs [a] and quasi-experiments [b]

RCTs	Risk of bias (High, low, unclear)					
Source	Randomisation sequence Allocation	Concealment	Performance bias	Ascertainment bias [c]	Incomplete outcome data	selective reporting
Bassey et al. 1987	Unclear	Unclear	High	Low	High	Low
Sharpe et al. 1992	High	Unclear	High	High	High	Low
Granacher et al. 2011	High	Unclear	High	Low	Unclear	Low
Hughes et al. 2011	Low	Unclear	High	High	High	Low
Strijk et al. 2012	Low	Low	High	Low	High	High
Palumbo et al. 2012	Unclear	Unclear	High	low	High	High
Cook et al. 2015	Low	Low	High	High	High	Low
Low 2015	Unclear	Unclear	High	high	Unclear	High
Kocur et al. 2017	Unclear	Low	High	Low	High	Low
Chopp-Hurley et al. 2017	Low	Low	High	Low	Low	Low
Bergman et al. 2018	Low	unclear	High	Low	Low	Low
Qi et al. 2019	unclear	Low	High	low	High	Low
Non-RCT: Confounding						
Chen et al., 2016	High	Low	High	High	High	Low
Arao, 2007	High	Low	High	Low	High	Low

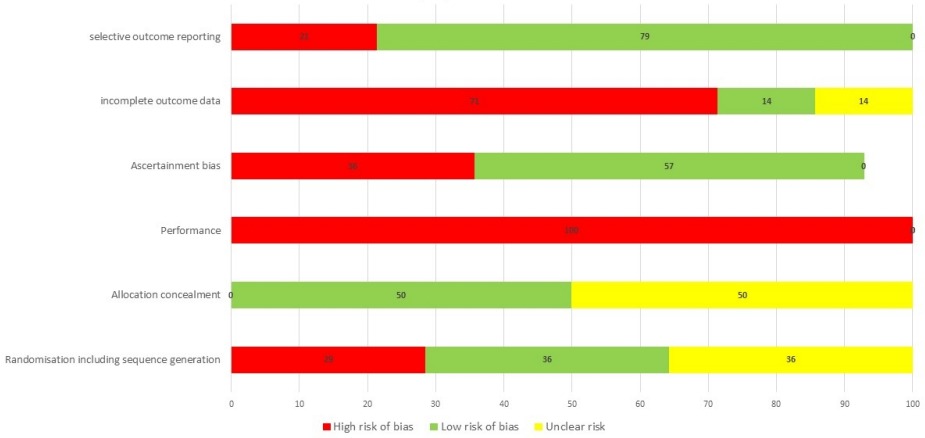
[a] Higgins JPT, Altman DG, Gotzsche PC, et al. Cochrane Bias Methods Group Cochrane Statistical Methods Group. The Cochrane Collaboration's tool for assessing risk of bias in randomised trials. *BMJ* 2011;343:d5928. doi:10.1136/bmj.d5928

[b] Sterne JA, Hernán MA, Reeves BC, et al. ROBINS-I: a tool for assessing risk of bias in non-randomised studies of interventions. BMJ. 2016 Oct 12;355:i4919. doi: 10.1136/bmj.i4919. PMID: 27733354; PMCID: PMC5062054.

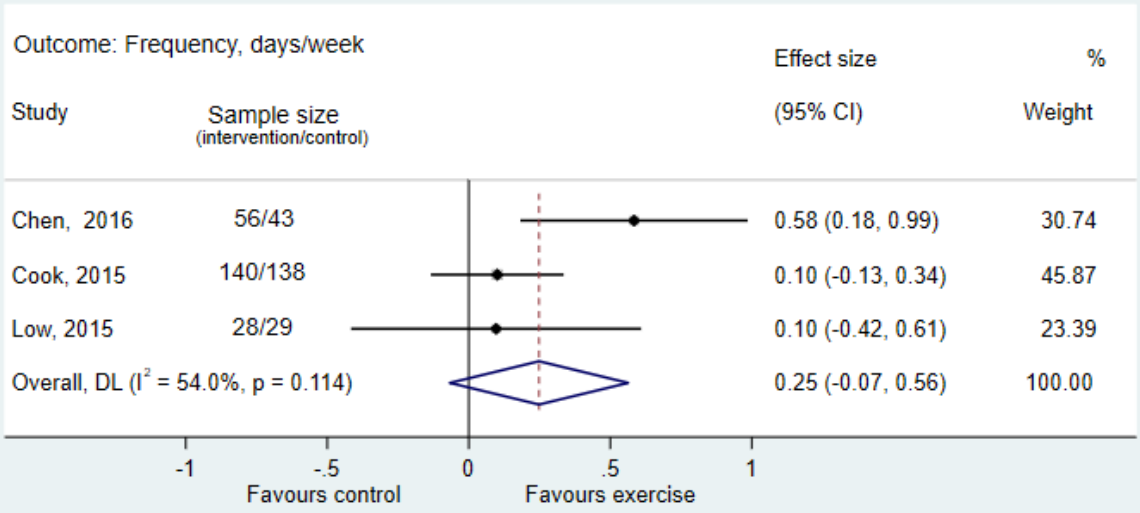
[c] Studies that had included objective physical activity outcomes and subjective outcomes were classified as low

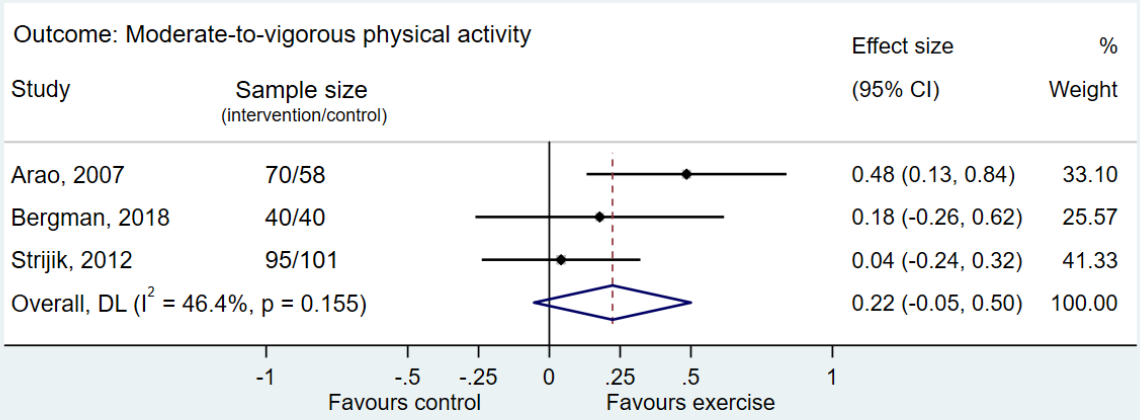
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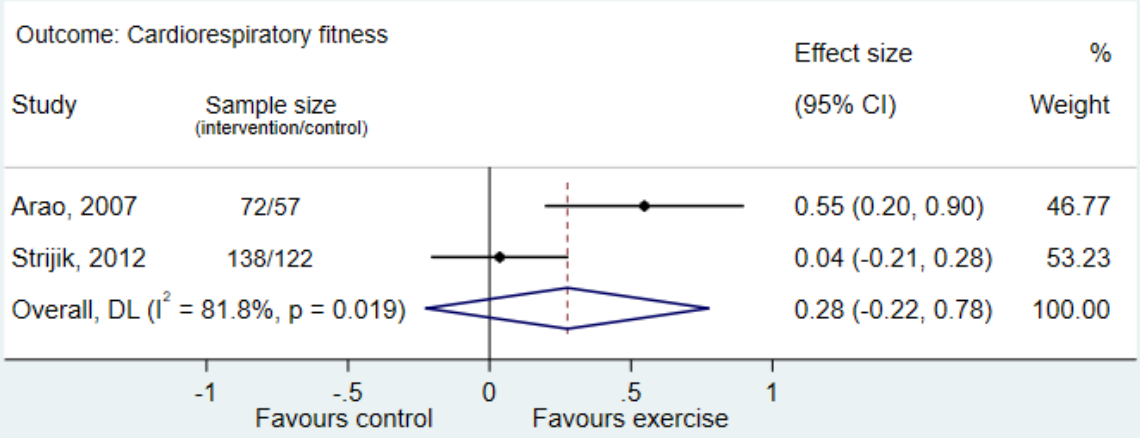
Supplementary Figure 1 : Risk of bias summary of each item in 14 worksite interventions delivered to older employees

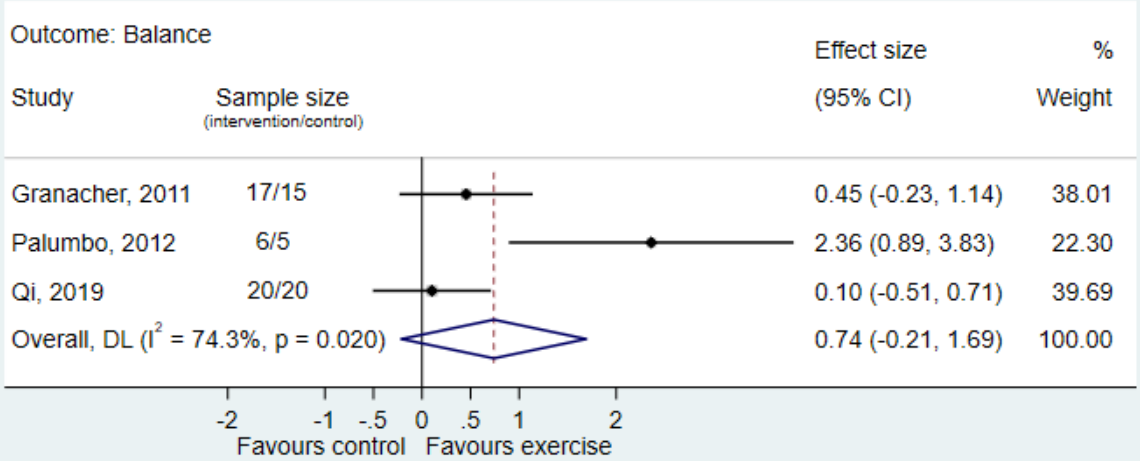


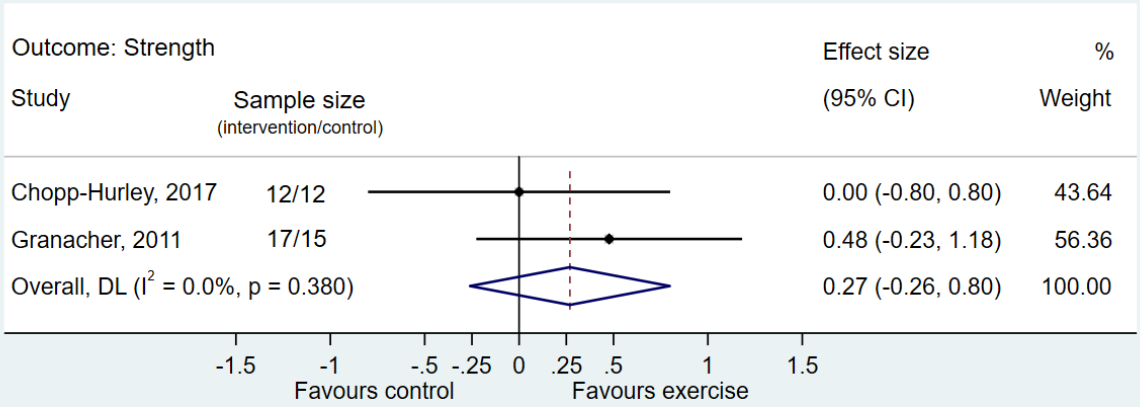
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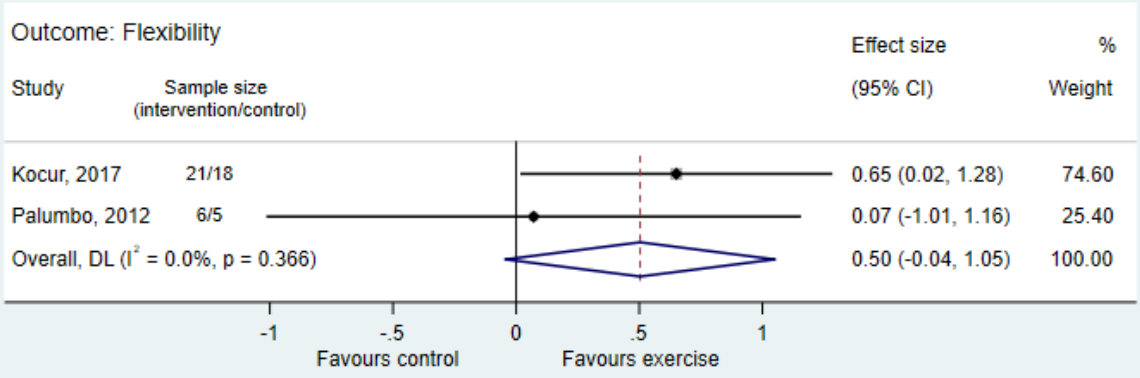

Outcome: Frequency, days/week
Study
Sample size
(intervention/control)
Effect size
(95% CI)
%
Weight
Chen, 2016
56/43
0.58 (0.18, 0.99)
30.74
Cook, 2015
140/138
0.10 (-0.13, 0.34)
45.87
Low, 2015
28/29
0.10 (-0.42, 0.61)
23.39
Overall, DL (I² = 54.0%, p = 0.114)
0.25 (-0.07, 0.56)
100.00
-1
-.5
0
.5
1
Favours control
Favours exercise


Outcome: Moderate-to-vigorous physical activity
Study
Sample size
(intervention/control)
Effect size
(95% CI)
%
Weight
Arao, 2007
70/58
0.48 (0.13, 0.84)
33.10
Bergman, 2018
40/40
0.18 (-0.26, 0.62)
25.57
Strijik, 2012
95/101
0.04 (-0.24, 0.32)
41.33
Overall, DL (I² = 46.4%, p = 0.155)
0.22 (-0.05, 0.50)
100.00
-1
-.5
-.25
0
.25
.5
1
Favours control
Favours exercise


Outcome: Cardiorespiratory fitness
Study
Sample size (intervention/control)
Effect size (95% CI)
% Weight
Arao, 2007 72/57 0.55 (0.20, 0.90) 46.77
Strijik, 2012 138/122 0.04 (-0.21, 0.28) 53.23
Overall, DL (I^2 = 81.8%, p = 0.019) 0.28 (-0.22, 0.78) 100.00
-1 -.5 0 .5 1
Favours control
Favours exercise


Outcome: Balance
Study
Sample size (intervention/control)
Effect size (95% CI)
% Weight
Granacher, 2011 17/15 0.45 (-0.23, 1.14) 38.01
Palumbo, 2012 6/5 2.36 (0.89, 3.83) 22.30
Qi, 2019 20/20 0.10 (-0.51, 0.71) 39.69
Overall, DL (I^2 = 74.3%, p = 0.020) 0.74 (-0.21, 1.69) 100.00
-2 -1 -.5 0 .5 1 2
Favours control
Favours exercise


Outcome: Strength
Study
Sample size
(intervention/control)
Effect size
(95% CI)
%
Weight
Chopp-Hurley, 2017 12/12 0.00 (-0.80, 0.80) 43.64
Granacher, 2011 17/15 0.48 (-0.23, 1.18) 56.36
Overall, DL (I² = 0.0%, p = 0.380) 0.27 (-0.26, 0.80) 100.00
-1.5
-1
-.5
-.25
0
.25
.5
1
1.5
Favours control
Favours exercise


Outcome: Flexibility
Study
Sample size
(intervention/control)
Effect size
(95% CI)
%
Weight
Kocur, 2017 21/18 0.65 (0.02, 1.28) 74.60
Palumbo, 2012 6/5 0.07 (-1.01, 1.16) 25.40
Overall, DL (I² = 0.0%, p = 0.366) 0.50 (-0.04, 1.05) 100.00
-1
-.5
0
.5
1
Favours control
Favours exercise

PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
TITLE			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	1
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	2
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	4-5
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	5
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	6
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	6
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	6-7
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	Supplementary file
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	6-7
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	7
Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	6
Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	7-8
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	8
Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I^2) for each meta-analysis.	8 Table 3

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PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	8, GRADE Table 3
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	NA
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	8 -9 & Figure 1
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	Table 1 & Supp T1
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	19, & Supp T2
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	18 Table 2
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	19-21 Table 3
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	Table 3
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	NA
DISCUSSION			
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	23
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	25-26
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	26 & Table 4
FUNDING			
Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	NA

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PRISMA 2009 Checklist

From: Moher D, Liberati A, Tetzlaff J, Altman DG, The PRISMA Group (2009). Preferred Reporting Items for Systematic Reviews and Meta-Analyses: The PRISMA Statement. PLoS Med 6(7): e1000097. doi:10.1371/journal.pmed1000097

For more information, visit: **www.prisma-statement.org**.

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Physical Activity Worksite Interventions Delivered To Older Employees: A Systematic Review

Medline Search Strategy – limit final results by age (45 and up), humans, not reviews, English

Work Terms	Physical Activity	Intervention
MeSH terms		
Workplace	Exercise (exp)	Health Education
Occupations	Walking	Inservice Training (exp)
Employment (exp)	Postural balance	Occupational Health Services
	Physical fitness (exp)	Occupational Health
	Physical exertion	Health Promotion (exp)
	Exercise Therapy (exp) (inc. resistance training, Muscle Stretching Exercises)	"Physical education and training"
	Quality of Life	
	Life style (exp) (inc. Healthy Lifestyle)	
	Health Behavior	
	Yoga	
	Sports (exp)	
Keywords		
Worksite.mp	Aerobics.mp	Intervention.mp
Work environment.mp	Physical activity.mp	Program.mp
Job site.mp	Wellness.mp	Prevent*.mp
Employee*.mp	Fitness.mp	Promot*.mp
Worker*.mp	Pilates.mp	
Economically active.mp	Exercise.mp	
	Tai chi.mp	

CINAHL Search Strategy – limit final results by age (45 and up), humans, not reviews, English

Work Terms	Physical Activity	Intervention

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MeSH terms?		
Work Environment	Exercise (exp)	Health Education
Occupations	Walking	
Employment (exp)	Postural balance	Occupational Health Services
Work Environment	Physical fitness (exp)	Occupational Health
	Exertion	Health Promotion (exp)
	Therapeutic Exercise	"Physical education and training"
	Quality of Life	
	Life style (exp)	
	Resistance Training	
	Yoga	
	Sports (exp)	
	Group exercise	
Keywords		
Worksite.mp	Aerobic Exercises	Intervention.mp
	Physical activity	Program.mp
Job site.mp	Wellness	Prevent*.mp
Employee*.mp	Psychological wellbeing	Promot*.mp
Worker*.mp	Pilates	Inservice training.mp
Economically active.mp	Exercise.mp	
Workplace.mp	Tai chi.mp	

PsycInfo Search Strategy – limit final results by age (40 and up), humans, English

Work Terms	**Physical Activity**	**Intervention**
MeSH terms		
	Exercise (exp)	Health Education
Occupations	Walking	Inservice Training (exp)
Employment (exp)	Physical activity	
	Physical fitness (exp)	Occupational Health

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	Movement Therapy	Health Promotion (exp)
	Lifestyle changes	Workplace Intervention
	Quality of Life	
	Aerobic exercise	
	Lifestyle (exp)	
	Health Behavior	
	Yoga	
	Sports (exp)	
Keywords		
Worksite.mp	Aerobics.mp	Intervention.mp
Work environment.mp	Physical activity.mp	Program.mp
Job site.mp	Wellness.mp	Prevent*.mp
Employee*.mp	Fitness.mp	Promot*.mp
Worker*.mp	Pilates.mp	
Economically active.mp	Exercise.mp	
Workplace.mp	Tai chi.mp	
Employment.mp	Resistance training.mp	

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▼ Search History (47) Vie

# ▲	Searches	Results	Type	Actions	Annotation
1	Workplace/	23992	Advanced	Display Results | More	
2	occupations/	23208	Advanced	Display Results | More	
3	exp Employment/	88952	Advanced	Display Results | More	
4	worksite.mp.	2972	Advanced	Display Results | More	
5	work environment.mp.	9401	Advanced	Display Results | More	
6	job site.mp.	77	Advanced	Display Results | More	
7	employee*.mp.	64566	Advanced	Display Results | More	
8	worker*.mp.	201742	Advanced	Display Results | More	
9	economically active.mp.	606	Advanced	Display Results | More	
10	exp exercise/	203707	Advanced	Display Results | More	
11	walking/	34042	Advanced	Display Results | More	
12	postural balance/	24102	Advanced	Display Results | More	
13	exp physical fitness/	31436	Advanced	Display Results | More	
14	physical exertion/	56619	Advanced	Display Results | More	
15	exp exercise therapy/	52973	Advanced	Display Results | More	
16	quality of life/	204059	Advanced	Display Results | More	
17	exp life style/	96894	Advanced	Display Results | More	
18	health behavior/	51575	Advanced	Display Results | More	
19	yoga/	2968	Advanced	Display Results | More	
20	exp sports/	188295	Advanced	Display Results | More	
21	aerobics.mp.	645	Advanced	Display Results | More	
22	physical activity.mp.	118123	Advanced	Display Results | More	
23	wellness.mp.	10986	Advanced	Display Results | More	
24	fitness.mp.	96101	Advanced	Display Results | More	
25	pilates.mp.	542	Advanced	Display Results | More	
26	exercise.mp.	366702	Advanced	Display Results | More	
27	tai chi.mp.	1893	Advanced	Display Results | More	
28	health education/	61303	Advanced	Display Results | More	
29	exp inservice training/	29418	Advanced	Display Results | More	
30	Occupational Health Services/	10609	Advanced	Display Results | More	
31	occupational health/	34329	Advanced	Display Results | More	
32	exp health promotion/	78692	Advanced	Display Results | More	
33	physical education.mp. and training/ [mp=title, abstract, original title, name of substance word, subject heading word, floating	0	Advanced	Save | More	

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